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# **BQLX <GO> Help on BQL in Excel Help Page**

Enter BQLX <GO>, then press <Help>

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# Introduction to BQL

BQLX <GO> provides information on using the Bloomberg Query Language (BQL) to retrieve data and perform analysis in Microsoft® Excel.

BQL is an API based on normalized, curated, point-in-time data. Using BQL, you can perform custom calculations directly in the Bloomberg data center. Analyses that previously required you to download thousands of data points and perform complicated Microsoft® Excel manipulation now require only one formula. You just specify the data you want and the calculation you want to perform—including, but not limited to, arithmetic and statistical operations, filtering, grouping, scoring, and dates-based analysis—so you can synthesize large amounts of data and extract the exact information you need.

In a single BQL query, you construct a universe of tickers (such as a list of peers) and define the data you want (such as earnings per share). You can apply analysis and calculations on the data directly on the server side, such as calculating an average or multiplying two values. You can also add optional parameters that fit the data to your model, such as the time period (if you want historical data) and currency (if you want to convert it).

There are two different formulas you can use in Excel to perform a BQL query:

- **BQL():** Uses the standard Bloomberg formula structure (i.e., "Ticker", "Field", "[optional parameters]"). BQL() is the simplified formula recommended for data retrieval. For more: [BQL\(\) Formulas](#).
- **BQL.Query():** Uses a raw string of BQL code, containing *get()* and *for()* clauses. BQL.Query() exposes the full capabilities of BQL in terms of screening, aggregation, and server-side analysis. For more: [BQL.Query\(\) Formulas](#).

## Getting Started with BQL

The following videos present an introduction to using BQL to perform security analysis.

### BQL Essentials Series

| Type                                                                                | Title                                          | Description                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Episode 1: What is BQL? (5:45)</a> | An introduction to using BQL in Excel.                                                                                                                                                                 |
|  | <a href="#">Episode 2: BQL Syntax (11:59)</a>  | An overview of using BQL syntax to construct queries and return data.<br><br> <a href="#">Companion Spreadsheet</a> |

| Type                                                                              | Title                                                                | Description                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Episode 3: Data Retrieval &amp; Manipulation</a> (17:12) | <p>An overview of retrieving current and historical data, as well as using BQL functions and universe functions to screen and aggregate data in the Bloomberg cloud.</p> <p> <a href="#">Companion Spreadsheet</a></p>                    |
|  | <a href="#">Episode 4: BQL for Equity Analysis</a> (8:41)            | <p>An overview of performing equity analysis in BQL, incorporating equity fundamentals, estimates, and revisions into queries.</p> <p> <a href="#">Companion Spreadsheet</a></p>                                                          |
|  | <a href="#">Episode 5: BQL for Fixed Income Analysis</a> (6:50)      | <p>An overview of performing fixed income analysis in BQL, incorporating fixed income fundamentals, corporate structures, issuance analysis, and index-level aggregation into queries.</p> <p> <a href="#">Companion Spreadsheet</a></p> |

## Getting Started

The following videos help you get started using the Bloomberg Query Language (BQL) and the BQL Builder to analyze Bloomberg data in Microsoft® Excel.

| Type                                                                                | Title                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">How to Use BQL</a> (33:04) | <p>A video tutorial that shows you how to use the BQL Builder to request historical data, aggregate data by sector, screen for securities, and more.</p> <p> <a href="#">Companion Spreadsheet 1</a></p> <p> <a href="#">Companion Spreadsheet 2</a></p> |

| Type                                                                              | Title                                                                             | Description                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Learning by Example: Using the BQL Builder Example Library (2:42)</a> | A short video that shows you how to load examples in the BQL Builder and customize them to fit your own analysis.                                                                                                                         |
|  | <a href="#">Spreadsheet Builder: Screening an Index for Default Risk (2:46)</a>   | A short video that shows you how to use the Spreadsheet Builder to screen for default risk.                                                                                                                                               |
|  | <a href="#">Getting Started: Fund Holdings (4:26)</a>                             | <p>A short video that shows you how you use BQL and the BQL Builder to easily download fund holdings and weights, as well as aggregate fund holdings by sector.</p> <p>For more on funds: <a href="#">BQL for Institutions/Funds</a>.</p> |
|  | <a href="#">Get Bloomberg Data in Excel through BQL (0:38)</a>                    | A short video that shows you how to drag and drop a BQL query from a Bloomberg Terminal® function into Excel.                                                                                                                             |

## BQL Builder

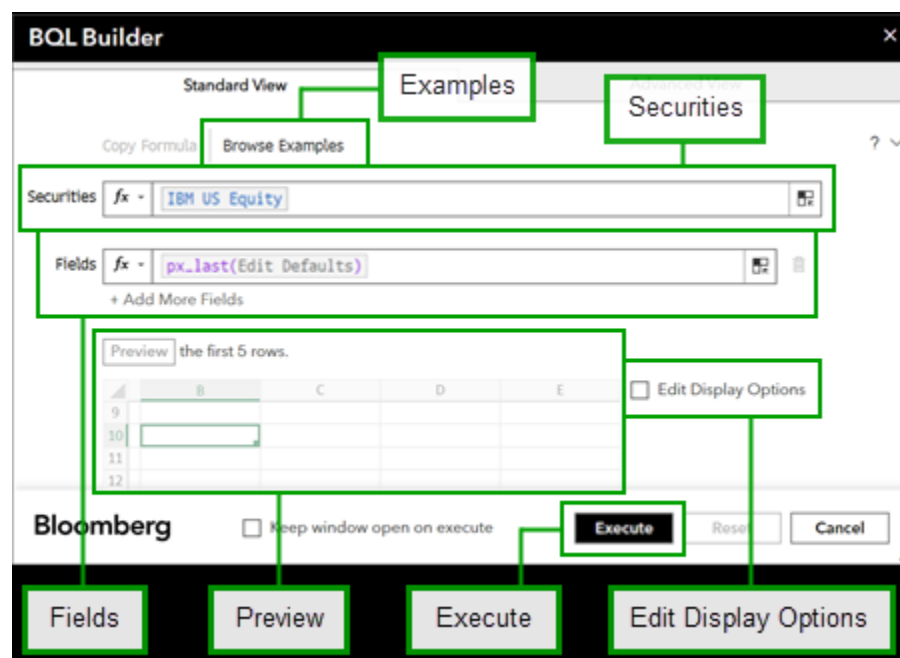
The BQL Builder harnesses the power of BQL to make query writing easier. It is organized into two tabs, *Standard View* and *Advanced View*, which provide you with different levels of granularity for building BQL queries and returning financial data. You can click the tabs at the top of the BQL Builder to navigate between both views.

On both views, you define a query by entering securities and data fields, then specifying mathematical operations, statistical analysis, and/or grouping options to return the data you need. Both views offer you smart security and field suggestions as you type in each field. The tab you select remains selected the next time you launch the BQL Builder.

**Note:** Depending on their level of complexity, some queries are too complex to load in the *Standard View*. A warning appears and the query loads in the *Advanced View* instead.

## Standard View

The *Standard View* tab allows you to browse query examples, write or edit a query, and preview the query output before inserting and running the query in your spreadsheet.

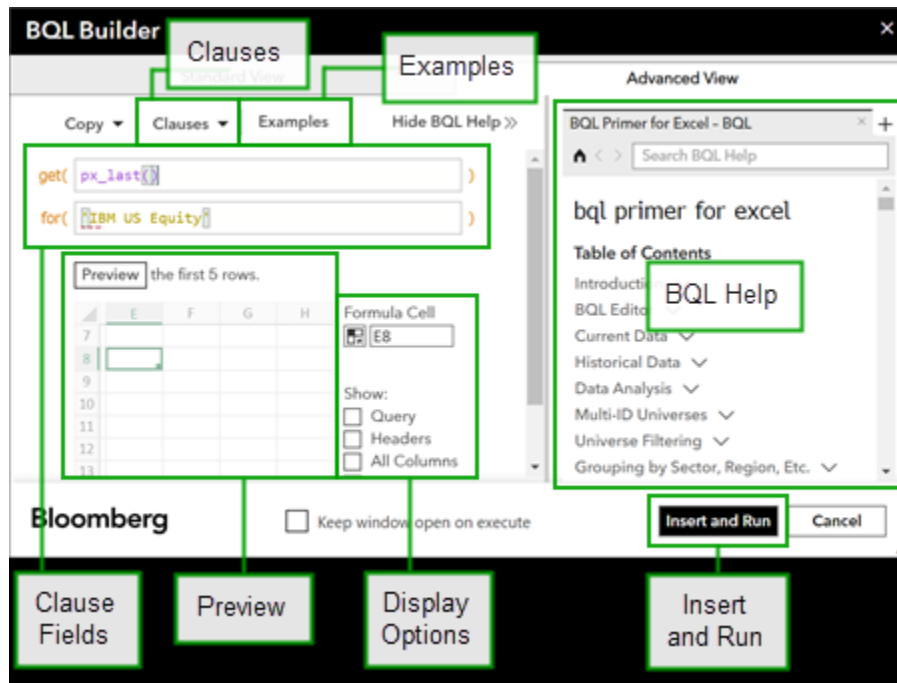


- **Examples:** Choose from a list of common queries that you can load in the BQL Builder to review, customize, and execute in your spreadsheet.
- **Securities:** Specify one or more securities you want to include in your query. If you want to get data for the members of an index, holdings of a fund, or all the bonds from an issuer, you can select an option from the *fx* drop-down menu.
- **Fields:** Enter the data fields and mathematical operators that define the data you want to get. You can click each field to change the properties of the data, like its currency or as of date. From the *fx* drop-down menu, you can apply common statistical analysis to the data, like averages or sums, as well as group the data for analysis by country, sector, or across all the securities in your query.
- **Preview:** Click the button to see the first five rows of the query's output and check if it contains the data you want.
- **Edit Display Options:** Adjust how the output of your query appears in the preview and your spreadsheet. For example, you can show or hide dates, security identifiers, and column headers.
- **Execute:** Click the button to run the query and return the results in your spreadsheet.

## Advanced View

The *Advanced View* allows you to build BQL queries using features of the BQL language that may not be available in the

Standard View.



- **Clauses:** Add or remove optional clauses that help you simplify your query or adjust how it runs.  
**Note:** The `get()` and `for()` clauses are required and cannot be removed.
- **Examples:** Choose from a list of common queries that you can load in the Builder to review, customize, and run in your spreadsheet.
- **Clause Fields:** Construct each clause in your query. In the `for()` clause, you can specify a security, list of securities, or universe function that returns securities you want to get data for. In the `get()` clause, you can specify the data items, parameters, functions, and data grouping options that define the data you want. In each field, you get smart code suggestions as you type.  
**Note:** If you select optional clauses from the *Clauses* menu, they appear here for you to define.
- **Preview:** Click the button to see the first five rows of the query's output and check if it contains the data you want.
- **Display Options:** Adjust how the output of your query appears in your spreadsheet. For example, you can show or hide column headers.
- **Insert and Run:** Click the button to run the query and return the results in your spreadsheet.
- **BQL Help:** Access help documentation that you can use to learn more about the functions and data fields available for your query. Definitions, parameter values, and use-case examples are available for a growing number of BQL elements.

## Trending Files

The following table displays the BQL documents, spreadsheets, and videos that are most popular with users in the last quarter.

| File Type                                                                           | Category     | Name and Description                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Multi-Asset  | <a href="#">Function Reference</a><br><br>A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL.                                                      |
|    | Multi-Asset  | <a href="#">Bloomberg Liquidity Assessment (LOA) in BQL</a><br><br>A spreadsheet tutorial that shows how to evaluate liquidity with the Bloomberg Liquidity Assessment (LOA) model using BQL.               |
|    | Equities     | <a href="#">BQL Company Fundamentals</a><br><br>An overview of the benefits of using BQL to get company financials data in Excel, as well as an overview of the calendarization and blending methodologies. |
|  | Fixed Income | <a href="#">BQL for Fixed Income</a><br><br>A spreadsheet tutorial that shows how to download fixed income data to Excel using BQL.                                                                         |
|  | Equities     | <a href="#">BQL for Universe Analysis</a><br><br>A spreadsheet tutorial of screening security universes in BQL using the intersect(), setdiff(), and union() universe functions.                            |

| File Type                                                                           | Category    | Name and Description                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Portfolios  | <p><a href="#">BQL for Portfolios</a></p> <p>A spreadsheet tutorial that shows how to access portfolio data using BQL.</p>                                                                                                                                                                                                                                                                                                                                                   |
|    | Equities    | <p><a href="#">BQL for Fundamentals</a></p> <p>A spreadsheet tutorial that shows how to download company financials and estimates to Excel using BQL.</p>                                                                                                                                                                                                                                                                                                                    |
|    | Multi-Asset | <p><a href="#">How to Use BQL</a> (33:04)</p> <p>A video tutorial that shows you how to use the BQL Builder to request historical data, aggregate data by sector, screen for securities, and more.</p> <p> <a href="#">Companion Spreadsheet 1</a></p> <p> <a href="#">Companion Spreadsheet 2</a></p> |
|  | Multi-Asset | <p><a href="#">Episode 1: What is BQL?</a> (5:45)</p> <p>An introduction to using BQL in Excel.</p>                                                                                                                                                                                                                                                                                                                                                                          |

| File Type                                                                         | Category     | Name and Description                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Fixed Income | <a href="#">BQL for Funds</a><br><br>A spreadsheet tutorial that shows you how to download funds data to Excel using BQL.                                                                                                      |
|  | Equities     | <a href="#">BQL Company Financials: Parameters, Functions &amp; Associative Columns</a><br><br>A reference guide to the parameters and associative columns you can use to download company financials data to Excel using BQL. |

## Asset Class Reference

### BQL for Equities

BQL is an efficient and flexible method for retrieving a wide range of equity data from the Bloomberg database, including company financials, estimate, and pricing data.

### Equities Docs and Media

The following tables contain spreadsheets, tutorials, and videos created by Bloomberg to assist you with performing data analysis for equities using BQL.

### Equities Reference

| File Type                                                                           | Name                                                                                    | Description                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Bloomberg Equity Index Financials</a>                                       | A reference guide to the accounting and calculations methodology, parameters, and associative columns available for downloading equity index financials data to Excel using BQL. |
|  | <a href="#">BQL Company Financials: Parameters, Functions &amp; Associative Columns</a> | A reference guide to the parameters and associative columns you can use to download company financials data to Excel using BQL.                                                  |

| File Type                                                                         | Name                               | Description                                                                                                  |
|-----------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Function Reference</a> | A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL. |

### Equities Tutorials and Workflows

| File Type                                                                           | Name                                                        | Description                                                                                                                                                |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">Supply Chain in BQL</a>                         | A spreadsheet tutorial that shows how to analyze supply chain relationships using the <i>suppliers()</i> and <i>customers()</i> universe functions in BQL. |
|    | <a href="#">Factor Exposures in BQL</a>                     | A spreadsheet tutorial that shows how to analyze factor exposures for equity securities using BQL.                                                         |
|    | <a href="#">Bloomberg Liquidity Assessment (LOA) in BQL</a> | A spreadsheet tutorial that shows how to evaluate liquidity with the Bloomberg Liquidity Assessment (LOA) model using BQL.                                 |
|   | <a href="#">BQL Company Financials - Segment Explorer</a>   | A spreadsheet tutorial that shows how to analyze financials for a company's segments in Excel using BQL.                                                   |
|  | <a href="#">BQL for Broker Level Estimates</a>              | A spreadsheet tutorial that shows how to download broker-level estimates to Excel using BQL and carry out both aggregation and screening analysis.         |
|  | <a href="#">BQL for Actual and Projected Dividends</a>      | A spreadsheet tutorial that shows how to access historical and projected dividends in Excel using BQL.                                                     |
|  | <a href="#">BQL for Fundamentals</a>                        | A spreadsheet tutorial that shows how to download company financials and estimates to Excel using BQL.                                                     |
|  | <a href="#">BQL for Historical Corporate Actions</a>        | A spreadsheet tutorial that shows how to download historical corporate actions to Excel using BQL.                                                         |

| File Type                                                                           | Name                                                               | Description                                                                                                                                                               |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">BQL for Index Financials</a>                           | A spreadsheet tutorial that shows how to use BQL for index financials for equity workflows, such as relative valuation, earnings revisions, and market surveillance.      |
|    | <a href="#">BQL for Technical Analysis Tutorial</a>                | A spreadsheet tutorial that shows how to perform technical analysis for scoring, screening, and aggregating data using BQL.                                               |
|    | <a href="#">BQL for Universe Analysis</a>                          | A spreadsheet tutorial of screening security universes in BQL using the intersect(), setDiff(), and union() universe functions.                                           |
|    | <a href="#">BQL Fundamentals</a>                                   | An overview of the benefits of using BQL to get company financials data in Excel, as well as an overview of the calendarization and blending methodologies.               |
|    | <a href="#">Equities Scoring, Screening, and Aggregation</a>       | A spreadsheet tutorial that shows how to screen for, perform aggregated analysis of, and score equities in Excel using BQL.                                               |
|  | <a href="#">Equity Peer Groups in BQL</a>                          | A spreadsheet tutorial that shows how to use BQL to retrieve the peers of a company to your spreadsheet, so you can compare their performance and analyze relative value. |
|  | <a href="#">Monitoring Short-Term Warning Signals for Equities</a> | A customizable spreadsheet that lets you monitor significant downward outlook revisions across a range of datasets using BQL.                                             |
|  | <a href="#">Screening for Equities by Piotroski Score</a>          | A customizable spreadsheet that lets you screen for equities with the highest Piotroski score using BQL.                                                                  |

## Equities Videos

| File Type                                                                           | Name                                                                                         | Description                                                                                                                                                       |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">Screen, Score, and Calculate Aggregate Stats from an Equity Universe</a> (50:38) | A video introduction to BQL with a focus on equities, showcasing the new functionality released for equities in Excel that allows for custom analysis in seconds. |
|    | <a href="#">BQL for Technical Analysis</a> (20:59)                                           | A video introduction to BQL and how it can be used to request technical analysis data, create scores and screens, and perform relative valuation.                 |
|    | <a href="#">Retrieving Maximum Drawdowns (Covid-19 v GFC)</a> (3:07)                         | A short video that shows you how to use the BQL Builder to quickly compare the maximum drawdown of specified securities or indices in a specified time period.    |
|    | <a href="#">Screen, Score, and Calculate Aggregate Stats from an Equity Universe</a> (50:38) | A video introduction to BQL with a focus on equities, showcasing the new functionality released for equities in Excel that allows for custom analysis in seconds. |
|   | <a href="#">Retrieving Maximum Drawdowns (Covid-19 v GFC)</a> (3:07)                         | A short video that shows you how to use the BQL Builder to quickly compare the maximum drawdown of specified securities or indices in a specified time period.    |
|  | <a href="#">MODL Drag and Drop to BQL</a> (2:11)                                             | A short video that shows you how to drag and drop BQL formulas from the <i>Company Financials</i> (MODL) function into Excel.                                     |

## Factor Exposure

The `factor_exposure()` data item in BQL allows you to analyze the sensitivity of an investment to market trends based on security characteristics such as the size of the company, its momentum, or valuation.

Factors are expressed as a Z-score relative to the security's respective market-cap weighted average factor exposure, following the methodology from the PORT MAC3 Global Risk Model and the *Tilts* (TLTS) function.

**Signature:** `factor_exposure([dates], [fill], factor_name)`

where:

- `dates` is the as-of date to retrieve historical data. The default is today (OD).
- `fill` allows you to fill missing values with another value. The default is previous (PREV).

- *factor\_name* (required) is the name of the factor for which you want to retrieve exposures data (e.g., VALUE). The default is BETA.

For more on methodology and factor definitions: [TLTS Help Page > Methodology](#).

To download a spreadsheet tutorial on analyzing factor exposure: [Factor Exposures in BQL](#).

## Parameters

The following topics provide information on parameters for equity data sets.

### Company Financials Parameters

Company financials parameters allow you to specify the desired content for any analysis that relies on company financials data:

| Name & Mnemonics                                                          | Description                                                                                                                                            | Default & Other Values                  |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <a href="#">Accounting Standard</a><br><br>fa_acct_std<br><br>std         | Choose the accounting standard for the fundamental data, where available.                                                                              | <b>MIXED</b> (default), IFRS, US, LOCAL |
| <a href="#">Actuals or Estimates</a><br><br>fa_act_est_data,<br><br>ae    | Choose whether the data you return is actual (reported) only, estimate only, or actual data for reported periods and estimate data for future periods. | <b>AE</b> (default), A, E               |
| <a href="#">Actuals and Estimates Mapping</a><br><br>act_est_mapping      | Choose whether the estimates data you return is generically or precisely aligned with actuals data.                                                    | <b>GENERIC</b> (default), PRECISE       |
| <a href="#">Adjusted for Abnormal Items</a><br><br>fa_adjusted<br><br>adj | Choose between adjusted or as reported data. Adjusting cleans one-off and recurring charges from the data.                                             | Y, <b>N</b> (default)                   |

| Name & Mnemonics                                                                    | Description                                                                                                                              | Default & Other Values                                       |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <u>Adjusted for Capital Changes</u><br><br>capital_changes_adjust<br><br>ca_adj     | Adjust price history for capital changes, such as spin-offs, stock splits, stock dividends, and rights offerings.                        | <b>SPLITS</b> (default), RAW                                 |
| <u>As of Date</u><br><br>as_of_date<br><br>dates                                    | Specify a historical date or date range to retrieve the publicly available data that was observed "as of" those date(s).                 | <b>OD</b> (default), absolute or relative date or date range |
| <u>Consolidated Data</u><br><br>fa_consolidated<br><br>consl                        | Choose between consolidated or parent data.                                                                                              | <b>Y</b> (default), N                                        |
| <u>Currency</u><br><br>currency                                                     | Convert values to a different currency.                                                                                                  | <b>None</b> (default), any three-character ISO currency code |
| <u>Currency Conversion Method</u><br><br>currency_conversion_method<br><br>cur_meth | Choose the method that the <i>Currency</i> parameter uses to convert values from one currency to another.                                | <b>ACC</b> (default), AOD                                    |
| <u>Estimate Source</u><br><br>est_source<br><br>brok                                | Choose the consensus type or broker source for estimate data. For example, the default source is the Bloomberg Standard Consensus (BST). | <b>BST</b> (default), BLI, BPE, CGD                          |

| Name & Mnemonics                                                             | Description                                                                                                                                                                                                                                                                                      | Default & Other Values                                                        |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <u>Exclude Impact of Cap Oper Leases</u><br><br>excl_oper_leases<br><br>exol | Choose whether to include or exclude the impact of operating leases under the new Operating Lease Accounting Standard (ASC 842 and IFRS 16).                                                                                                                                                     | <b>N</b> (default), Y                                                         |
| <u>Filing Status</u><br><br>fa_filing_status<br><br>fs                       | Get data from a specific report version, such as the least recent or most recent filing.                                                                                                                                                                                                         | <b>MR</b> (default), MRXP, LR                                                 |
| <u>Fill When Data is Missing</u><br><br>fill                                 | Fill missing values in a data series with either NA or the previous value.                                                                                                                                                                                                                       | <b>NA</b> (default), PREV                                                     |
| <u>Period Offset</u><br><br>fa_period_offset<br><br>fpo                      | Choose a relative fiscal period or period range for which you want data. The anchor period "0" is either today or the end of the period(s)/year(s) entered for the <i>Period Reference</i> parameter. For example, "-2, 2" gets data for the last three reported periods and next two estimates. | <b>OR</b> (default – no offset), see link for other possible values           |
| <u>Period Reference</u><br><br>fa_period_reference<br><br>fpr                | Select a fixed date or fiscal period for the fundamental data you want to retrieve, based on the company's reporting calendar. You can choose a single date/period or a range of dates/periods. When no <i>Period Reference</i> is specified, the <i>Period Reference</i> is today.              | <b>None</b> (default), see link for other possible values                     |
| <u>Period Type</u><br><br>fa_period_type<br><br>fpt                          | Select the type of fiscal period returned, which could be a fiscal period reported by the company (such as an Annual, Quarterly or Semiannual fiscal period) or a synthetic fiscal period created by Bloomberg (such as a last-12-months, calendar, or blended period).                          | <b>LTM</b> (default), A, Q, S, P, YTD, BA, BQ, BS, BT, Q1, Q2, Q3, Q4, S1, S2 |

| Name & Mnemonics                                                                   | Description                                                                                                                                                                                                                                                                        | Default & Other Values            |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <u>Source for Calculated Period Types</u><br><br>fa_period_type_source<br><br>fpts | If you chose a synthetic fiscal periods for the <i>Period Type</i> parameter, such as the Last 12 Months (LTM) or Blended Trailing (BT) periods, select the reporting periodicity used to calculate the values. The default source is the company's primary reporting periodicity. | <b>P</b> (default), A, S, Q       |
| <u>Year End Alignment</u><br><br>fa_period_end_year<br><br>fye                     | Realign a company's fiscal period reported values to a custom calendar. You can enter "C" to realign the fiscal year with the calendar year or "C[MMDD]" (e.g., "C0925") to end the fiscal year on a custom date.                                                                  | <b>F</b> (default), C, C + <MMDD> |

For more on how the parameters interact: [Fundamental Parameter Interactions](#).

## Accounting Standard

For BQL() and BQL.Query() formulas in Excel, the *Accounting Standard* (fa\_acct\_std) parameter lets you choose the accounting standard for the company financial data you are retrieving, when that accounting standard is available.

**Parameter Mnemonic:** fa\_acct\_std

**Parameter Mnemonic Alias:** std

**Valid Values:**

| Value | Description                                                                                                                                                                                                                                                                                                                                                                              |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MIXED | Mixed Accounting Standards (depending on data availability) <b>(Default)</b><br><br>Retrieves data in the most widely used accounting standard, depending on data availability for the company. If data sets in more than one accounting standard are available for a fiscal period, the priority waterfall of the retrieved accounting standard is IFRS, then US GAAP, then Local GAAP. |

| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IFRS  | <p>IFRS</p> <p>Retrieves the data set reported by the company in the International Financial Reporting Standards (IFRS) accounting standard, if available.</p> <p>This standard is set by the International Accounting Standards Board (IASB) and provides a reliable and transparent framework for financial reporting to support economic decisions by investors, lenders, and other users of general purpose financial statements.</p> |
| US    | <p>US GAAP</p> <p>Retrieves the data set reported by the company in the US Generally Accepted Accounting Principles (GAAP) accounting standard, if available.</p> <p>GAAP procedures and rules govern accepted accounting practices as defined and supervised by the Financial Accounting Standards Board (FASB), a self-regulatory organization.</p>                                                                                     |
| LOCAL | <p>Local GAAP</p> <p>Retrieves the data set reported by company in the Local GAAP accounting standard, if available.</p>                                                                                                                                                                                                                                                                                                                  |

**Example Formulas:**

- Retrieve Toyota's earnings per share (EPS) for the latest Last 12 Months (LTM) period, using data based on the IFRS accounting standard (if not available, the formula attempts to download data based on US GAAP, then local GAAP):  
`=BQL("TOYOF US Equity", "is_eps")`  
`=BQL.Query("get(is_eps) for(['TOYOF US Equity'])")`
- Retrieve Toyota's EPS for the latest Last 12 Months (LTM) period, using data based on the local GAAP accounting standard, if available:  
`=BQL("TOYOF Equity", "is_eps(fa_acct_std=LOCAL)")`  
`=BQL.Query("get(is_eps(std=LOCAL)) for(['TOYOF US Equity'])")`
- Retrieve Toyota's EPS for the latest Last 12 Months (LTM) period, using data based on the IFRS accounting standard (which is not available for Toyota):  
`=BQL("TOYOF US Equity", "is_eps(std=IFRS)")`

```
=BQL.Query("get(is_eps(std=IFRS)) for(['TOYOF US Equity'])")
```

## Actuals or Estimates

For BQL() and BQL.Query() formulas in Excel, the *Actual or Estimate* (fa\_act\_est\_data) parameter lets you choose whether to retrieve actual or estimated company financials data. For example, you can use the parameter to retrieve actual data for historical periods and estimate data for future periods, or estimate data for both historical and future periods.

**Parameter Mnemonic:** fa\_act\_est\_data

**Parameter Mnemonic Alias:** ae

**Valid Values:**

| Value | Description                                                                                                                                                                                                                             |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AE    | Actuals for Reported Periods; Estimates for Future Periods ( <b>Default</b> )<br><br>Retrieves actual data reported by the company for historical fiscal periods and estimates data for fiscal periods not yet reported by the company. |
| A     | Actuals<br><br>Retrieves actual data reported by the company for historical fiscal periods.                                                                                                                                             |
| E     | Estimates<br><br>Retrieves estimates data both for historical fiscal periods reported by the company and for fiscal periods not yet reported by the company.                                                                            |

## Example Formulas:

- Retrieve the 2017 fiscal year estimate revisions for IBM's earnings per share (EPS), as forecast every day between May 1, 2017 and May 5, 2017:  

```
=BQL("IBM US Equity", "is_eps(dates=range(2017-05-01, 2017-05-05), fpr='2017', fa_act_est_data=E)")
```

```
=BQL.Query("get(is_eps(dates=range(2017-05-15, 2017-05-19), fpr='2017', fa_act_est_data=E)) for(['IBM US Equity'])")
```
- Retrieve IBM's actual EPS for the last five quarters and estimate EPS for the next two quarters:  

```
=BQL("IBM US Equity", "is_eps(fpt=Q, fpo=range('-4', '2'), ae=AE)")
```

```
=BQL.Query("get(is_eps(fpt=Q, fpo=range('-4', '2'), ae=AE)) for(['IBM US Equity'])")
```

## Actuals and Estimates Mapping

For BQL() and BQL.Query() formulas in Excel, the *Actuals and Estimates Mapping* (act\_est\_mapping) parameter lets you choose whether to retrieve estimates data that is generically or precisely aligned with actuals data.

**Parameter Mnemonic:** act\_est\_mapping

**Valid Values:**

| Value   | Description                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GENERIC | <p>Generic Mapping (<b>Default</b>)</p> <p>Returns estimates data that is generically aligned with actuals following the same methodology for all companies. Provides better coverage and history than the PRECISE option, but GENERIC does not guarantee a precise alignment. GENERIC is the recommended option for analysis that requires more history and coverage, such as backtesting.</p> |
| PRECISE | <p>Precise Mapping</p> <p>Returns estimates data that is aligned to actuals following a company-specific methodology. PRECISE provides a precise alignment but with less coverage and history since it requires deeper estimates content from sell-side research contributors. PRECISE is the recommended option for uses such as company modeling.</p>                                         |

**Example Formula:**

Return Alphabet's revenue estimate for the next year using PRECISE data:

```
=BQL("GOOGL US Equity", "sales_rev_turn(fpo=1, fpt=A, act_est_mapping=PRECISE)")
```

```
=BQL.Query("get(sales_rev_turn) for('GOOGL US Equity') with(FPO=1, FPT=A, ACT_EST_MAPPING=PRECISE)")
```

## Adjusted for Abnormal Items

For BQL() and BQL.Query() formulas in Excel, the *Adjusted for Abnormal Items* (fa\_adjusted) parameter lets you choose between adjusted or as-reported data for company financials data items. Adjusting cleans the data for one off and non-recurring charges.

**Note:** This parameter only impacts data for companies where the value for the data item adjusted\_financials\_indicator is Y.

**Parameter Mnemonic:** fa\_adjusted

**Parameter Mnemonic Alias:** adj

**Valid Values:**

| Value | Description                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------------------|
| Y     | Adjusted<br><br>Retrieves data that is adjusted for abnormal items.                                             |
| N     | Not Adjusted ( <b>Default</b> )<br><br>Retrieves data that is not adjusted for abnormal items, where available. |

**Example Formulas:**

- Retrieve IBM's EPS for the latest LTM period, not adjusted for abnormal items:  
`=BQL("IBM US Equity", "is_eps")`  
`=BQL.Query("get(is_eps) for(['IBM US Equity'])")`
- Retrieve IBM's earnings per share (EPS) for the latest Last 12 Months (LTM) period, adjusted for abnormal items:  
`=BQL("IBM US Equity", "is_eps(adj=Y)")`  
`=BQL.Query("get(is_eps(adj=Y)) for(['IBM US Equity'])")`

**Adjusted for Capital Changes**

For BQL() and BQL.Query() formulas in Excel, when using company financials data items that take into account the company's stock price, the *Adjusted for Capital Changes* (capital\_changes\_adjust) parameter lets you adjust the stock's price history for capital changes, such as spin-offs, stock splits, stock dividends, and rights offerings.

**Parameter Mnemonic:** capital\_changes\_adjust

**Parameter Mnemonic Alias:** ca\_adj

**Valid Values:**

| Value  | Description                                                                                                                                             |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| SPLITS | Adjusted ( <b>Default</b> )<br><br>Retrieves data adjusted for capital changes, such as spin-offs, stock splits, stock dividends, and rights offerings. |

| Value | Description                                                              |
|-------|--------------------------------------------------------------------------|
| RAW   | Not Adjusted<br><br>Retrieves data not adjusted for any capital changes. |

**Example Formulas:**

- Retrieve Argus Therapeutics' earnings per share (EPS) for the latest Last 12 Months (LTM) period, adjusted for capital changes:  
`=BQL("ARGS US Equity", "is_eps")`  
  
`=BQL.Query("get(is_eps) for(['ARGS US Equity'])")`
- Retrieve Argus Therapeutics' EPS for the latest Last 12 Months (LTM) period, not adjusted for capital changes:  
`=BQL("IBM US Equity", "is_eps(ca_adj=RAW)")`  
  
`=BQL.Query("get(is_eps(ca_adj=RAW)) for(['ARGS US Equity'])")`

**As of Date**

For BQL() and BQL.Query() formulas in Excel, the *As Of Date* (as\_of\_date) parameter lets you perform point-in-time analysis as of a specific date or date range. For example, for company financials data, you can use the parameter to request publically available data that was observed "as of" a given date in the past.

**Parameter Mnemonic:** as\_of\_date

**Parameter Mnemonic Alias:** dates

**Default Value:** OD (today's date)

**Valid Values:**

| Value  | Description                                                                                                                                                                                                                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Date> | Absolute As Of Date<br><br>Retrieves the data that would have been known to an observer having perfect access to public information on a fixed date. The date must be in the format <YYYY-MM-DD>.<br><br>For example, "dates=2016-09-15" retrieves the data that was publicly available as of September 15, 2016. |

| Value                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Date range>                                              | <p>Range of Absolute As Of Dates</p> <p>Retrieves the data that would have been known to an observer having perfect access to public information between two fixed observation dates. The dates must be specified as an argument to the <i>range()</i> function and in the &lt;YYYY-MM-DD&gt; format.</p> <p>For example, "dates=range(2016-01-01, 2017-12-31)" retrieves the series of data that was publicly available as of every date from January 1, 2016 to December 31, 2017.</p> |
| <Non-positive integer><br>+<br><br>D, W, M, Q, S, or Y    | <p>Relative As Of Date</p> <p>Retrieves the data that would have been known to an observer having perfect access to public information on a relative, rolling observation date.</p> <p>For example, "dates=-1Y" retrieves the data that was publicly available exactly one year ago, measured from today.</p>                                                                                                                                                                            |
| <Non-positive integer range> +<br><br>D, W, M, Q, S, or Y | <p>Range of Relative As Of Dates</p> <p>Retrieves the data that would have been known to an observer having perfect access to public information between two relative, rolling observation dates. The relative date range must be specified as an argument to the <i>range()</i> function.</p> <p>For example, "dates=range(-10D, 0D)" retrieves the stream of data that was publicly available as of every day between ten days ago and today.</p>                                      |

**Note:** The value you assign to *As Of Date* interacts with the [Period Offset](#), [Period Reference](#), and [Period Type](#) parameters. For more information, see [Fundamental Parameter Interactions](#).

#### Example Formulas:

- Retrieve IBM's earnings per share (EPS) for the Last 12 Months (LTM) period reported as of September 15, 2016:  
 =BQL("IBM US Equity", "is\_eps(dates=2016-09-15)")  
  
 =BQL.Query("get(is\_eps(dates=2016-09-15)) for(['IBM US Equity'])")
- Retrieve IBM's EPS for the LTM period on the first date in the range, then the new value on each day it was reported between January 1, 2016 to December 31, 2017. Output is blank when no new value was reported:

```
=BQL("IBM US Equity", "is_eps(dates=range(2016-01-01, 2017-12-31))")
```

```
=BQL.Query("get(is_eps(dates=range(2016-01-01, 2017-12-31))) for(['IBM US Equity'])")
```

- Retrieve IBM's EPS for the Last 12 Months (LTM) period reported as of ten days ago:

```
=BQL("IBM US Equity", "is_eps(dates=-10D)")
```

```
=BQL.Query("get(is_eps(dates=-10D)) for(['IBM US Equity'])")
```

- Retrieve IBM's EPS for the last reported quarter every day between ten days ago and today. The last reported EPS is retrieved for the first date in the range, then on every day a new value was reported. Output is NA when no new value was reported:

```
=BQL("IBM US Equity", "is_eps(dates=range(-10D, 0D), fpt=Q)")
```

```
=BQL.Query("get(is_eps(dates=range(-10D, 0D), fpt=Q)) for(['IBM US Equity'])")
```

## Consolidated Data

For BQL() and BQL.Query() formulas in Excel, the *Consolidated Data* (fa\_consolidated) parameter allows you to choose between consolidated or unconsolidated data.

A consolidated financial statement covers the activities of a parent company and its subsidiaries in a single report, as if they were a single company operating under one roof. Unconsolidated financial statements, treat each entity as if it were entirely separate - the parent unrelated to the subsidiaries, and the subsidiaries unrelated to one another. For example, if a subsidiary earned \$1 in income, that \$1 would show up on the parent's consolidated statement and the subsidiary's unconsolidated statement, but not the parent's unconsolidated statement.

**Parameter Mnemonic:** fa\_consolidated

**Parameter Mnemonic Alias:** consl

**Valid Values:**

| Value | Description                                                                                                                                                                                          |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Y     | <p>Consolidated (<b>Default</b>)</p> <p>Retrieves consolidated data.</p>                                                                                                                             |
| N     | <p>Unconsolidated</p> <p>Retrieves unconsolidated data, when available.</p> <p>Unconsolidated data is available only for companies where the value for the data item eqy_consol_fund_avail is Y.</p> |

**Example Formulas:**

- Retrieve Toyota's parent sales figures for Q2 2016:  
`=BQL("TOYOF US Equity", "sales_rev_turn(fpt=Q, fpr='2016Q2', consl=N)")`  
`=BQL.Query("get(sales_rev_turn(fpt=Q, fpr='2016Q2', consl=N)) for(['TOYOF US Equity'])")`
- Retrieve sales figures for Q2 2016 for the parent company (Toyota) and all its subsidiaries:  
`=BQL("TOYOF US Equity", "sales_rev_turn(fpt=Q, 'fpr=2016Q2')")`  
`=BQL.Query("get(sales_rev_turn(fpt=Q, fpr='2016Q2')) for(['TOYOF US Equity'])")`

**Currency**

For BQL() and BQL.Query() formulas in Excel, the *Currency* (currency) parameter allows you to convert monetary values into another currency. For example, you can use the parameter to compare company financials between companies that report in different currencies.

**Parameter Mnemonic:** currency

**Default Value:** None (local currency)

**Valid Values:** Three-character ISO currency codes. For a list of codes: [CURR Help Page > Currency Codes](#).

**Example Formulas:**

- Retrieve IBM's earnings per share (EPS) for the latest Last 12 Months (LTM) period converted from USD to EUR:  
`=BQL("IBM US Equity", "is_eps(currency=EUR)")`  
`=BQL.Query("get(is_eps(currency=EUR)) for(['IBM US Equity'])")`
- Retrieve IBM's EPS for the latest Last 12 Months (LTM) period, as reported on February 15, 2018, converted from USD to EUR using the conversion rate on that date:  
`=BQL("IBM US Equity", "is_eps(currency=EUR, cur_meth=AOD, dates=2018-02-15)")`  
`=BQL.Query("get(is_eps(currency=EUR, cur_meth=AOD, dates=2018-02-15)) for(['IBM US Equity'])")`

**Currency Conversion Method**

For BQL() and BQL.Query() formulas in Excel, the *Currency Conversion Method* (currency\_conversion\_method) parameter lets you choose the method that the [Currency](#) (currency) parameter uses to convert monetary values from one currency to another.

**Parameter Mnemonic:** currency\_conversion\_method

**Parameter Mnemonic Alias:** cur\_meth

**Valid Values:**

| Value | Description                                                                                                                                                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACC   | <p>Account Method (<b>Default</b>)</p> <p>Converts data to another currency using the average conversion rate during the fiscal period for flow accounting concepts (such as Revenue) and the currency conversion rate at the end of the fiscal period for stock accounting concepts (such as Assets).</p> |
| AOD   | <p>As Of Date Method</p> <p>Converts fundamental data to another currency using the currency conversion rate on a date specified via the <a href="#">As of Date</a> parameter.</p>                                                                                                                         |

**Example Formulas:**

- Retrieve IBM's earnings per share (EPS) for the Last 12 Months (LTM) period as reported on February 15, 2018, converted from USD to EUR using the average conversion rate during the period:  
`=BQL("IBM US Equity", "is_eps(currency=EUR, dates=2018-02-15)")`  
  
`=BQL.Query("get(is_eps(currency=EUR, dates=2018-02-15)) for(['IBM US Equity'])")`
- Retrieve IBM's EPS for the Last 12 Months (LTM) period as reported on February 15, 2018, converted from USD to EUR using the conversion rate on that date:  
`=BQL("IBM US Equity", "is_eps(currency=EUR, cur_meth=AOD, dates=2018-02-15)")`  
  
`=BQL.Query("get(is_eps(currency=EUR, cur_meth=AOD, dates=2018-02-15)) for(['IBM US Equity'])")`

**Estimate Source**

For BQL() and BQL.Query() formulas in Excel, the *Estimate Source* (est\_source) parameter lets you specify your preferred consensus type or broker source for estimates data.

**Parameter Mnemonic:** est\_source

**Parameter Mnemonic Alias:** brok

**Valid Values:**

| Value | Description                                                                                                                                                                                                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BST   | <p>Bloomberg Standard (<b>Default</b>)</p> <p>Returns the estimates data from the Bloomberg standard estimates consensus.</p>                                                                                                                                                                                                                              |
| BLI   | <p>Bloomberg Leading Indicator Consensus</p> <p>Returns the estimates data from the Bloomberg leading indicator estimates consensus, which includes only the estimates updated within the last four weeks.</p>                                                                                                                                             |
| BPE   | <p>Bloomberg Post Event</p> <p>Returns the estimates data from the Bloomberg post-event estimates consensus, which includes only the estimates updated after the last earnings announcement date.</p>                                                                                                                                                      |
| CMPY  | <p>Company Reported</p> <p>Returns estimates data provided by the company (i.e., guidance).</p>                                                                                                                                                                                                                                                            |
| [BBB] | <p>Broker Code</p> <p>Returns the estimates data from an individual sell-side research contributor.</p> <p><b>Note:</b> Entitlement for the research for the selected contributor is required. To view the entitlements or request entitlement for a particular research contributor, use the <i>Research Entitlements</i> (ENTC &lt;GO&gt;) function.</p> |

| Value               | Description                                                                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ['BBB','CCC', ... ] | <p>List of Broker Codes</p> <p>Returns the estimates data from a list of sell-side research contributors.</p>                                                                      |
| BROKERS_ALL         | <p>All Brokers</p> <p>Returns all brokers who have provided Bloomberg an estimate for the company.</p>                                                                             |
| BROKERS_ALL_ACTIVE  | <p>All Active Brokers</p> <p>Returns all brokers who have provided Bloomberg an estimate for the company within the past 500 calendar days of the specified <u>As Of Date</u>.</p> |

**Example Formulas:**

- Retrieve Toyota's estimate earnings per share (EPS) from the Bloomberg Standard consensus for the next fiscal period:  
 =BQL("TOYOF US Equity", "is\_eps(fpo=1)")  
  
 =BQL.Query("get(is\_eps(fpo=1)) for(['TOYOF US Equity'])")
- Retrieve Toyota's estimate EPS from Company Reported for the next fiscal period:  
 =BQL("TOYOF US Equity", "is\_eps(fpo=1', est\_source=CMFY)")  
  
 =BQL.Query("get(is\_eps(fpo=1', est\_source=CMFY)) for(['TOYOF US Equity'])")
- Retrieve Toyota's estimate EPS from the Bloomberg Leading Indicator consensus for the next fiscal period:  
 =BQL("TOYOF US Equity", "is\_eps(fpo=1', est\_source=BLI)")  
  
 =BQL.Query("get(is\_eps(fpo=1', est\_source=BLI)) for(['TOYOF US Equity'])")

**Exclude Impact of Cap Oper Leases**

For BQL() and BQL.Query() formulas in Excel, the *Exclude Impact of Capitalized Operating Leases* (excl\_oper\_leases) parameter lets you choose whether to include or exclude the impact of operating leases under the new Operating Lease Accounting Standard (ASC 842 and IFRS 16).

**Parameter Mnemonic:** excl\_oper\_leases

**Parameter Mnemonic Alias:** exol

**Valid Values:**

| Value | Description                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------|
| N     | Includes the Impact <b>(Default)</b><br><br>Returns data that includes the impact of operating leases. |
| Y     | Excludes the Impact<br><br>Returns data that excludes the impact of operating leases.                  |

**Example Formulas:**

- Retrieve The Gap's net debt for the most recently reported Last 12 Months (LTM) period, including the impact of operating leases in the value:  
 =BQL("GPS US Equity", "net\_debt")  
  
 =BQL.Query("get(net\_debt) for(['GPS US Equity'])")
- Retrieve The Gap's net debt for the most recently reported Last 12 Months (LTM) period, excluding the impact of operating leases in the value:  
 =BQL("GPS US Equity", "net\_debt(excl\_oper\_leases=Y)")  
  
 =BQL.Query("get(net\_debt(excl\_oper\_leases=Y)) for(['GPS US Equity'])")

## Filing Status

For BQL() and BQL.Query() formulas in Excel, the *Filing Status* (fa\_filing\_status) parameter lets you get company financials data from a specific report version, such as the least recent or most recent filing. You can also filter out filings with less complete data, such as preliminary filings.

**Parameter Mnemonic:** fa\_filing\_status

**Parameter Mnemonic Alias:** fs

**Valid Values:**

| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MR    | <p>Most Recent Filing (<b>Default</b>)</p> <p>Retrieves the data from the most recent filing available for the fiscal period you specify in your formula.</p> <p><b>Note:</b> This option is best used when building complete financial statements.</p>                                                                                                                                                                                                                                                                                                                                      |
| MRXP  | <p>Most Recent Filing, Excluding Earnings or Preliminary Filings</p> <p>Retrieves the data from the most recent filing available for the fiscal period you specify in your formula, excluding earnings or preliminary records. In other words, the filing could be the most recent 10K, 10Q, Annual Report, Semi-Annual Report, or similar, but not a preliminary filing, such as an 8K, Earnings Press Release, etc.</p> <p><b>Note:</b> This option is best used when building complete financial statements, backtesting, or screening, but you want to exclude any preliminary data.</p> |
| MRC   | <p>Most Recent Composite</p> <p>Retrieves the most data for the fiscal period you specify in your formula from all filings available. This option allows you to retrieve the most recent data available for all metrics.</p> <p><b>Note:</b> This option is best used when backtesting, screening, or financial modeling.</p>                                                                                                                                                                                                                                                                |
| LR    | <p>Least Recent (First) Filing</p> <p>Retrieves the least recent data from the different versions of filings available for the fiscal period you specify in your formula (i.e., the first version of data that a company reported for the fiscal period's "Earnings Data").</p> <p><b>Note:</b> This option is best used when computing surprise and analyzing the first reported filing.</p>                                                                                                                                                                                                |

Example Formulas:

- Retrieve IBM's earning per share (EPS) for the latest Last 12 Months (LTM) period, using reported data from the most recent filing:

```
=BQL("IBM US Equity", "is_eps")
```

```
=BQL.Query("get(is_eps) for(['IBM US Equity'])")
```

- Retrieve IBM's EPS for the latest Last 12 Months (LTM) period, using reported data from the least recent filing:

```
=BQL("IBM US Equity", "is_eps(fs=LR)")
```

```
=BQL.Query("get(is_eps(fs=LR)) for(['IBM US Equity'])")
```

### Fill When Data is Missing

For BQL() and BQL.Query() formulas in Excel, the *Fill When Data is Missing* (fill) parameter lets you fill missing values in a data series of company financials with the last available data, so you can get a continuous stream of data even when there is no data available for certain observations.

**Parameter Mnemonic:** fill

**Valid Values:**

| Value | Description                                            |
|-------|--------------------------------------------------------|
| NA    | No Data <b>(Default)</b><br><br>Returns NA.            |
| PREV  | Previous Data<br><br>Returns the last available value. |

### Example Formulas:

- Retrieve a time series of estimates for IBM's LTM price to earnings (P/E) ratio, as reported between January 1, 2014 and December 31, 2014; for non-trading days, "NaN" appears:

```
=BQL("IBM US Equity", "pe_ratio(dates=range(2014-01-01, 2014-12-31), ae=E)")
```

```
=BQL.Query("get(pe_ratio(dates=range(2014-01-01, 2014-12-31), ae=E)) for(['IBM US Equity'])")
```

- Retrieve a time series of estimates for IBM's LTM price to earnings (P/E) ratio, as reported between January 1, 2014 and December 31, 2014; for non-trading days, the value from the previous day appears:

```
=BQL("IBM US Equity", "pe_ratio(dates=range(2014-01-01, 2014-12-31), ae=E, fill=PREV)")
```

```
=BQL.Query("get(pe_ratio(dates=range(2014-01-01, 2014-12-31), ae=E, fill=PREV)) for(['IBM US Equity'])")
```

Period Offset

For BQL() and BQL.Query() formulas in Excel, the *Period Offset* (fa\_period\_offset) parameter lets you choose relative fiscal period(s) for which you want data.

The data starts either from today or from the end of the period or year you specify with the Period Reference (fa\_period\_reference) parameter. Offsets are entered as positive and/or negative integers, with the reference date (today or another date) defined as period "0".

For example, "fpo='1'" retrieves data for the next period forward. "fpo=range('-2', '2')" retrieves data for the last three reported periods and the next two estimates.

**Parameter Mnemonic:** fa\_period\_offset

**Parameter Mnemonic Alias:** fpo

**Default Value:** OR (no offset)

**Valid Values:**

| Value                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Integer>                                    | <p>Integer</p> <p>A negative integer retrieves data for a fiscal period in the past, while a positive integer pulls data for a fiscal period in the future. For example, "fpo='-1'" pulls data from one fiscal period prior to period 0, while "fpo='1'" pulls from one fiscal period after period 0.</p> <p><b>Note:</b> When using this syntax, the length of one offset period is determined by the <u>Period Type</u> parameter.</p> |
| <Integer> + R<br><br>or<br><br><Integer> + G | <p>Rolling Integer</p> <p>Retrieves data on a rolling basis when you do not specify a reference date with the <u>Period Reference</u> parameter. For example, "fpo='-1R'" always pulls data from the fiscal period prior to the last reported period before the observation date (when you specify a date or date range with the <u>As of Date</u> parameter or today [when a value for dates is not specified]).</p>                    |

| Value                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Integer> + F                                                  | <p>Fixed Integer</p> <p>Retrieves data on a fixed basis. For example, "fpo='-1F'" retrieves data from the fiscal period prior to the last reported fiscal period on the specified date.</p> <p><b>Note:</b> When using this syntax, you must use the <a href="#">As of Date</a> parameter to specify an observation date or date range.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <Integer> +<br>A, S, or Q<br><br>< Integer> +<br>RA, RS, or RQ | <p>Rolling Integer with Step Length</p> <p>Retrieves data for a fiscal period in the past or future. The letter indicates the length of the step: annual (A), semi-annual (S), or quarterly (Q) periods. For example:</p> <ul style="list-style-type: none"> <li>• "fpo='1A'" retrieves data looking forward one annual period.</li> <li>• "fpo='-1S'" retrieves data looking back one semi-annual period prior to the last reported period.</li> <li>• "fpo='-3Q'" retrieves data looking back three quarters prior to the last reported period.</li> </ul> <p><b>Note:</b> If you do not specify a value for the <a href="#">Period Type</a> parameter, this syntax also determines the periodicity for the data. For example, "fpo='1A'" specifies that you retrieve data for an annual period.</p> |
| < Integer> +<br>FA, FS, or FQ                                  | <p>Fixed Integer with Step Length</p> <p>Retrieves data for annual, semi-annual, or quarterly periods on a fixed basis. For example:</p> <ul style="list-style-type: none"> <li>• "fpo='-1FA'" retrieves data looking back one annual period prior to the last reported period on the specified date.</li> <li>• "fpo='-1FS'" retrieves data looking back one semi-annual period prior to the last reported period on the specified date.</li> <li>• "fpo='-1FQ'" retrieves data looking back one quarter prior to the last reported period on the specified date.</li> </ul> <p><b>Note:</b> When using this syntax, you must use the <a href="#">As of Date</a> parameter to specify an observation date or date range.</p>                                                                          |

| Value                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Integer range>              | <p>Integer Range</p> <p>Retrieves data for a range of periods relative to the specified date. The integer range must be specified as an argument to the <i>range()</i> function. For example, "fpo=range('-3', '3')" retrieves seven points of data from fiscal periods -3, -2, -1, 0, 1, 2 and 3.</p> <p><b>Note:</b> When using this syntax, the length of one relative period is determined by the <a href="#">Period Type</a> parameter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <Integer range> + A, S, or Q | <p>Integer Range with Step Length</p> <p>Retrieves data for a range of periods of a specified length. The letter indicates the length of the steps: annual (A), semi-annual (S), or quarterly (Q) periods. The range must be specified as an argument to the <i>range()</i> function. For example:</p> <ul style="list-style-type: none"> <li>"fpo=range('-3A', '3A')" retrieves seven points of data from fiscal periods -3A, -2A, -1A, 0, 1A, 2A and 3A, where the length of the offset between the fiscal periods is one annual period.</li> <li>"fpo=range('-2S', '3S')" retrieves six points of data from fiscal periods -2S, -1S, 0, 1S, 2S and 3S, where the length of the offset between the fiscal periods is one semi-annual period.</li> <li>"fpo=range('-2Q', '2Q')" retrieves five points of data from fiscal periods -2Q, -1Q, 0, 1Q, and 2Q, where the length of the offset between the fiscal periods is one quarter.</li> </ul> |

**Note:** The value you assign to *Period Offset* interacts with the [Period Reference](#), [Period Type](#), and [As of Date](#) parameters. For more: [Fundamental Parameter Interactions](#).

#### Example Formulas:

- Retrieve IBM's estimate earnings per share (EPS) for the next twelve-month period:  
=BQL("IBM US Equity", "is\_eps(fpo='1')")

=BQL.Query("get(is\_eps(fpo='1')) for(['IBM US Equity'])")

**Note:** Because the [Period Type](#) (FPT) parameter is not included in the formula, the default period Last Twelve Months (LTM) is used.

- Retrieve IBM's actual EPS for the last four Last 12 Month (LTM) periods and estimate EPS for the next three LTM periods:  
=BQL("IBM US Equity", "is\_eps(fpo=range('-3', '3'))")

=BQL.Query("get(is\_eps(fpo=range('-3', '3')) for(['IBM US Equity']))")

- Retrieve IBM's actual EPS for the last five quarters and estimate EPS for the next two quarters:

```
=BQL("IBM US Equity", "is_eps(fpt=Q, fpo=range('-4', '2'))")
```

```
=BQL.Query("get(is_eps(fpt=Q, fpo=range('-4', '2')) for(['IBM US Equity']))")
```

- Retrieve IBM's next LTM estimate EPS:

```
=BQL("IBM US Equity", "is_eps(fpt=BT, fpo='1')")
```

```
=BQL.Query("get(is_eps(fpt=BT, fpo='1')) for(['IBM US Equity']))")
```

- Retrieve IBM's estimate EPS for the LTM period that ends at the same time as the next reported quarter:

```
=BQL("IBM US Equity", "is_eps(fpo='1Q', fpt=LTM)")
```

```
=BQL.Query("get(is_eps(fpo='1Q', fpt=LTM)) for(['IBM US Equity']))")
```

- Retrieve IBM's LTM EPS, looking back from the end of Q2 2016 and Q2 2017:

```
=BQL("IBM US Equity", "is_eps(fpr=range('2015Q4', '2016Q4'), fpo='2Q', fpt=LTM)")
```

```
=BQL.Query("get(is_eps(fpr=range('2015Q4', '2016Q4'), fpo='2Q', fpt=LTM)) for(['IBM US Equity']))")
```

## Period Reference

For BQL() and BQL.Query() formulas in Excel, the *Period Reference* (fa\_period\_reference) parameter lets you choose a fixed fiscal period for company financials data based on the company's reporting calendar. You can choose a single period or a range of periods. For example, "2015 " retrieves data for the year 2015.

If you also use the [Period Offset](#) (fa\_period\_offset) parameter in your query, the offset periods for which you want data are relative to the reference period.

**Parameter Mnemonic:** fa\_period\_reference

**Parameter Mnemonic Alias:** fpr

**Default Value:** None (the reference period is the last reported period prior to today or the observation date you specify with the [As of Date](#) (dates) parameter)

**Valid Values:**

| Value  | Description                                                                                                                                                                                                                                               |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Date> | <p>Sets the reference to the fiscal period that ends on or directly before the date you specify. The date must be in the YYYY-MM-DD format.</p> <p>For example, "fpr=2015-12-31" sets the reference to the fiscal period ending on December 31, 2015.</p> |

| Value                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Date range>                                 | <p>Sets the reference to a range of fiscal periods with one year jumps, where the first reference is the fiscal period that ends on or directly before the start date and the last reference is the fiscal period that ends on or directly before the end date. The dates must be specified as an argument to the <i>range()</i> function and in the YYYY-MM-DD format.</p> <p>For example, "fpr=range(2013-12-31, 2017-12-31)" sets the first reference to the fiscal period with an end date on or directly before 12/31/2013 and the last reference to the fiscal period with an end date on or directly before 12/31/2017.</p> |
| <YYYY><br><br>or<br><br><YYYY> + A           | <p>Sets the reference to the fiscal period that ends at the same time as the year you specify.</p> <p>For example, "fpr='2015'" or "fpr='2015A'" sets the reference to the fiscal period with the same end date as 2015A.</p>                                                                                                                                                                                                                                                                                                                                                                                                      |
| <YYYY> +<br><br>S1, S2, Q1, Q2, Q3, or<br>Q4 | <p>Sets the reference to the fiscal period that ends at the same time as the semi-annual or quarterly period you specify.</p> <p>For example:</p> <ul style="list-style-type: none"> <li>"fpr='2015S2'" sets the reference to the fiscal period that ends at the same time as S2 2015.</li> <li>"fpr='2015Q1'" sets the reference to the fiscal period with an end date on or directly before the end of Q1 2015.</li> </ul>                                                                                                                                                                                                       |

| Value                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>&lt;YYYY Range&gt;</p> <p>+ A, S1, S2, Q1, Q2, Q3, or Q4</p> | <p>Sets the reference to a range of periods, with the letter representing the length of the steps between reference periods in the range. The range must be specified as an argument to the <i>range()</i> function.</p> <p>For example:</p> <ul style="list-style-type: none"> <li>"fpr=range('2015A', '2016A')" specifies a range of reference periods, with one year jumps between the reference periods.</li> <li>"fpr=range('2015S1', '2016S2')" specifies a range of reference periods, with one semi-annual period jumps between the reference periods: S1 2015, S2 2015, S1 2016, and S2 2016.</li> <li>"fpr=range('2015Q1', '2016Q2')" specifies a range of reference periods, with one quarter jumps between the reference periods: Q1 2015, Q2 2015, Q3 2015, Q4 2015, Q1 2016, Q2 2016.</li> </ul> |

**Note:** The value you assign to *Period Reference* interacts with the [Period Offset](#), [Period Type](#), and [As of Date](#) parameters. For more: [Fundamental Parameter Interactions](#).

#### Example Formulas:

- Retrieve IBM's earnings per share (EPS) for the Last 12 Months (LTM) period ending at the same time as Q3 2015:  
 =BQL("IBM US Equity", "is\_eps(fpr='2015Q3')")  
  
 =BQL.Query("get(is\_eps(fpr='2015Q3')) for(['IBM US Equity'])")
- Retrieve IBM's EPS for the Last 12 Months (LTM) period ending at the same time as the fiscal period that ends on or directly before September 30, 2015 (i.e., Q3 2015):  
 =BQL("IBM US Equity", "is\_eps(fpr=2015-09-30)")  
  
 =BQL.Query("get(is\_eps(fpr=2015-09-30)) for(['IBM US Equity'])")
- Retrieve IBM's EPS at the end of each quarter between Q3 2015 and Q3 2017:  
 =BQL("IBM US Equity", "is\_eps(fpr=range('2015Q3', '2017Q3'))")  
  
 =BQL.Query("get(is\_eps(fpr=range('2015Q3', '2017Q3'))) for(['IBM US Equity'])")
- Retrieve IBM's EPS for each fiscal period that ends on or between September 30, 2015 and September 30, 2017 (the fiscal periods for data depend on the primary periodicity that the company uses for reporting):  
 =BQL("IBM US Equity", "is\_eps", "fpr=range(2015-09-30, 2017-09-30)")  
  
 =BQL.Query("get(is\_eps(fpr=range(2015-09-30, 2017-09-30))) for(['IBM US Equity'])")
- Retrieve IBM's EPS for each trailing twelve-month period ending at the same time as each quarter between Q2 2016 to Q4 2017:  
 =BQL("IBM US Equity", "is\_eps", "fpr=range('2016Q2', '2017Q4')", "fpt=BA")

```
=BQL.Query("get(is_eps(fpr=range('2016Q2', '2017Q4'), fpt=BA)) for(['IBM US Equity']))")
```

## Period Type

For BQL() and BQL.Query() formulas in Excel, the *Period Type* (fa\_period\_type) parameter lets you control the type of fiscal period returned, which could be a fiscal period reported by the company (such as an Annual, Quarterly or Semiannual fiscal period) or a synthetic fiscal period created by Bloomberg (such as a last-12-months, calendar, or blended period).

The source for reported data can be the company (from periodic financial filings) or sell-side research (from research reports).

**Parameter Mnemonic:** fa\_period\_type

**Parameter Mnemonic Alias:** fpt

**Valid Values:**

| Value | Description                                                                                                                                                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | Annual<br><br>Retrieves data for annual fiscal periods, as reported by the source.                                                                                                                                                                                               |
| Q     | Quarter<br><br>Retrieves data for quarterly fiscal periods, as reported by the source.                                                                                                                                                                                           |
| S     | Semi-Annual<br><br>Retrieves data for semi-annual fiscal periods, as reported by the source.                                                                                                                                                                                     |
| P     | Primary<br><br>Retrieves data for the primary periodicity used by the company to report its financials (e.g., quarters, annuals, etc.).<br><br><b>Note:</b> For the primary reporting periodicity of the company, see the company's value for the data item primary_periodicity. |

| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LTM   | <p>Last Twelve Months <b>(Default)</b></p> <p>Retrieves data for a synthetic, Bloomberg-calculated fiscal period that covers the previous 12 months. The calculation may use quarterly, semi-annual, or annual data reported by the company, depending on the company's primary reporting periodicity.</p> <p>LTM can be calculated as either:</p> <ul style="list-style-type: none"> <li>• The sum of four quarterly or two semi-annual periods (depending on company disclosure) for flow accounting concepts (such as Revenue)</li> <li>• The average of four quarterly or two semi-annual periods (depending on company disclosure) for average accounting concepts (such as Average Shares Outstanding)</li> <li>• The data reported in the latest fiscal period for stock accounting concepts (such as Total Assets)</li> <li>• The data reported at the beginning of the period for any beginning-of-period accounting concepts (such as Cash at the beginning of the period)</li> </ul> <p>For any last-12-month fiscal period coinciding with the annual fiscal period, the annual data is returned without any additional calculations.</p> |
| YTD   | <p>Year to Date</p> <p>Retrieves data for a synthetic, Bloomberg-calculated fiscal period that covers the last three, six, nine, or 12 months of the current calendar year. The calculation may use quarterly, semi-annual, or annual data reported by a company, depending on the company's primary reporting periodicity.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BA    | <p>Blended Annual</p> <p>Retrieves data for a synthetic, 12-month fiscal period calculated by Bloomberg that ends on a custom date and is independent of any observation date you specify with the <a href="#">As of Date</a> parameter. The calculation blends data from quarterly, semi-annual, or annual data reported by a company (depending on the company's primary reporting periodicity), using a time-weighted average.</p> <p>The default fiscal year end date is the latest December 31 date that precedes the end date of the last fiscal period reported by the company. However, you can customize the end date with the <a href="#">Year End Alignment</a> parameter.</p>       |
| BQ    | <p>Blended Quarter</p> <p>Retrieves data for a synthetic, three-month fiscal period calculated by Bloomberg that ends on a custom date and is independent of any observation date you specify with the <a href="#">As of Date</a> parameter. The calculation blends data from quarterly, semi-annual, or annual data reported by a company (depending on the company's primary reporting periodicity), using a time-weighted average.</p> <p>The default fiscal year end date is the latest December 31 date that precedes the end date of the last fiscal period reported by the company. However, you can customize the end date with the <a href="#">Year End Alignment</a> parameter.</p>   |
| BS    | <p>Blended Semi-Annual</p> <p>Retrieves data for a synthetic, six-month fiscal period calculated by Bloomberg that ends on a custom date and is independent of any observation date you specify with the <a href="#">As of Date</a> parameter. The calculation blends data from quarterly, semi-annual, or annual data reported by a company (depending on the company's primary reporting periodicity), using a time-weighted average.</p> <p>The default fiscal year end date is the latest December 31 date that precedes the end date of the last fiscal period reported by the company. However, you can customize the end date with the <a href="#">Year End Alignment</a> parameter.</p> |

| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BT    | <p>Blended Trailing</p> <p>Retrieves data for a synthetic 12-month fiscal period calculated by Bloomberg that ends on a custom date. The calculation blends data from quarterly, semi-annual, or annual data reported by a company (depending on the company's primary reporting periodicity), using a time weighted average.</p> <p>The end date is determined by the value for the <a href="#">As of Date</a> parameter.</p> |
| Q1    | <p>Quarter 1</p> <p>Returns data for a Q1 fiscal period, as reported by the source.</p>                                                                                                                                                                                                                                                                                                                                        |
| Q2    | <p>Quarter 2</p> <p>Returns data for a Q2 fiscal period, as reported by the source.</p>                                                                                                                                                                                                                                                                                                                                        |
| Q3    | <p>Quarter 3</p> <p>Returns data for a Q3 fiscal period, as reported by the source.</p>                                                                                                                                                                                                                                                                                                                                        |
| Q4    | <p>Quarter 4</p> <p>Returns data for a Q4 fiscal period, as reported by the source.</p>                                                                                                                                                                                                                                                                                                                                        |
| S1    | <p>Semiannual 1</p> <p>Returns data for a Semiannual 1 fiscal period, as reported by the source.</p>                                                                                                                                                                                                                                                                                                                           |

| Value | Description                                                                                   |
|-------|-----------------------------------------------------------------------------------------------|
| S2    | Semiannual 2<br><br>Returns data for a Semiannual 2 fiscal period, as reported by the source. |

**Note:** The value you assign to *Period Type* interacts with the [Period Offset](#), [Period Reference](#), and [As of Date](#) parameters. For more: [Fundamental Parameter Interactions](#).

#### Example Formulas:

- Retrieve IBM's earnings per share (EPS) for the last reported quarter:  
`=BQL("IBM US Equity", "is_eps(fpt=Q)")`  
  
`=BQL.Query("get(is_eps(fpt=Q)) for(['IBM US Equity'])")`
- Retrieve IBM's EPS for a trailing 12-month period that ends on February 1, 2018:  
`=BQL("IBM US Equity", "is_eps(fpt=BT, dates=2018-02-01)")`  
  
`=BQL.Query("get(is_eps(fpt=BT, dates=2018-02-01)) for(['IBM US Equity'])")`
- Retrieve IBM's estimate EPS for the next quarter:  
`=BQL("IBM US Equity", "is_eps(fpo='1', fpt=Q)")`  
  
`=BQL.Query("get(is_eps(fpo='1', fpt=Q)) for(['IBM US Equity'])")`
- Retrieve IBM's estimate EPS for the trailing 12-month period ending on October 20, 2018, as forecast each day from February 01, 2018 to April 15, 2018. Output is blank when no new estimate was made:  
`=BQL("IBM US Equity", "is_eps(fpt=BA, dates=range(2018-02-01, 2018-04-15), fye=C1020)")`  
  
`=BQL.Query("get(is_eps(fpt=BA, dates=range(2018-02-01, 2018-04-15), fye=C1020)) for(['IBM US Equity'])")`

#### Source for Calculated Period Types

For BQL() and BQL.Query() formulas in Excel, the *Source for Calculated Period Types* (fa\_period\_type\_source) parameter lets you choose the fiscal periodicity used to calculate values for synthetic periods you select with the [Period Type](#) (fa\_period\_type) parameter, such as the Last 12 Months (LTM) and Blended Trailing (BT) periods. By default, the source is the Primary (P) periodicity.

**Parameter Mnemonic:** fa\_period\_type\_source

**Parameter Mnemonic Alias:** fpts

**Valid Values:**

| Value | Description                                                                                                                                                                                                                                                                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P     | <p>Primary (<b>Default</b>)</p> <p>Calculates the data for synthetic periods using the primary fiscal periodicity used by the company for reporting. Companies may report data annually, semi-annually, and/or quarterly.</p> <p><b>Note:</b> For the primary reporting periodicity of the company, see the company's value for the data item primary_periodicity.</p> |
| A     | <p>Annual</p> <p>Calculates the data for synthetic fiscal periods from annual periods reported by the company.</p>                                                                                                                                                                                                                                                     |
| S     | <p>Semi-Annual</p> <p>Calculates the data for synthetic fiscal periods from semi-annual periods reported by the company.</p>                                                                                                                                                                                                                                           |
| Q     | <p>Quarter</p> <p>Calculates the data for synthetic fiscal periods from quarterly periods reported by the company.</p>                                                                                                                                                                                                                                                 |

**Example Formulas:**

- Retrieve IBM's earnings per share (EPS) for the latest Last 12 months (LTM) period, calculated from the annual reported EPS from the company:  
`=BQL("IBM US Equity", "is_eps(fpts=A)")`  
`=BQL.Query("get(is_eps(fpts=A)) for(['IBM US Equity'])")`
- Retrieve IBM's EPS for the trailing 12-month period ending on September 15, 2017, calculated from the quarterly reported EPS from the company:  
`=BQL("IBM US Equity", "is_eps(fpts=Q, fpt=BT, dates=2017-09-15)")`  
`=BQL.Query("get(is_eps(fpts=Q, fpt=BT, dates=2017-09-15)) for(['IBM US Equity'])")`

## Year End Alignment

For BQL() and BQL.Query() formulas, the *Year End Alignment* (fa\_period\_year\_end) parameter allows you to realign a company's fiscal periods to a custom calendar for reported values. You can specify 'C' to align the fiscal year with the calendar year or C + MMDD (e.g., 'C0925') to end the fiscal year on a custom date.

**Note:** If you do not use a blended period for the *Period Type* (fa\_period\_type) parameter, such as Blended Annual (BA), you are just shifting values to a different date and not recalculating them.

**Parameter Mnemonic:** fa\_period\_end\_year

**Parameter Mnemonic Alias:** fye

**Valid Values:**

| Value      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F          | <p>Fiscal Year End <b>(Default)</b></p> <p>Defines the fiscal periods of a company according to the standard fiscal calendar of that company.</p> <p><b>Note:</b> Bloomberg defines the standard fiscal calendar for a company based on the period end date of a company's annual report. If the period end date of a company's annual report falls between 01/15/&lt;YYYY&gt; to 01/14/&lt;YYYY&gt;+1, Bloomberg defines that annual report as Fiscal Year &lt;YYYY&gt;. Interim (semiannual or quarterly) reports are then defined based on the definition of the annual report, which includes the time frames covered by those interim reports.</p> |
| C          | <p>Calendar Year End</p> <p>Realigns the fiscal periods of a company based on a calendar year (ending on December 31).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| C + <MMDD> | <p>Custom Year End</p> <p>Realigns the fiscal periods of a company based on a fiscal year ending on a custom month and date.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

For example, Company X has a fiscal year ending on September 30 and Company Y has a fiscal year ending on June 30. Both companies report Annual and Quarterly filings. The fye parameters of "C" or "C0531" make fiscal period names consistent for both companies:

| Period end   | Company   | fye='F' | fye='C' | fye='C0531' |
|--------------|-----------|---------|---------|-------------|
| March 31     | Company X | Q2      | Q1      | Q3          |
|              | Company Y | Q3      | Q1      | Q3          |
|              |           |         |         |             |
| June 30      | Company X | Q3      | Q2      | Q4/A        |
|              | Company Y | Q4/A    | Q2      | Q4/A        |
|              |           |         |         |             |
| September 30 | Company X | Q4/A    | Q3      | Q1          |
|              | Company Y | Q1      | Q3      | Q4          |
|              |           |         |         |             |
| December 31  | Company X | Q1      | Q4/A    | Q2          |
|              | Company Y | Q2      | Q4/A    | Q2          |

#### Example Formulas:

- Retrieve IBM's earnings per share (EPS) for the last reported fiscal year, realigning the fiscal year to end at the end of the calendar year:  
`=BQL("IBM US Equity", "is_eps(fye='C', fpt=A)")`  
`=BQL.Query("get(is_eps(fye='C', fpt=A)) for(['IBM US Equity'])")`
- Retrieve IBM's EPS for the last reported fiscal year, realigning the fiscal year to end on September 15:  
`=BQL("IBM US Equity", "is_eps(fye='C0915', fpt=A)")`  
`=BQL.Query("get(is_eps(fye='C0915', fpt=A)) for(['IBM US Equity'])")`

#### Fundamental Parameter Interactions

When using equity fundamental parameters to retrieve company financials and estimates for specific and/or relative time periods, it is important to understand how the parameters interact and what data you will get. The following topics explain key fundamental parameter interactions.

#### Type and Offset

Period Type (fa\_period\_type; abbrev: fpt) and Period Offset (fa\_period\_offset; abbrev: fpo):

| Example                                                                                                                                                                                                 | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity Only</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpt=Q)")</pre><br><pre>=BQL.Query("get(is_eps(fpt=Q)) for(['IBM US Equity'])")</pre>                                                  | <p>The value for <i>fpt</i> is "Q", so the query retrieves quarterly data. Because you are not using <i>fpo</i> to specify an offset, the query gets data for the last reported period.</p><br><p>Therefore, the query retrieves the EPS for the last reported quarter.</p>                                                                                                                                                                                                           |
| <p><b>Offset Only</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpo='1')")</pre><br><pre>=BQL.Query("get(is_eps(fpo='1')) for(['IBM US Equity'])")</pre>                                                   | <p>Because you are not using <i>fpt</i> to specify a periodicity, the default Last 12 Months ("LTM") period is implied. The value for <i>fpo</i> is a positive integer ("1"), so the query looks forward one period. However, <i>fpo</i> does not include a length for the "1" offset, so the <i>fpt</i> value – "LTM" – is also used as the offset length.</p><br><p>Therefore, the query retrieves the estimate EPS for the next LTM period after the last reported LTM period.</p> |
| <p><b>Periodicity and Offset (Integer Only)</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpo='1', fpt=Q)")</pre><br><pre>=BQL.Query("get(is_eps(fpo='1', fpt=Q)) for(['IBM US Equity'])")</pre>           | <p>The value for <i>fpo</i> is a positive integer ("1"), so the query looks forward one period. However, <i>fpo</i> ("1") does not specify the length of the offset, so the value for <i>fpt</i> determines the offset length in addition to determining the periodicity.</p><br><p>Since <i>fpt</i>=Q, the query retrieves the estimated EPS for the next quarter.</p>                                                                                                               |
| <p><b>Periodicity and Offset (Integer + Period)</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpo='1Q', fpt=LTM)")</pre><br><pre>=BQL.Query("get(is_eps(fpo='1Q', fpt=LTM)) for(['IBM US Equity'])")</pre> | <p>The value for <i>fpo</i> specifies the length of the offset step as one quarter ("1Q") forward. The value for <i>fpt</i> specifies the periodicity as Last 12 Months ("LTM").</p><br><p>Therefore, the query retrieves the estimated EPS for the LTM period that ends at the same time as the next reported quarter.</p>                                                                                                                                                           |

| Example                                                                                                                                                                                                                                  | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity and Offset (Integer Range + Period)</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpo=range('-1Q', '1Q'), fpt=LTM)")</pre> <pre>=BQL.Query("get(is_eps(fpo = range('-1Q', '1Q'), fpt=LTM)) for(['IBM US Equity'])")</pre> | <p>The value for <i>fpo</i> ("range(-1Q, 1Q)") indicates an offset range with steps of one quarter (the quarter before the last reported quarter, the last reported quarter, and the next reported quarter). The value for <i>fpt</i> is Last 12 Months ("LTM").</p> <p>Therefore, the query retrieves the actual EPS for the LTM period that ended at the same time as the quarter before the last reported quarter; the actual EPS for the LTM period that ended at the same time as the last reported quarter; and the estimate EPS for the LTM period that ends at the same time as the next reported quarter.</p> |

For additional information: [BQL Bloomberg Fundamentals: Data Parameters & Associative Columns](#).

## Type and Reference

[Period Type](#) (fa\_period\_type; abbrev: fpt) and [Period Reference](#) (fa\_period\_reference; abbrev: fpr):

| Example                                                                                                                                                         | Explanation                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity Only</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpt=Q)")</pre> <pre>=BQL.Query("get(is_eps(fpt=Q)) for(['IBM US Equity'])")</pre>             | <p>Because no value is specified for <i>fpr</i>, the default value (today) is implied, so data is retrieved for the last fiscal period that ended on or directly before today. The value for <i>fpt</i> indicates the periodicity is quarterly ("Q").</p> <p>Therefore, the query retrieves the EPS for the last reported quarter.</p> |
| <p><b>Reference Only</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr='2016Q4')")</pre> <pre>=BQL.Query("get(is_eps(fpr='2016Q4')) for(['IBM US Equity'])")</pre> | <p>Because no value is specified for <i>fpt</i>, the default Last 12 Months ("LTM") period is implied. The value for <i>fpr</i> defines the end of the fiscal period for which you want data as the end of Q4 2016.</p> <p>Therefore, the query retrieves the EPS for the LTM period that ended at the same time as Q4 2016.</p>       |

| Example                                                                                                                                                                                                                                                    | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity and Reference</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr='2016Q4', fpt=Q)")</pre><br><pre>=BQL.Query("get(is_eps(fpr= '2016Q4', fpt=Q)) for(['IBM US Equity']))")</pre>                                                              | <p>The value for <i>fpr</i> is "2016Q4", which indicates that the end of the fiscal period for which you want data is the end of Q4 2016. The value for <i>fpt</i> is quarterly ("Q").</p> <p>Therefore, the query retrieves the Q4 2016 EPS.</p>                                                                                                                                                                                                                                            |
| <p><b>Periodicity and Reference Range (Years)</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr=range('2015', '2016'), fpt=Q)")</pre><br><pre>=BQL.Query("get(is_eps(fpr = range('2015', '2016'), fpt=Q)) for(['IBM US Equity']))")</pre>                     | <p>The value for <i>fpr</i> is a range ("range('2015', '2016')"), so the query retrieves data between the end of 2015 and the end of 2016. By default, when you specify only a year for <i>fpr</i> (like "2015"), an annual fiscal period is implied. That means each step in the <i>fpr</i> range is one year. The value for <i>fpt</i> is quarterly ("Q").</p> <p>Therefore, the query retrieves the Q4 2015 EPS and Q4 2016 EPS.</p>                                                      |
| <p><b>Periodicity and Reference Range (Years + Period)</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr=range('2016Q2', '2016Q4'), fpt=LTM)")</pre><br><pre>=BQL.Query("get(is_eps(fpr= range('2016Q2', '2016Q4'), fpt=LTM)) for(['IBM US Equity']))")</pre> | <p>The value for <i>fpr</i> is a range ("range('2016Q2', '2016Q4')"), so the query retrieves data between the end of Q2 2016 and the end of Q4 2016. By including a periodicity in the <i>fpr</i> values ("Q"), you specify the length of each step (one quarter) in the <i>fpr</i> range. The value for <i>fpt</i> is Last 12 Months ("LTM").</p> <p>Therefore, the query retrieves the LTM EPS values for periods ending at the same time as the end of Q2 2016, Q3 2016, and Q4 2016.</p> |

| Example                                                                                                                                                                                                                                            | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity and Reference Range (Dates)</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr=range(2015-12-31, 2016-12-31), fpt=Q)")</pre> <pre>=BQL.Query("get(is_eps(fpr= range(2015-12-31, 2016-12-31), fpt=Q)) for(['IBM US Equity']))")</pre> | <p>The value for <i>fpr</i> is "range(2015-12-31, 2016-12-31)", which specifies the range for which you want data begins with the fiscal period ending on or directly before December 31, 2015 and ends with the period that ends on or directly before December 31, 2016. Because the <i>fpr</i> range only specifies fiscal period end dates and no fiscal period type (e.g., "2015Q2" or "2016"), the length of the steps is determined by the value for <i>fpt</i>, which is quarterly ("Q"). <i>fpt=Q</i> also indicates that you want quarterly data.</p> <p>Therefore, the query retrieves the Q4 2015, Q1 2016, Q2 2016, Q3 2016, and Q4 2016 EPS.</p> |

For additional information: [BQL Bloomberg Fundamentals: Data Parameters & Associative Columns](#).

### Type, Offset, and Reference

[Period Type](#) (fa\_period\_type; abbrev: fpt), [Period Offset](#) (fa\_period\_offset; abbrev: fpo), and [Period Reference](#) (fa\_period\_reference; abbrev: fpr):

| Example                                                                                                                                                                                                                                                                                                                                                                                                    | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity, Offset, and Reference - No Ranges</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr='2016Q1', fpo='1', fpt=Q)")</pre> <pre>=BQL.Query("get(is_eps(fpr= '2016Q1', fpo='1', fpt=Q)) for(['IBM US Equity']))")</pre> <pre>=BQL("IBM US Equity", "is_eps(fpr=2016-03-31, fpo='1', fpt=Q)")</pre> <pre>=BQL.Query("get(is_eps(fpr= 2016-03-31, fpo='1', fpt=Q)) for(['IBM US Equity']))")</pre> | <p>The value for <i>fpr</i> ("2016Q1" or March 31, 2016) specifies that you want data looking forward or backward from the end of Q1 2016. The offset you specify with <i>fpo</i> ("1") is a positive integer, so the query looks forward one step from Q1 2016. <i>fpo</i> does not specify the length of the offset step, so the <i>fpt</i> value ("Q") determines the length of the offset step as well as the periodicity for the data.</p> <p>Therefore, the query retrieves the Q2 2016 EPS.</p> |

| Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity, Reference, and Offset Range (Integers Only)</b></p> <p><code>=BQL("IBM US Equity",<br/>"is_eps(fpr='2017Q1', fpo=range('-1', '1'),<br/>fpt=Q)")</code></p> <p><code>=BQL.Query("get(is_eps(fpr='2017Q1',<br/>fpo=range('-1', '1'), fpt=Q)) for(['IBM US<br/>Equity']))")</code></p> <p><code>=BQL("IBM US Equity", "is_eps(fpr=2017-<br/>03-31, fpo=range('-1', '1'), fpt=Q)")</code></p> <p><code>=BQL.Query("get(is_eps(fpr= 2017-03-31,<br/>fpo=range('-1', '1'), fpt=Q)) for(['IBM US<br/>Equity']))")</code></p>                        | <p>The value for <i>fpr</i> ("2017Q1" or March 31, 2017) specifies that you want data looking forward or backward from the end of Q1 2017. The offset range you specify with <i>fpo</i> ("range('-1', '1')") gets data for a range starting with the previous period and ending with the next period. Because <i>fpo</i> does not specify the length of the offset steps, the <i>fpt</i> value ("Q") determines the length of the steps as well as the periodicity of the data. Each offset step is one quarter: one quarter before Q1 2017, Q1 2017, and one quarter after Q1 2017.</p> <p>Therefore, the query retrieves the EPS for Q4 2016, Q1 2017, and Q2 2017.</p> |
| <p><b>Periodicity, Reference, and Offset Range (Integers + Periods)</b></p> <p><code>=BQL("IBM US Equity",<br/>"is_eps(fpr='2017Q1', fpo=range('-1Q',<br/>'1Q'), fpt=LTM)")</code></p> <p><code>=BQL.Query("get(is_eps(fpr= '2017Q1',<br/>fpo=range('-1Q', '1Q' ), fpt=LTM)) for(['IBM<br/>US Equity']))")</code></p> <p><code>=BQL("IBM US Equity", "is_eps(fpr=2017-<br/>03-31, fpo=range('-1Q', '1Q'), fpt=LTM)")</code></p> <p><code>=BQL.Query("get(is_eps(fpr= 2017-03-31,<br/>fpo=range('-1Q', '1Q'), fpt=LTM)) for(['IBM<br/>US Equity']))")</code></p> | <p>The value for <i>fpr</i> ("2017Q1" or March 31, 2017) specifies that you want data looking forward or backward from the end of Q1 2017. The offset range you specify with <i>fpo</i> ("range('-1Q', '1Q')") gets data for a range starting with the previous period and ending with the next period. It also specifies the length of each offset step as one quarter. The value for <i>fpt</i> is Last 12 Months ("LTM"), so for each offset step, the query retrieves data for LTM periods.</p> <p>Therefore, the query retrieves the LTM EPS for the periods that end at the same time as the end of Q4 2016, Q1 2017, and Q2 2017.</p>                              |

| Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity, Offset, and Reference Range</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr=range('2015', '2016'), fpo='1', fpt=LTM)")</pre> <pre>=BQL.Query("get(is_eps(fpr=range('2015', '2016'), fpo='1', fpt=LTM)) for(['IBM US Equity']))")</pre> <pre>=BQL("IBM US Equity", "is_eps(fpr=range(2015-12-31, 2016-12- 31), fpo='1', fpt=LTM)")</pre> <pre>=BQL.Query("get(is_eps(fpr= range(2015- 12-31, 2016-12-31), fpo='1', fpt=LTM)) for(['IBM US Equity']))")</pre>                        | <p>Because the <i>fpr</i> range only specifies a fiscal year/date and does not specify a fiscal period type ("range('2015', '2016')"), it is implied that you want data looking forwards or backwards from the end of each year. Since the offset you specify with <i>fpo</i> ("1") does not specify the length of the offset step, the value for <i>fpt</i> ("LTM") defines the length of the offset step as one LTM period. <i>fpo='1'</i> specifies that the period for data should be one step forward from the end of each year. <i>fpt=LTM</i> specifies that you want Last 12 Month EPS.</p> <p>Therefore, the query retrieves the LTM EPS from the end of 2016 and 2017.</p> |
| <p><b>Periodicity, Offset (Integer + Period), and Reference Range</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr=range('2015', '2016'), fpo='2Q', fpt=LTM)")</pre> <pre>=BQL.Query("get(is_eps(fpr=range('2015', '2016'), fpo='2Q', fpt=LTM)) for(['IBM US Equity']))")</pre> <pre>=BQL("IBM US Equity", "is_eps(fpr=range(2015-12-31, 2016-12- 31), fpo='2Q', fpt=LTM)")</pre> <pre>=BQL.Query("get(is_eps(fpr= range(2015- 12-31, 2016-12-31), fpo='2Q', fpt=LTM)) for(['IBM US Equity']))")</pre> | <p>Because the <i>fpr</i> range only specifies a fiscal year/date and does not specify a fiscal period type ("range('2015', '2016')"), it is implied that you want data looking forwards or backwards from the end of each year. The offset you specify with <i>fpo</i> ("2Q") sets the length of the offset step as two quarters forward. <i>fpt=LTM</i> indicates that you want Last 12 Month EPS.</p> <p>Therefore, the query retrieves the LTM EPS from the end of Q2 2016 and Q2 2017.</p>                                                                                                                                                                                      |

| Example                                                                                                                                                                                                                                                                                                | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity, Offset (Integer + Period), and Reference Range (Year + Period)</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr=range('2015Q4', '2016Q4'), fpo='2Q', fpt=LTM)")</pre> <pre>=BQL.Query("get(is_eps(fpr= range('2015Q4', '2016Q4'), fpo='2Q', fpt=LTM)) for(['IBM US Equity']))")</pre> | <p>Because the <i>fpr</i> range ("range('2015Q4', '2016Q4')") specifies a quarterly fiscal period, the query retrieves data looking forwards or backwards from a range between Q4 2015 and Q4 2016. By specifying the quarterly fiscal period with <i>fpr</i>, you specify that the <i>fpr</i> range is in quarterly steps: Q4 2015, Q1 2016Q1, Q2 2016, Q3 2016, and Q4 2016. The <i>fpo</i> value ("2Q") specifies the length of an offset step that is two quarters forward from the end of each quarter in the <i>fpr</i> range. The value for <i>fpt</i> ("LTM") specifies that you want the Last 12 Month EPS.</p> <p>Therefore, the query retrieves the LTM EPS from the end of Q2 2016, Q3 2016, Q4 2016, Q1 2017, and Q2 2017.</p> |
| <p><b>Periodicity, Offset Range, and Reference Range</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr=range('2015Q4', '2016Q4'), fpo=range('-2Q', '1Q'), fpt=LTM)")</pre> <pre>=BQL.Query("get(is_eps(fpr= range('2015Q4', '2016Q4'), fpo=range('-2Q', '1Q'), fpt=LTM)) for(['IBM US Equity']))")</pre>  | <p>This query generates an error. You cannot specify a range for both the <i>fpr</i> and the <i>fpo</i> parameters.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

For additional information: [BQL Bloomberg Fundamentals: Data Parameters & Associative Columns](#).

## Type, Offset, Reference, and Dates

Period Type (fa\_period\_type; abbrev: fpt), Period Offset (fa\_period\_offset; abbrev: fpo), Period Reference (fa\_period\_reference; abbrev: fpr), and As of Date (as\_of\_date; abbrev: dates):

| Example                                                                                                                                                                                           | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>No Parameters</b></p> <p>=BQL("IBM US Equity", "is_eps")</p> <p>=BQL.Query("get(is_eps) for(['IBM US Equity'])")</p>                                                                        | <p>Because no value is specified for <i>fpr</i>, <i>fpo</i>, <i>fpt</i>, or <i>dates</i>, the default values are implied (<i>fpr=today</i>, <i>fpo='O'</i>, <i>fpt=LTM</i>, and <i>dates=today's date</i>).</p> <p>Therefore, the formula retrieves EPS for the last LTM fiscal period reported as of today.</p>                                                                                                                   |
| <p><b>As of Date Only</b></p> <p>=BQL("IBM US Equity",<br/>"is_eps(dates=2016-09-01)")</p> <p>=BQL.Query("get(is_eps(dates=2016-09-01)) for(['IBM US Equity'])")</p>                              | <p>Because no value is specified for <i>fpr</i>, <i>fpo</i>, or <i>fpt</i>, the default values are implied (<i>fpr=none</i>, <i>fpo='O'</i>, and <i>fpt=LTM</i>). The value for <i>dates</i> ("2016-09-01") specifies that the observation date for the value is September 1, 2016.</p> <p>Therefore, the formula retrieves the EPS for the last LTM period, as forecast on September 1, 2016.</p>                                 |
| <p><b>Periodicity and As Of Date</b></p> <p>=BQL("IBM US Equity", "is_eps(fpt=Q,<br/>dates=2016-09-01)")</p> <p>=BQL.Query("get(is_eps(fpt=Q,<br/>dates=2016-09-01)) for(['IBM US Equity'])")</p> | <p>Because no value is specified for <i>fpr</i> or <i>fpo</i>, the default values are implied (<i>fpr=none</i> and <i>fpo='O'</i>). The value for <i>dates</i> ("2016-09-01") specifies that the observation date for the value is September 1, 2016. The value for <i>fpt</i> specifies that you want quarterly data.</p> <p>Therefore, the formula retrieves the EPS for the last quarter, as forecast on September 1, 2016.</p> |

| Example                                                                                                                                                                                                                                                                                                                                                                                                                                   | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Reference and As Of Date</b></p> <p>=BQL("IBM US Equity",<br/>"is_eps(fpr='2016Q4', dates=2016-11-30)")</p> <p>=BQL.Query("get(is_eps(fpr='2016Q4',<br/>dates=2016-11-30)) for(['IBM US Equity'])")</p> <p>=BQL("IBM US Equity", "is_eps(fpr=2016-12-31, dates=2016-11-30)")</p> <p>=BQL.Query("get(is_eps(fpr=2016-12-31, dates=2016-11-30)) for(['IBM US Equity'])")</p>                                                          | <p>Because no value is specified for <i>fpo</i> or <i>fpt</i>, the default values are implied (<i>fpo</i>='0' and <i>fpt</i>=<i>LTM</i>). The value for <i>fpr</i> is "2016Q4" or "2016-12-31", so the end date for the fiscal period for which you want data is Q4 2016 (December 31, 2016). The value for <i>dates</i> ("2016-11-30") specifies that the observation date for the value is November 30, 2016.</p> <p>Therefore, the formula retrieves the estimate EPS for the LTM period ending on December 31, 2016, as forecast on November 30, 2016.</p>                                                                                     |
| <p><b>Reference, Offset, and As Of Date</b></p> <p>=BQL("IBM US Equity",<br/>"is_eps(fpr='2016Q4', fpo='-1Q',<br/>dates=2016-09-01)")</p> <p>=BQL.Query("get(is_eps(fpr='2016Q4',<br/>fpo='-1Q', dates=2016-09-01)) for(['IBM US Equity'])")</p> <p>=BQL("IBM US Equity", "is_eps(fpr=2016-12-31, fpo='-1Q', dates=2016-09-01)")</p> <p>=BQL.Query("get(is_eps(fpr=2016-12-31, fpo='-1Q', dates=2016-09-01)) for(['IBM US Equity'])")</p> | <p>Because no value is specified for <i>fpt</i>, the default value is implied (<i>fpt</i>=<i>LTM</i>). The value for <i>fpr</i> is "2016Q4" or "2016-12-31", so the end date for the fiscal period for which you want data is Q4 2016 (December 31, 2016). The value for <i>fpo</i> ("-1Q") specifies that you want data for one quarter backward from the end of Q4 2016. The value for <i>dates</i> ("2016-09-01") specifies that the observation date for the value is September 1, 2016.</p> <p>Therefore, the formula retrieves the estimate EPS for the LTM period ending at the same time as Q3 2016, as forecast on September 1, 2016.</p> |

| Example                                                                                                                                                                                                                                                                     | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Periodicity, Offset (Rolling), and As Of Date Range</b></p> <p>=BQL("IBM US Equity", "is_eps(fpt=Q, fpo='1', dates=range(2016-09-03, 2016-12-03)))")</p> <p>=BQL.Query("get(is_eps(fpt=Q, fpo='1', dates=range(2016-09-03, 2016-12-03))) for(['IBM US Equity'])")</p> | <p>Because no value is specified for <i>fpr</i>, the default value is implied (today). The value for <i>fpo</i> ("1") indicates that you want data looking forward one step from today. Additionally, because "R" or "F" is not specified with the integer value for <i>fpo</i>, "R" (rolling) is implied. Therefore, the reference date for offset steps rolls every day, so on every date in the <i>dates</i> range ("range(2016-09-03, 2016-12-03)"), the query looks forward one period. The value of <i>fpt</i> ("Q") determines the length of the step as well as the fiscal period type for the data.</p> <p>Therefore, the formula retrieves the next quarter EPS for every day between September 3 and December 3, 2016.</p>                                                   |
| <p><b>Periodicity, Offset (Fixed), and As Of Date Range</b></p> <p>=BQL("IBM US Equity", "is_eps(fpt=Q, fpo='1F', dates=range(2016-09-03, 2016-12-03)))")</p> <p>=BQL.Query("get(is_eps(fpt=Q, fpo='1F', dates=range(2016-09-03, 2016-12-03))) for(['IBM US Equity'])")</p> | <p>Because no value is specified for <i>fpr</i>, the default value is implied (today). The value for <i>fpo</i> ("1") indicates that you want data looking forward one step from today. <i>fpo</i> does not specify the length of the offset step, so the value for <i>fpt</i> ("Q") determines the length of the step as well as the fiscal periodicity of the data. Additionally, because "R" or "F" is not specified with the integer value for <i>fpo</i>, "R" (rolling) is implied. Because no value is specified for <i>fpr</i>, the end date for the period for which you want data rolls to match every day in the <i>dates</i> range.</p> <p>Therefore, the formula retrieves the next quarter forward EPS as forecast every day between September 3 and December 3, 2016.</p> |

| Example                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Reference and As Of Date Range</b></p> <pre>=BQL("IBM US Equity", "is_eps(fpr='2017Q1', dates=range(2016-08-03,2016-12-03)))")</pre> <pre>=BQL.Query("get(is_eps(fpr='2017Q1', dates=range(2016-08-03,2016-12-03))) for(['IBM US Equity'])")</pre> <pre>=BQL("IBM US Equity", "is_eps(fpr=2017-03-31, dates=range(2016-08-03, 2016-12-03)))")</pre> <pre>=BQL.Query("get(is_eps(fpr=2017-03-31, dates=range(2016-08-03,2016-12-03))) for(['IBM US Equity'])")</pre> | <p>Because no value is specified for <i>fpo</i> or <i>fpt</i>, the default values are implied (<i>fpo</i>='0' and <i>fpt</i>=<i>LTM</i>). <i>fpr</i>='2017Q1', which defines the end of the fiscal period for which you want data as the end of Q1 2017.</p> <p>Therefore, the formula retrieves the EPS for the LTM period ending at the same time as Q1 2017, as forecast on each date between August 3 and December 3, 2016.</p> |

For additional information: [BQL Bloomberg Fundamentals: Data Parameters & Associative Columns](#).

## Pricing Data Parameters

Pricing parameters let you fit data such as bids, offers, VWAP, and turnover to your models:

## Corporate Action Adjustment

For BQL() and BQL.Query() formulas in Excel, the *Corporate Action Adjustment* (*ca\_adj*) parameter lets you choose whether equity data is adjusted for corporate actions that impact dividends and the number of shares outstanding.

**Parameter Mnemonic:** *ca\_adj*

**Valid Values:**

| Value  | Description                                                                       |
|--------|-----------------------------------------------------------------------------------|
| SPLITS | Adjusted for Splits ( <b>Default</b> )<br><br>Adjusts data for stock splits only. |

| Value | Description                                                                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FULL  | <p>Full Adjustment</p> <p>Adjusts data for corporate actions that impact pricing and the number of shares outstanding, including splits, stock dividends, spin-offs, rights offerings, etc.</p> |
| RAW   | <p>Not Adjusted</p> <p>Leaves data unadjusted for corporate actions.</p>                                                                                                                        |

**Example Formulas:**

- Retrieve a time series of daily last prices for PG&E Corp from April 1, 2013 to June 1, 2013, unadjusted for corporate actions:  
`=BQL("PCG US Equity", "px_last(dates=range(2013-04-01, 2013-06-01), ca_adj=RAW)")`  
`=BQL.Query("get(px_last(dates=range(2013-04-01,2013-06-01), ca_adj=RAW)) for(['PCG US Equity']))")`
- Retrieve a time series of daily last prices for PG&E Corp from April 1, 2013 to June 1, 2013, adjusted for all corporate actions:  
`=BQL("PCG US Equity", "px_last(dates=range(2013-04-01, 2013-06-01), ca_adj=FULL)")`  
`=BQL.Query("get(px_last(dates=range(2013-04-01,2013-06-01), ca_adj=FULL)) for(['PCG US Equity']))")`

**Currency**

For BQL() and BQL.Query() formulas in Excel, the *Currency* (currency) parameter allows you to convert currency-based data into another currency.

**Parameter Mnemonic:** currency

**Default Value:** None (local currency is used)

**Valid Values:** Three-character ISO currency codes. For a list of codes: [CURR Help Page > Currency Codes](#).

**Example Formulas:**

- Retrieve IBM's last price converted from USD to EUR:  
`=BQL("IBM US Equity", "px_last(currency=EUR)")`  
`=BQL.Query("get(px_last(currency=EUR)) for(['IBM US Equity']))")`
- Retrieve a time series of IBM's daily prices from February 1, 2016 to February 28, 2016, converted from USD to JPY:  
`=BQL("IBM US Equity", "px_last(dates=range(2016-02-01, 2016-02-28), currency=JPY)")`  
`=BQL.Query("get(px_last(dates=range(2016-02-01, 2016-02-28), currency=JPY)) for(['IBM US Equity']))")`

As Of Date

For BQL() and BQL.Query() formulas in Excel, the *As Of Date* (dates) parameter lets you perform historical analysis on a single date or a time series of data. You can use the parameter to request data that was observed 'as of' a given date in the past.

When retrieving data for a historical time series, add the *range()* function to specify the date range.

**Parameter Mnemonic:** dates

**Default Value:** OD (today's date)

**Valid Values:**

| Value                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Date>                                 | <p>Absolute as of date</p> <p>Retrieves data as of a specific, fixed date. The date must be in the YYYY-MM-DD format.</p> <p>For example, "dates=2016-09-15" retrieves the data from September 15, 2016.</p>                                                                                                                                                                                                                                         |
| <Date range>                           | <p>Range of Absolute As Of Dates</p> <p>Retrieves a historical time series of data as of specific, fixed date range. Each of the two dates must be in the YYYY-MM-DD format, with the start date first and the dates separated by commas. Additionally, the dates must be nested in the <i>range()</i> function.</p> <p>For example, "dates=range(2016-01-01, 2017-12-31)" retrieves a series of data from January 1, 2016 to December 31, 2017.</p> |
| <Integer> +<br><br>D, W, M, Q, S, or Y | <p>Relative As Of Date</p> <p>Retrieves data as of a relative, rolling date.</p> <p>For example, "dates=-1M" retrieves the data from one month ago, looking back from today.</p>                                                                                                                                                                                                                                                                     |

| Value                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Non-positive integer range> +<br><br>D, W, M, Q, S, or Y | <p>Range of Relative As Of Dates, expressed in days, weeks, months, quarters, semi-annuals, or years</p> <p>Retrieves a time series of data between two specified relative, rolling observation dates. The start date must be first and the two dates must be separated by commas. Additionally, the dates must be nested in the <i>range()</i> function.</p> <p>For example, "dates=range(-10D,0D)" retrieves a stream of data between ten days ago and today.</p> |

**Example Formulas:**

- Retrieve IBM's last price on February 10, 2017:  
 =BQL("IBM US Equity", "px\_last(dates=2017-02-10)")  
  
 =BQL.Query("get(px\_last(dates=2017-02-10)) for(['IBM US Equity'])")
- Retrieve a time series of daily prices for IBM between January 7, 2014 and January 15, 2014:  
 =BQL("IBM US Equity", "px\_last(dates=range(2014-01-07, 2014-01-15))")  
  
 =BQL.Query("get(px\_last(dates=range(2014-01-07, 2014-01-15))) for(['IBM US Equity'])")

**Fill When Data is Missing**

For BQL() and BQL.Query() formulas in Excel, the *Fill When Data is Missing* (fill) parameter specifies the filler value for non-trading days when downloading a historical time series of data. The parameter gives you the ability to get a continuous stream of data even when there is no data available for certain observation dates.

**Parameter Mnemonic:** fill

**Valid Values:**

| Value | Description                                                    |
|-------|----------------------------------------------------------------|
| NA    | <p>No Data <b>(Default)</b></p> <p>Shows "NA" in the cell.</p> |

| Value | Description                                            |
|-------|--------------------------------------------------------|
| PREV  | Previous Data<br><br>Returns the last available value. |
| NEXT  | Next Data<br><br>Returns the next available value.     |

**Example Formulas:**

- Retrieve a time series of IBM's daily prices between January 1, 2014 and December 31, 2014; for non-trading days, "NA" appears:  
`=BQL("IBM US Equity", "px_last(dates=range(2014-01-01, 2014-12-31))")`  
  
`=BQL.Query("get(px_last(dates=range(2014-01-01, 2014-12-31))) for(['IBM US Equity'])")`
- Retrieve a time series of IBM's daily prices between January 1, 2014 and December 31, 2014; for non-trading days, the value from the previous day appears:  
`=BQL("IBM US Equity", "px_last(dates=range(2014-01-01, 2014-12-31), fill=PREV)")`  
  
`=BQL.Query("get(px_last(dates=range(2014-01-01, 2014-12-31), fill=PREV)) for(['IBM US Equity'])")`

**Periodicity**

For BQL() and BQL.Query() formulas in Excel, the *Periodicity* (per) parameter specifies the sampling periodicity of data when downloading a historical times series.

**Parameter Mnemonic:** per

**Valid Values:**

| Value | Description              |
|-------|--------------------------|
| D     | Daily ( <b>Default</b> ) |
| W     | Weekly                   |
| M     | Monthly                  |
| Q     | Quarterly                |

| Value | Description   |
|-------|---------------|
| S     | Semi-Annually |
| Y     | Yearly        |

**Example Formulas:**

- Retrieve a time series of IBM's weekly prices between January 1, 2014 and December 31, 2014:  
 =BQL("IBM US Equity", "px\_last(dates=range(2014-01-01, 2014-12-31), per=W)")  
  
 =BQL.Query("get(px\_last(dates=range(2014-01-01, 2014-12-31), per=W)) for(['IBM US Equity'])")
- Retrieve a time series of IBM's yearly prices between December 31, 2005 and December 31, 2015:  
 =BQL("IBM US Equity", "px\_last(dates=range(2005-12-31, 2015-12-31), per=Y)")  
  
 =BQL.Query("get(px\_last(dates=range(2005-12-31, 2015-12-31), per=Y)) for(['IBM US Equity'])")

**Consensus Statistics**

For data items that return company financials, you can retrieve specific consensus or revision statistics, based on the contributions included in the consensus estimate for the data item. For example, you can retrieve the count of all contributors included in the consensus, the highest estimate, or the number of contributors who dropped coverage within a particular time frame.

**contributor\_count()**

In BQL() and BQL.Query() formulas in Excel, *contributor\_count()* allows you to specify a company financials data item and return the number of contributors included in the Bloomberg Consensus who are providing an estimate for that data item. The number of contributors helps you assess whether the consensus estimate is robust.

**Signature:** contributor\_count(*data\_item*)

where: *data\_item* is a company financials data item (e.g., *is\_eps* or *net\_income*) and its named parameters. Note that data items calculated through BQL expressions (e.g., *pe\_ratio*) are not supported at this time.

**Note:** The data item must include the parameter Period Type (*fa\_period\_type* or *fpt*) with the parameter value A (Annual), S (Semi-Annual), or Q (Quarterly). For example, "*is\_eps(fpt=Q)*".

**Example Formulas:**

- Count the number of contributors whose estimates are included in the calculation of the Bloomberg Consensus estimate for earnings per share (EPS) for the current annual fiscal period:  
 =BQL("IBM US Equity", "contributor\_count(is\_eps(adj=Y, fpo='1', fpt=A))")  
  
 =BQL.Query("get(contributor\_count(is\_eps(adj=Y, fpo='1', fpt=A))) for(['IBM US Equity'])")
- Count the number of contributors whose estimates are included in the calculation of the Bloomberg Consensus estimate for operating income for the current quarter:

```
=BQL("IBM US Equity", "contributor_count(is_oper_inc(adj=Y, fpo='I', fpt=Q))")
```

```
=BQL.Query("get(contributor_count(is_oper_inc(adj=Y, fpo='I', fpt=Q))) for(['IBM US Equity'])")
```

### contributor\_revisions()

In BQL() and BQL.Query() formulas in Excel, *contributor\_revisions()* allows you to specify a company financials data item and count the number of estimate revisions or coverage changes for all contributors to the Bloomberg Consensus estimate for that data item. For example, you can retrieve the number of upgrades to IBM's net income estimates or the number of brokers who dropped coverage within a specified time frame. Revision statistics provide insight into sentiment around a particular measure.

**Signature:** contributor\_revisions(*data\_item*, *revision\_type*, *revision\_window*)

where:

- *data\_item* is a company financials data item (e.g., *is\_eps* or *net\_income*) and its named parameters. Note that data items calculated through BQL expressions (e.g., *pe\_ratio*) are not supported at this time.
- *revision\_type* is the type of estimate revision data you want to retrieve.
- *revision\_window* is the time frame for counting the revisions.

**Note:** The data item must include the parameter Period Type (*fa\_period\_type* or *fpt*) with the parameter value A (Annual), S (Semi-Annual), or Q (Quarterly). For example, "*is\_eps(fpt=Q)*".

**Valid Parameter Values for *revision\_type*:**

| Value   | Description                                                                                                                                               |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| NUMUP   | <p>Number of Upgrades</p> <p>Retrieves the number of upgrade events from contributors included in the consensus estimate during the window.</p>           |
| NUMDN   | <p>Number of Downgrades</p> <p>Retrieves the number of downgrade events from contributors included in the consensus estimate during the window.</p>       |
| NUMCONF | <p>Number of Confirmations</p> <p>Retrieves the number of confirmation events from contributors included in the consensus estimate during the window.</p> |

| Value    | Description                                                                                                                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NUMADD   | <p>Number of Additions</p> <p>Retrieves the number of coverage initiation events from contributors included in the consensus estimate during the window.</p>                                                 |
| NUMDROP  | <p>Number of Drops</p> <p>Retrieves the number of coverage drop events from contributors included in the consensus estimate during the window.</p>                                                           |
| NUMCHG   | <p>Number of Changes</p> <p>Retrieves the number of revision events changing a value from contributors included in the consensus estimate during the window.</p>                                             |
| NUMUNCHG | <p>Number of No Changes</p> <p>Retrieves the number of days for which each contributor included in the consensus estimate did not change any values during the window.</p>                                   |
| NETUP    | <p>Number of Net Upgrades</p> <p>Retrieves the number of contributors included in the consensus whose estimate at the end of the the window was higher than the estimate at the beginning of the window.</p> |
| NETDN    | <p>Number of Net Downgrades</p> <p>Retrieves the number of contributors included in the consensus whose estimate at the end of the window was lower than the estimate at the beginning of the window.</p>    |

| Value    | Description                                                                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NETCONF  | <p>Number of Net Confirmations</p> <p>Retrieves the number of contributors included in the consensus who confirmed that their estimate at the end of the window was the same as the estimate at the beginning of the window.</p> |
| NETADD   | <p>Number of Net Additions</p> <p>Retrieves the number of contributors included in the consensus at the end of the window who were not covering the security at the beginning of the window.</p>                                 |
| NETDROP  | <p>Number of Net Drops</p> <p>Retrieves the number of contributors included in the consensus at the beginning of the window who were not covering the security at the end of the window.</p>                                     |
| NETCHG   | <p>Number of Net Changes</p> <p>Retrieves the number of contributors included in the consensus whose estimate at the beginning of the window was different from the estimate at the end of the window.</p>                       |
| NETUNCHG | <p>Number of Net No Changes</p> <p>Retrieves the number of contributors included in the consensus whose estimate at the beginning of the window was the same as the estimate at the end of the window.</p>                       |

Valid Parameter Values for revision\_window:

| Value                         | Description                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------|
| <Positive integer> +<br><br>D | <p>Number of Days</p> <p>Allows you to set the time period for the calculation to a specific number of days.</p> |

| Value                         | Description                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| <Positive integer> +<br><br>W | Number of Weeks<br><br>Allows you to set the time period for the calculation to a specific number of weeks. |

**Example Formulas:**

- Retrieve the number of upgrades made over the last 13 weeks to estimates for IBM's earnings per share (EPS) for the current annual fiscal period by contributors to the Bloomberg Consensus estimate:  
`=BQL("IBM US Equity", "contributor_revisions(is_eps(adj=Y, fpo='1', fpt=A), NETUP, 13W)")`  
  
`=BQL.Query("get(contributor_revisions(is_eps(adj=Y, fpo='1', fpt=A), NETUP, '13W')) for(['IBM US Equity'])")`
- Retrieve the number of contributors to the Bloomberg Consensus estimate who added coverage of IBM's current quarter EPS over the last ten weeks:  
`=BQL("IBM US Equity", "contributor_revisions(is_eps(adj=Y, fpt=Q, fpo='1'), NUMADD, '10W')")`  
  
`=BQL.Query("get(contributor_revisions(is_eps(adj=Y, fpt=Q, fpo='1'), NUMADD, '10W')) for(['IBM US Equity'])")`
- Retrieve the number of upgrade events in the last eight days for the current quarter's estimates for Toyota's net income, based on all contributors to the Bloomberg Consensus estimate:  
`=BQL("TOYOF US Equity", "contributor_revisions(net_income(adj=Y, fpt=Q, fpo='1'), NUMUP, '8D')")`  
  
`=BQL.Query("get(contributor_revisions(net_income(adj=Y, fpt=Q, fpo='1'), NUMUP, '8D')) for(['TOYOF US Equity'])")`

**contributor\_stats()**

In BQL() and BQL.Query() formulas in Excel, *contributor\_stats()* allows you to specify a company financials data item and return a statistic based on contributions to the Bloomberg Consensus estimate for that data item. You can retrieve the average, median, maximum, or minimum estimate, or the standard deviation between estimates.

**Signature:** `contributor_stats(data_item, stat_type)`

where:

- data\_item* is a company financials data item (e.g., *is\_eps* or *net\_income*) and its named parameters. Note that data items calculated through BQL expressions (e.g., *pe\_ratio*) are not supported at this time.
- stat\_type* is the consensus statistic you want to retrieve.

**Note:** The data item must include the parameter Period Type (*fa\_period\_type* or *fpt*) with the parameter value A (Annual), S (Semi-Annual), or Q (Quarterly). For example, "*is\_eps(fpt=Q)*".

**Valid Parameter Values for *stat\_type*:**

| Value  | Description                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------|
| AVG    | <p>Average (<b>Default</b>)</p> <p>Retrieves the average of the values from contributors included in the consensus estimate.</p>  |
| MEDIAN | <p>Median</p> <p>Retrieves the median of the values from contributors included in the consensus estimate.</p>                     |
| MAX    | <p>Maximum</p> <p>Retrieves the highest value among contributors included in the consensus estimate.</p>                          |
| MIN    | <p>Minimum</p> <p>Retrieves the lowest value among contributors included in the consensus estimate.</p>                           |
| STD    | <p>Standard Deviation</p> <p>Retrieves the standard deviation of values from contributors included in the consensus estimate.</p> |

**Example Formulas:**

- Retrieve the average of operating income estimates for IBM for the current quarter from all contributors to the Bloomberg Consensus estimate:  
`=BQL("IBM US Equity", "contributor_stats(is_oper_inc(adj=Y, fpo='1', fpt=Q), AVG)")`  
`=BQL.Query("get(contributor_stats(is_oper_inc(adj=Y, fpo='1', fpt=Q), AVG)) for(['IBM US Equity'])")`
- Retrieve the median estimate for earnings per share for the current annual period, based on all contributors to the Bloomberg Consensus estimate:  
`=BQL("IBM US Equity", "contributor_stats(adj=Y, is_eps(fpo='1', fpt=A), MEDIAN)")`  
`=BQL.Query("get(contributor_stats(is_eps(adj=Y, fpo='1', fpt=A), MEDIAN)) for(['IBM US Equity'])")`

- Retrieve the standard deviation in net income estimates for IBM for the current quarter, based on contributors to the Bloomberg Consensus estimate:  
=BQL("IBM US Equity", "contributor\_stats(net\_income(adj=Y, fpo='1', fpt=Q), STD)")  
  
=BQL.Query("get(contributor\_stats(net\_income(adj=Y, fpo='1', fpt=Q), STD)) for ([IBM US Equity'])")

Company Financials Segments

You can use BQL to retrieve actuals and estimates across individual segments, such as geographic locations, business units, end users, or products.

segments() Universe Function

The segments() universe function allows you to discover the segments of a company and retrieve details about those segments.

**Signature:** segments(*symbols*, *type*, *dates*, [*hierarchy*], [*level*], [*lookback*])

Parameters for segments():

| Parameter | Description and Valid Values                                                                                                                                                                                      |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| symbols   | Equity ticker <b>(Required)</b><br><br>Allows you to specify the equity ticker of the company for which you want to retrieve segments.                                                                            |
| type      | Segments universe type<br><br>Adjusts the type of segments returned. Available types are:<br><br>REPORTED <b>(Default)</b> : Returns the company-reported segments subject to the defaults of the sub-parameters. |

| Parameter | Description and Valid Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| hierarchy | <p data-bbox="458 253 782 282">Segment hierarchy breakdown</p> <p data-bbox="458 359 1264 417">Allows you to specify the desired hierarchical segment breakdown. Available hierarchies are:</p> <p data-bbox="458 494 1379 585"><b>PRODUCT (Default):</b> Returns the reported segment breakdown that best represents the company's business lines determined by Bloomberg. Most often it is a segment breakdown that represents products (goods/services) that the company provides.</p> <p data-bbox="458 662 1353 784">However, instead of reporting segments that represent products (goods/services), a company may report a segment breakdown that represents operating segments, end markets, customers, distribution channels, etc. Bloomberg decides which segments best represent the company's business based on the disclosure available.</p> <p data-bbox="458 861 1353 919"><b>Note:</b> Bloomberg prioritizes the availability of reported and estimate revenue metrics during the segment breakdown determination process.</p> <p data-bbox="458 996 1287 1054"><b>LOCATION:</b> Returns the reported segment breakdown that best represents the locations where the company sells products (goods/services).</p> <p data-bbox="458 1132 1384 1222"><b>REPORTABLE:</b> Returns the segments that are part of the reportable hierarchy (i.e., the hierarchy for which HIERARCHY_REPORTABLE=Y), including the hierarchical structure of the segments in the reportable hierarchy.</p> <p data-bbox="458 1300 1307 1329"><b>H1, H2, H3,...HN:</b> Returns the reported segment breakdown via the hierarchy ID.</p> <p data-bbox="458 1406 1353 1528"><b>ALL:</b> Returns the list of segments, including the company-reported hierarchy or hierarchies, of which they are a part. A segment can appear more than once with this type selected, since a segment can be part of one or more company-reported hierarchies.</p> |

| Parameter       | Description and Valid Values                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>level</b>    | <p>Level of segments in the hierarchy</p> <p>Allows you to adjust the level of segments in the hierarchy segment breakdown. Available values are:</p> <p><b>ALL (Default):</b> Returns all segment levels available in the hierarchy segment breakdown.</p> <p><b>1, 2, 3, 4,..N:</b> Returns the specific segment level selected or an array of segment-level segments in the hierarchical breakdown. The value specified must be a positive integer.</p> |
| <b>lookback</b> | <p>Lookback to retrieve defunct segments</p> <p>Allows you to specify a lookback period for defunct segments and retrieve those segments. The LOOKBACK parameter is optional and the default value, if not specified, is 0D (i.e., 0 days). You can specify any positive integer with days (D), weeks (W), months (M), or years (Y).</p>                                                                                                                   |
| <b>dates</b>    | <p>Observation date</p> <p>Specifies the date of the observation for point-in-time retrieval of the available segments as of a given date. The DATES parameter is optional and the default value, if not specified, is today's date.</p>                                                                                                                                                                                                                   |

**Example Formulas:**

- Retrieve the primary segment breakdown for Apple:  
`=BQL("segments('AAPL US Equity')", " segment_name,sales_rev_turn")`  
`=BQL.Query("get(segment_name,sales_rev_turn) for(segments('AAPL US Equity'))")`
- Retrieve the primary segment breakdown for Apple, by just the first level of segments:  
`=BQL("segments('AAPL US Equity', level=1)", " segment_name,sales_rev_turn")`  
`=BQL.Query("get(segment_name,sales_rev_turn) for(segments('AAPL US Equity', level=1))")`
- Retrieve the primary location segment breakdown for Apple:  
`=BQL("segments('AAPL US Equity', hierarchy=location)", " segment_name,sales_rev_turn")`  
`=BQL.Query("get(segment_name,sales_rev_turn) for(segments('AAPL US Equity', hierarchy=location))")`
- IBM re-segmented recently. Retrieve its defunct segments for use in a model:

```
=BQL("segments('IBM US Equity', hierarchy=product, level=1, lookback=5y)", "segment_name,sales_rev_turn")
```

```
=BQL.Query("get(segment_name,sales_rev_turn) for(segments('IBM US Equity',hierarchy=product,level=1,lookback=5y)))")
```

- Retrieve the reportable (as defined by the company) segment breakdown for Apple:

```
=BQL("segments('AAPL US Equity', hierarchy=reportable)", " segment_name,sales_rev_turn")
```

```
=BQL.Query("get(segment_name,sales_rev_turn) for(segments('AAPL US Equity', hierarchy=reportable)))")
```

## Segment IDs

You can use BQL to retrieve financial actuals and estimates for specific segments of a company, such as regions, business units, end users, or products.

**Note:** Bloomberg issues an immutable identifier to segments reported by companies, called the *Segment ID*. The identifier ensures the stability of the segment data retrieval to BQL users in case of a segment name change. New segment identifiers are issued if the company reports a segment definition change.

To get data for a segment, you write a query using its segment ID. Segment IDs consist of the three characters "SEG" plus a numeric code, appended by the word "Segment". Segment IDs are unique for each company.

Segment IDs work the same way as equity IDs in BQL For example:

- To get 2018 annual revenue data for Apple:

```
=BQL.Query("get(sales_rev_turn(fpr=2018, fpt=A) for(['AAPL US Equity'])))")
```

```
=BQL("AAPL US Equity", "sales_rev_turn(fpr=2018, fpt=A)")
```

- To get 2018 annual revenue data for iPhone:

```
=BQL.Query("get(sales_rev_turn(fpr=2018, fpt=A) for(['SEG0000155493 Segment'])))")
```

```
=BQL("SEG0000155493 Segment", "sales_rev_turn(fpr=2018, fpt=A)")
```

**Note:** You can easily export BQL formulas for segment data from the Bloomberg Terminal® to Excel from MODL. For more: [MODL Help Page > Exporting Data](#).

## Segment Names and Other Details

To return the name of a company segment, you can use the data item **segment\_name()**.

For example, to get the segment name for SEG0000155493 Segment (Apple's iPhone):

- =BQL.Query("get(segment\_name()) for(['SEG0000155493 Segment'])))")

- =BQL("SEG0000155493 Segment", "segment\_name()")

**segment\_name()** returns multiple columns with useful information about the segment:

| Column Name           | Description                                                                                                                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID                    | The unique identifier of the segment.                                                                                                                                                                                                                                                                            |
| segment_reportable    | Indicates whether the segment is reportable by the company. "Reportable" usually means that the segment represents more than 10% of the total revenue, profits, or assets of the company and is specifically disclosed by the company as being a reportable segment (or equivalent). Possible values are Y or N. |
| fundamental_ticker    | The ticker of company that holds the segment.                                                                                                                                                                                                                                                                    |
| segment_revision_date | The date the segment was last revised. For example, the segment name or ID could change over time.                                                                                                                                                                                                               |
| as_of_date            | The observation date of the segment data.                                                                                                                                                                                                                                                                        |
| name                  | A short description of the segment.                                                                                                                                                                                                                                                                              |
| segment_active        | Indicates whether the segment was active or inactive (defunct) on the as_of_date. Possible values are Y (active) or N (inactive).                                                                                                                                                                                |
| hierarchy_id          | The ID of the hierarchy the segment exists in.                                                                                                                                                                                                                                                                   |
| hierarchy_name        | The name of the hierarchy the segment exists in.                                                                                                                                                                                                                                                                 |
| level                 | A non-negative number that indicates the level of the segment in the hierarchy.                                                                                                                                                                                                                                  |
| parent_ID             | The ID of the parent segment of the requested segment. The value is null/NA if the level is 0.                                                                                                                                                                                                                   |
| parent_name           | The name of the parent segment of the requested segment. The value is null/NA if the level is 0.                                                                                                                                                                                                                 |
| order                 | The sort order of the segment in the hierarchy structure.                                                                                                                                                                                                                                                        |

## Equity Universe Functions

The following topics provide information on universe functions for equity data sets.

### peers()

In BQL() and BQL.Query() formulas in Excel, peers() allows you to specify a company and return securities with similar fundamentals and performance markers that you, your colleagues, or Bloomberg define.

**Signature:** peers(symbols, [type], [peer\_source])

where:

- *symbols* is the equity ticker (e.g., 'MSFT US Equity').
- *type* is the optional parameter that specifies the peer group you want to return (e.g., CUSTOM).
- *peer\_source* is the optional parameter that specifies the name of a custom peer group, expressed as a string, that you previously saved in the *Comparables Editor* (RVCE) function.

**Valid Parameter Values for type:**

| Value              | Description                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CUSTOM             | <p>Custom Peer Group (<b>Default</b>)</p> <p>A peer group created by you or received from a colleague in a shared community. peers() will default to the last custom peer group you selected in a relative analysis function such as <i>Relative Valuation</i> (RV) or <i>Equity Relative Valuation</i> (EQRV). For more on creating custom peer groups: <a href="#">EQRV Help Page &gt; Configuring Comps</a>.</p> |
| BLOOMBERG_BEST_FIT | <p>Bloomberg Best Fit</p> <p>A proprietary peer group curated by a Bloomberg analyst (BI) or a Bloomberg algorithm (BICS Best Fit) data source.</p>                                                                                                                                                                                                                                                                 |

**Example Formulas:**

- Retrieve Microsoft's peers as selected by Bloomberg:  
 =BQL("peers('MSFT US Equity', type=BLOOMBERG\_BEST\_FIT)", "id()")  
  
 =BQL.Query("get(id()) for(peers('MSFT US Equity', type=BLOOMBERG\_BEST\_FIT))")
- Retrieve IBM's peers based on your last custom peer group selection on RV <GO> or EQRV <GO>:  
 =BQL("peers('IBM US Equity')", "id()")  
  
 =BQL.Query("get(id()) for(peers('MSFT US Equity'))")

- Retrieve IBM's peers based on a peer group you saved in RVCE <GO> as *My Comp Group*:

```
=BQL("peers('IBM US Equity', peer_source='My Comp Group')", "id()")
```

```
=BQL.Query("get(id()) for(peers('IBM US Equity', peer_source='My Comp Group'))")
```

For more:  [Equity Peer Groups in BQL](#).

## BQL for Institutions/Funds

BQL is an efficient and flexible method for retrieving a wide range of fund data from the Bloomberg database, including screening, risk return metrics, and fund flow data.

## Institutions/Funds Docs and Media

The following tables contain spreadsheets, tutorials, and videos created by Bloomberg to assist you with performing data analysis for institutions and funds using BQL.

### Institutions/Funds Reference

| File Type                                                                           | Name                               | Description                                                                                                                                          |
|-------------------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">Fund Data Items</a>    | A reference guide of data items and data item values you can use for analyzing fund data and screening for institutions and funds with BQL in Excel. |
|  | <a href="#">Function Reference</a> | A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL.                                         |

### Institutions/Funds Tutorials and Workflows

| File Type                                                                           | Name                                                                     | Description                                                                                                                                                                    |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Holdings Based Fixed Income and Equities Style Analytics</a> | A spreadsheet tutorial that shows how to analyze the composition, style characteristics, and allocation exposures of equity-oriented funds based on their underlying holdings. |
|  | <a href="#">Bloomberg Liquidity Assessment (LOA) in BQL</a>              | A spreadsheet tutorial that shows how to evaluate liquidity with the Bloomberg Liquidity Assessment (LOA) model using BQL.                                                     |

| File Type                                                                         | Name                                                                                                          | Description                                                                                                                                            |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">BQL for Funds Tutorial</a>                                                                        | A spreadsheet tutorial that shows how to download funds data to Excel using BQL.                                                                       |
|  | <a href="#">BQL Risk-Return Metrics</a>                                                                       | A spreadsheet tutorial that shows how to retrieve customized risk–return metrics for funds using BQL.                                                  |
|  | <a href="#">Institution and Fund Holdings</a><br><br><a href="#">Institution and Fund Holdings (Japanese)</a> | A spreadsheet tutorial that shows how to use BQL to download institutions' 13F filings and fund holdings data in Excel and addresses common use cases. |
|  | <a href="#">Tracking Changes in ETF Flows</a>                                                                 | A customizable spreadsheet that lets you monitor changes in ETF flows between asset classes and fund classifications using BQL.                        |

### Institutions/Funds Videos

| File Type                                                                           | Name                                                                                      | Description                                                                                                                                                  |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <a href="#">Analyze Fund Flows, AUM, Peers, and Risk/Return Metrics using BQL (42:59)</a> | A video introduction to BQL, showcasing new functionality released for funds in Excel that allows for custom analysis in seconds.                            |
|  | <a href="#">Getting Started: Institution and Fund Holdings (4:26)</a>                     | A short video that shows you how you use BQL and the BQL Builder to easily download fund holdings and weights, as well as aggregate fund holdings by sector. |

### Institutions/Funds Universe Functions

The following topics provide information on universe functions for institutions/funds data sets.

#### holdings()

In BQL() and BQL.Query() formulas in Excel, *holdings()* allows you to see the positions of ETFs, mutual funds, and institutions with 13F filings for the most recent quarter or a single point in time.

**Note:** The *holdings()* universe function replaces the *type=holdings* parameter value formerly in the *members()* universe function.

**Signature:** holdings(symbols, [dates], [filingsource])

where:

- *symbols* is the fund or institution's ticker (e.g., 'ARKK US Equity'). *holdings()* supports one ticker at a time.
- *dates* is the optional parameter that specifies the date in the 13F filing period (e.g., 2020-01-01). *holdings()* supports a single date at a time.
- *filingsource* is the optional parameter that specifies the source of the filings data for an institution (e.g., "13F").

**Valid Parameter Values for filingsource:**

| Value | Description                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------|
| ALL   | All<br><br>Reporting disclosures as collected by Bloomberg.                                           |
| 13F   | 13F<br><br>Quarterly disclosures made by institutional investment managers as required by the US SEC. |

**Example Formulas:**

- Retrieve the securities held by Tiger Global Management as of the most recent 13F filing quarter:  
 =BQL("holdings('809339Z US Equity',"id()")  
  
 =BQL.Query("get(id()) for(holdings('809339Z US Equity'))")
- Retrieve the securities held by Tiger Global Management as of December 31, 2020:  
 =BQL("holdings('809339Z US Equity', dates=2020-12-31)", "id()")  
  
 =BQL.Query("get(id()) for(holdings('809339Z US Equity', dates=2020-12-31))")
- Retrieve the securities held by Tiger Global Management as reported in the company's 13F filings for the latest filing period:  
 =BQL("holdings('809339Z US Equity',"id()", "filingsource=13F")  
  
 =BQL.Query("get(id()) for(holdings('809339Z US Equity', filingsource=13F))")

## BQL for Fixed Income

BQL is an efficient and flexible method for retrieving a wide range of fixed income data from the Bloomberg database, including data on bonds, loans, mortgages, munis, and fixed income indices.

## Fixed Income Docs and Media

The following tables contain spreadsheets, tutorials, and videos created by Bloomberg to assist you with performing data analysis for fixed income using BQL.

### Fixed Income Reference

| File Type                                                                         | Name                               | Description                                                                                                  |
|-----------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Function Reference</a> | A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL. |

### Fixed Income Workflows

| File Type                                                                           | Name                                       | Description                                                                            |
|-------------------------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------|
|    | <a href="#">BQL for Primary Markets</a>    | A spreadsheet tutorial that shows how to leverage BQL for primary market analysis.     |
|    | <a href="#">BQL for Relative Value</a>     | A spreadsheet tutorial that demonstrates BQL capabilities for relative value analysis. |
|  | <a href="#">BQL for Liquidity Analysis</a> | A spreadsheet tutorial that shows how to use BQL to analyze liquidity.                 |

### Fixed Income Tutorials

| File Type                                                                           | Name                                                     | Description                                                                                                                              |
|-------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Collateral Data for Securitized Products</a> | A spreadsheet tutorial that shows how to retrieve and analyze collateral data for CLOs using BQL.                                        |
|  | <a href="#">SDR DTCC Trades</a>                          | A spreadsheet tutorial that shows how to retrieve and analyze credit and rates swap trade data from the DTCC Swap Data Repository (SDR). |
|  | <a href="#">Curves in BQL</a>                            | A spreadsheet tutorial that shows how to analyze sovereign, credit, and custom curves using BQL.                                         |

| File Type                                                                           | Name                                                                            | Description                                                                                                                                                            |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#"><u>Holdings Based Fixed Income and Equities Style Analytics</u></a> | A spreadsheet tutorial that shows how to analyze fund-level fixed income characteristics derived from underlying portfolio holdings and aligned with PORT methodology. |
|    | <a href="#"><u>BQL for Private Credit</u></a>                                   | A spreadsheet tutorial that shows you how to analyze the private credit market using BQL.                                                                              |
|    | <a href="#"><u>Cross-Currency Analytics in BQL</u></a>                          | A spreadsheet tutorial that shows you how to analyze historical cross-currency spreads and yields using BQL.                                                           |
|    | <a href="#"><u>BQL for Securitized Products</u></a>                             | A spreadsheet tutorial that shows you how to use BQL to analyze securitized products such as mortgages.                                                                |
|    | <a href="#"><u>Securitized Products Analytics</u></a>                           | A spreadsheet tutorial that shows how to perform custom analyses of fixed income securitized products in BQL.                                                          |
|  | <a href="#"><u>Securitized Products: Trades and Spreads</u></a>                 | A spreadsheet tutorial that shows how to perform relative value analysis on securitized products using trades and spreads data in BQL.                                 |
|  | <a href="#"><u>BQL for Fixed Income Indices</u></a>                             | A spreadsheet tutorial that shows you how to perform a customized analysis of fixed income indices using BQL.                                                          |
|  | <a href="#"><u>BQL for Bonds and Loans</u></a>                                  | A spreadsheet tutorial that shows you how to retrieve and analyze fixed income data for bonds and loans using BQL.                                                     |
|  | <a href="#"><u>Bloomberg Intelligence (BI) Credit Drivers in BQL</u></a>        | A spreadsheet tutorial that shows how to analyze Bloomberg Intelligence Credit Drivers in BQL.                                                                         |
|  | <a href="#"><u>Market Implied Probability of Default (MIPD) in BQL</u></a>      | A spreadsheet tutorial that shows how to analyze MIPD metrics in BQL.                                                                                                  |

| File Type                                                                           | Name                                                        | Description                                                                                                                                                           |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">Bloomberg Liquidity Assessment (LOA) in BQL</a> | A spreadsheet tutorial that shows how to evaluate liquidity with the Bloomberg Liquidity Assessment (LOA) model using BQL.                                            |
|    | <a href="#">BQL for CDS</a>                                 | A spreadsheet tutorial that shows you how to use BQL to perform credit default swap (CDS) analysis, including CDS spread and sector credit research.                  |
|    | <a href="#">Credit Ratings in BQL</a>                       | A spreadsheet tutorial that shows how to analyze credit ratings using BQL.                                                                                            |
|    | <a href="#">BQL for Issuance Pricing Analysis</a>           | A spreadsheet tutorial that shows you how to uncover, analyze, and aggregate issuance data for bonds using BQL.                                                       |
|    | <a href="#">YAS Custom Analytics in BQL</a>                 | A spreadsheet tutorial that shows how to perform custom bond calculations in BQL that are equivalent to those in the <i>Yield and Spread Analysis</i> (YAS) function. |
|  | <a href="#">BQL for Axes Data</a>                           | A spreadsheet tutorial that shows you how to use BQL to download and analyze the axes you receive from your dealers on fixed income securities.                       |
|  | <a href="#">BQL for Fixed Income Saved Searches</a>         | A spreadsheet tutorial that demonstrates how to use security lists saved on SRCH <GO> as the basis for security universes in BQL queries.                             |
|  | <a href="#">BQL for Corporate Structure Navigation</a>      | A spreadsheet tutorial that shows you how to navigate the current and historical corporate structure of a debt security using BQL.                                    |
|  | <a href="#">BQL for Green, Social, and Sustainable Debt</a> | A spreadsheet tutorial that shows you how to use BQL to analyze sustainable debt.                                                                                     |
|  | <a href="#">BQL for Bond Total Return Analysis</a>          | A spreadsheet tutorial detailing bond total return analysis in BQL.                                                                                                   |

| File Type                                                                         | Name                                       | Description                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">BQL for Municipal Bonds</a>    | A spreadsheet tutorial and list of the data items you can use to analyze fixed income municipal securities with BQL, including breakdowns for municipals(), municipalsUniv(), and municipal bond indices. |
|  | <a href="#">Municipal Credit Tearsheet</a> | An example spreadsheet that constructs a detailed historical financial analysis for a selected municipal credit issuer.                                                                                   |

### Fixed Income Videos

| File Type                                                                           | Name                                                                               | Description                                                                                                                                                                        |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">BQL for Fixed Income (Analysts &amp; PMs) (15:38)</a>                  | A video introduction to BQL and how it can be applied by analysts and portfolio managers on the buy side to build customized bond and loan screening and index analytics in Excel. |
|    | <a href="#">BQL for Fixed Income (Sell Side) (19:27)</a>                           | A video introduction to BQL and how it can be applied by market players on the sell side to build customized bond and loan screening and index analytics in Excel.                 |
|  | <a href="#">Mortgage, Preferred, and Muni Bond Screening and Analytics (22:05)</a> | A video introduction to BQL and how it can be applied to build customized screening and analytical queries for mortgages, preferreds, and municipals in Excel.                     |
|  | <a href="#">Screening for Default Risk with the Spreadsheet Builder (2:46)</a>     | A short video that shows you how to use the Spreadsheet Builder in Excel to identify companies with the highest probability of default.                                            |

### Fixed Income Universe Functions

The following topics provide information on universe functions for fixed income data sets.

#### Historical FI Screening

In BQL() and BQL.Query() formulas in Excel, you can find active fixed income securities as of a historical date by using the parameter *dates* with the following fixed income universe functions: *bonds()*, *loans()*, *preferreds()*, *bondsUniv()*, and *loansUniv()*.

Notes:

- Currently, only a single date is supported.
- You should apply the parameter *dates* to the filtering criteria as well as the universe.
- When using the *with()* clause to apply *dates* to both the filtering criteria and universe, you should ensure all data items you are using as filters are eligible for historical screening. You can check eligibility on the *Fixed Income Search* (SRCH) function. On SRCH, select the criteria that matches the data items you want to use and change the as of date. If one or more data items are not eligible, an error appears when you run the search.

#### Example Formulas:

- Return the total amount of floating rate debt pegged to LIBOR vs. SOFR in 2019:  
 =BQL("filter(filter(bondsUniv(ACTIVE), crncy()=='USD' and floater()=='Y'), reset\_index()=='US0003M Index' or reset\_index()=='SOFRRATE Index'), "sum(group(amt\_outstanding()/1M, by=reset\_index()))", "dates=2019-12-31")  
  
 =BQL.Query("get(sum(group(amt\_outstanding()/1M, by=reset\_index())) for(filter(filter(bondsUniv(ACTIVE), crncy()=='USD' and floater()=='Y'), reset\_index()=='US0003M Index' or reset\_index()=='SOFRRATE Index')) with(dates=2019-12-31)")
- Count the number of Russian corporate IG bonds that were active as of December 2021:  
 =BQL("filter(bondsUniv(ACTIVE), rtg\_sp() in ['AAA', 'AA+', 'AA', 'AA-', 'A+', 'A', 'A-', 'BBB+', 'BBB', 'BBB-'] and classification\_name(classification\_scheme=BCLASS)=='Corporate' and cntry\_of\_risk()=='RU'), "count(group(id()))", "dates=2021-12-01")  
  
 =BQL.Query("get(count(group(id())) for(filter(bondsUniv(ACTIVE), rtg\_sp() in ['AAA', 'AA+', 'AA', 'AA-', 'A+', 'A', 'A-', 'BBB+', 'BBB', 'BBB-'] and classification\_name(classification\_scheme=BCLASS)=='Corporate' and cntry\_of\_risk()=='RU')) with(dates=2021-12-01)")

#### cds()

In BQL() and BQL.Query() formulas in Excel, *cds()* returns an ID that represents a credit default swap (CDS) for a company you specify. You can use a debt or equity security to specify the company.

**Signature:** *cds(symbols, [type])*

where:

- *symbols* is an ID representing a debt or equity security (e.g., 'MSFT US Equity').
- *tenor* is the optional parameter that specifies the tenor of the CDS you want (e.g., 4Y).

**Valid Parameter Values for tenor:**

| Value                                           | Description                                                                                                                                                                                                                                                            |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6M, 1Y, 2Y, 3Y, 4Y, 5Y<br>(Default), 7Y, or 10Y | <p>Tenor</p> <p>The length of the CDS contract at its inception. For example, a 1Y tenor represents a swap that is tradeable for 1 year.</p> <p><b>Note:</b> When the universe function is used without a parameter argument, <i>cds()</i> returns the 5-year CDS.</p> |

**Example Formulas:**

- Retrieve the ID that represents the 5-year CDS for Daimler AG:

```
=BQL("cds('DAI GR Equity')", "id()")
```

```
=BQL.Query("get(id()) for(cds('DAI GR Equity'))")
```

- Retrieve the ID that represents the 3-year CDS for Daimler AG:

```
=BQL("cds('DAI GR Equity', tenor=3Y)", "id()")
```

```
=BQL.Query("get(id()) for(cds('DAI GR Equity', tenor=3Y))")
```

For more:  [BQL for CDS](#).

**equityPricingTicker()**

In BQL() and BQL.Query() formulas in Excel, *equityPricingTicker()* lets you specify a debt security to retrieve the equity ticker related to the issuer that will return pricing data. The universe function searches the debt security's current or historical corporate structure to identify the closest entity with available pricing data.

**Signature:** equityPricingTicker(*symbols*, [*dates*])

where:

- symbols* is a debt security identifier (e.g., "AR868839 Corp"). *equityPricingTicker()* supports one ticker at a time.
- dates* is an optional parameter that specifies the date or a range of dates in which to search a debt security's corporate structure (e.g., "2020-01-01" or "2020-01-01, 2021-01-01").

**Example Formula:**

Return the company with pricing data that, as of October 14, 2018, is closest to the issuer of the bond AR868839 Corp:

```
=BQL("equityPricingTicker(['AR868839 Corp'], dates=2018-10-14)", "id()")
```

```
=BQL.Query("get(id()) for(equityPricingTicker(['AR868839 Corp'], dates=2018-10-14))")
```

For more:  [Corporate Structure Navigation in BQL](#).

## fundamentalTicker()

In BQL() and BQL.Query() formulas in Excel, *fundamentalTicker()* lets you specify a debt security to retrieve the equity ticker related to the issuer that will return company financials data (e.g., revenue and EPS). The universe function searches the debt security's current or historical corporate structure to identify the closest entity with available company fundamentals.

**Signature:** fundamentalTicker(*symbols*, [*dates*])

where:

- *symbols* is a debt security identifier (e.g., "AR868839 Corp"). *fundamentalTicker()* supports one ticker at a time.
- *dates* is an optional parameter that specifies the date or a range of dates in which to search a debt security's corporate structure (e.g., "2020-01-01" or "2020-01-01, 2021-01-01").

### Example Formula:

Return the company with reported fundamentals that, as of October 14, 2019, is closest to the issuer of the bond AR868839 Corp:

```
=BQL("fundamentalTicker(['AR868839 Corp'], dates=2019-10-14)", "id()")
```

```
=BQL.Query("get(id()) for(fundamentalTicker(['AR868839 Corp'], dates=2019-10-14))")
```

For more:  [Corporate Structure Navigation in BQL](#).

## screenResults()

In BQL() and BQL.Query() formulas in Excel, *screenResults()* retrieves a list of securities saved on the *Fixed Income Search* (SRCH) function to establish a security universe in a BQL query.

You can retrieve custom or Bloomberg-generated security lists from SRCH <GO> and its aliases, including MTGS <GO>, PSCH <GO>, and MSRC <GO>, as well as LEAG <GO>.

**Note:** You can also copy a BQL formula directly from SRCH and paste it into your spreadsheet. For more: [Copying BQL from SRCH](#).

**Signature:** screenResults(*type*, *screen\_name*)

where:

- *type* is the source of a saved search and requires the value "SRCH".
- *screen\_name* is the name of a list of securities saved on SRCH (e.g., "@COCO"). The names of saved lists/searches are case-sensitive in BQL.

### Example Formula:

Calculate the weighted average Z-spread by sector (BICS, Level 1) for Green Bonds included in the @GREEN example search from SRCH <GO>:

```
=BQL("screenResults(SRCH, '@GREEN')", "wAvg(group(spread(), by=classification_name()),  
group(amt_outstanding(currency=USD), by=classification_name())) as #wAvg_Spread")
```

```
=BQL.Query("get(wAvg(group(spread(), by=classification_name()), group(amt_outstanding(currency=USD),  
by=classification_name())) as #wAvg_Spread) for(screenResults(SRCH, '@GREEN'))")
```

For more:  [BQL for Fixed Income Saved Searches](#).

## Copying BQL from SRCH

You can build a saved search in the *Fixed Income Search* (SRCH) function, copy it as a BQL formula, then paste it into your spreadsheet to quickly load the results in Excel without having to construct the query yourself. You can then modify the query with data items, filters, and more to create your own custom analysis.

For more: [SRCH Help Page > Copying BQL Formulas](#).

The query uses the universe function *screenResults()* to pull the search results from SRCH into your spreadsheet.

For more: [screenResults\(\)](#).

## axedUniv()

In BQL() and BQL.Query() formulas in Excel, *axedUniv()* retrieves all current axes you have received from your dealers on one or both market sides. You can use *axedUniv()* as the basis for a security universe in your query.

**Note:** *axedUniv()* lets you analyze axes based on any fixed income data available in BQL. To return axe-specific data, you can use the data items [is\\_axed\(\)](#) and [axes\(\)](#).

**Signature:** *axedUniv(type)*

where:

*type* is the market side for the axes (BID, ASK, or ALL).

### Example Formulas:

- Return a list of all EUR denominated bonds axed on the ask side:  

```
=BQL("filter(axedUniv(ASK), crncy()==EUR)", "name()")
```

```
=BQL.Query("get(name()) for(filter(axedUniv(ASK), crncy()==EUR))")
```
- Get axes data aggregated by security on all investment grade bonds that are currently axed:  

```
=BQL("filter(axedUniv(ALL), rtg_sp()>='BBB-' or rtg_moody()>='Baa3' or rtg_fitc()>='BBB-')", "axes()", "showAllCols=TRUE")
```

```
=BQL.Query("get(axes()) for(filter(axedUniv(ALL), rtg_sp()>='BBB-' or rtg_moody()>='Baa3' or rtg_fitc()>='BBB-'))",  
"showAllCols=TRUE")
```

For more:  [BQL for Axes Data.](#)

## Fixed Income Data Items

### is\_axed()

In BQL() and BQL.Query() formulas in Excel, *is\_axed()* returns a Boolean that indicates whether a security is currently axed (TRUE) or not (FALSE). You can use *is\_asked()* to display the current axe status of a list of securities, filter a security universe by each security's axe status, and more.

**Note:** To retrieve all the current axes you received from dealers as the basis for a security universe in your query, you can use the universe function *axedUniv()*. You can use the data item *axes()* to get current axe information such as side, size, price, yield, spread, depth, and dealer on any set of securities.

**Signature:** *is\_axed(side)*

where:

*side* is an optional parameter that specifies the market side for the axes (BID, ASK, or ALL). By default, *is\_axed()* checks for axes on both the bid and ask side.

### Example Formulas:

- Identify any axed bonds in the Bloomberg Pan-European High Yield Index (LP01TREU Index):  
 =BQL("members(['LP01TREU Index']), "is\_axed()")  
  
 =BQL.Query("get(is\_axed()) for(members(['LP01TREU Index']))")
- Identify axed bonds on the bid side in the Bloomberg Pan-European High Yield Index (LP01TREU Index):  
 =BQL("members(['LP01TREU Index']), "is\_axed(side=BID)")  
  
 =BQL.Query("get(is\_axed(side=BID)) for(members(['LP01TREU Index']))")
- Return axes data aggregated by security for the ask axed bonds in the example search from SRCH <GO> named "@GREEN":  
 =BQL("filter(screenResults(SRCH, '@GREEN'), is\_axed(side=ASK)==TRUE)", "axes(currency=USD)", "aggregateby=SECURITY", "showallcols=TRUE")  
  
 =BQL.Query("get(axes(currency=USD)) for(filter(screenResults(SRCH, '@GREEN'), is\_axed(side=ASK)==TRUE))", "showallcols=TRUE")

For more:  [BQL for Axes Data.](#)

### axes()

In BQL() and BQL.Query() formulas in Excel, *axes()* returns aggregated axe data on one or more fixed income securities. The data includes side, size, price, yield, spread, depth, dealer, and more.

You can aggregate the axe data by security, or by both security and dealer.

**Note:** To retrieve all the current axes you received from dealers as the basis for a security universe in your query, you can use the universe function `axedUniv()`. You can use the data item `is_axed()` on any set of securities to determine whether each security is axed or not.

**Signature:** `axes(aggregateBy, currency)`

where:

- *aggregateBy* is an optional parameter that lets you choose how granularly you want to aggregate the axes data.
- *currency* is an optional parameter that converts the size of all axes into the same currency. Valid parameter values are any two-character ISO currency code. By default, each axe size is based on the security's currency of issuance.

**Note:** It is important to normalize the currency with the parameter *currency* when the securities you are analyzing were issued in different currencies.

**Valid Parameter Values for aggregateBy:**

| Value           | Description                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SECURITY        | <p>Aggregate by Security(<b>Default</b>)</p> <p>Returns the depth and total size of each axed security, on both the bid and ask sides.</p> <p>For additional data you can return, see the spreadsheet linked below.</p>                                                                                                                                       |
| SECURITY_DEALER | <p>Aggregate by Security, then by Dealer</p> <p>Provides a more granular view of the axes data. When you add the parameter <i>showAllCols=TRUE</i> to your formula, returns the price, dealer, size, spread, and yield for each axed security, on both the bid and ask side.</p> <p>For additional data you can return, see the spreadsheet linked below.</p> |

**Example Formulas:**

- Return axes data aggregated by security for the ask axed bonds in the example search from SRCH <GO> named "@GREEN":  
`=BQL("filter(screenResults(SRCH, '@GREEN'), is_axed(side=ASK)==TRUE)", "axes(currency=USD)", "aggregateby=SECURITY", "showallcols=TRUE")`  
  
`=BQL.Query("get(axes(currency=USD)) for(filter(screenResults(SRCH, '@GREEN'), is_axed(side=ASK)==TRUE))", "showallcols=TRUE")`
- Return axes data aggregated more granularly (by security, then by dealer) for the ask axed bonds in the example search from SRCH <GO> named "@GREEN":

```
=BQL("filter(screenResults(SRCH, '@GREEN'), is_axed(side=ASK)==TRUE)", "axes(currency=USD)",
"aggregateby=SECURITY_DEALER", "showallcols=TRUE")
```

```
=BQL.Query("get(axes(currency=USD, aggregateby=SECURITY_DEALER)) for(filter(screenResults(SRCH, '@GREEN'),
is_axed(side=ASK)==TRUE))", "showallcols=TRUE")
```

For more:  [BQL for Axes Data.](#)

## Sanctions Flags

A set of related BQL data items allows you to build BQL() and BQL.Query() formulas in Excel that identify fixed income securities subject to sanctions by the UK, EU, OFAC, and other entities.

These data items help you perform custom analysis on sanctioned debt, including screening for impacted securities, aggregating the sanctioned debt to assess its market impact, and tracking fund exposure to the sanctions over time. For example, you can determine how many Russian securities are subject to UK, EU, and OFAC and sectoral sanctions, as well as the total amount of debt subject to each type of sanction.

The data items are:

- european\_union\_sanction\_status()
- ofac\_sanctioned\_security()
- ofac\_sectoral\_sanc\_co()
- sectoral\_sanctioned\_security()
- un\_sanctioned\_security()
- dod\_sanctioned\_security()
- uk\_sanctioned\_security()

### Example Formula:

Screen for bonds subject to EU sanctions that were issued by entities whose country of risk is Russia and that have a coupon coming due in the next month:

```
=BQL("filter(bondsUniv(ACTIVE), cntry_of_risk()=='RU' and european_union_sanction_status()=='Y' and nxt_cpn_date()<=1M)",
"nxt_cpn_date()")
```

```
=BQL.Query("get(nxt_cpn_date()) for(filter(bondsUniv(ACTIVE), cntry_of_risk()=='RU' and
european_union_sanction_status()=='Y' and nxt_cpn_date()<=1M))")
```

For more:  [BQL for Fixed Income Sanctions.](#)

## BQL for Economics

BQL is an efficient and flexible method for retrieving a wide range of economic data from Bloomberg, including national accounts indicators, labor market statistics, and monetary sector data.

## Economics Docs and Media

The following tables contain spreadsheets, tutorials, and videos created by Bloomberg to assist you with performing data analysis for economics using BQL.

### Economics Reference

| File Type                                                                         | Name                                              | Description                                                                                                               |
|-----------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Product Manual: BQL for Economics</a> | A reference guide to the syntax, concepts, and parameters that allow you to retrieve and analyze economic data using BQL. |
|  | <a href="#">Function Reference</a>                | A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL.              |

### Economics Tutorials and Workflows

| File Type                                                                           | Name                                                             | Description                                                                                                                                                                             |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">BQL Tutorial for Economics</a>                       | A spreadsheet tutorial that shows how to download economic data to Excel using BQL and lists the concept fields, country tickers, other parameters, and data coverage available.        |
|  | <a href="#">BQL Workflow for Economics</a>                       | A workflow tool demonstrating how to access economics and country data using BQL, with detailed workflow examples of country dashboards, country scoring, and regional analytics.       |
|  | <a href="#">Countries Universe Function</a>                      | A spreadsheet tutorial that shows how to use the <i>countries()</i> universe function to retrieve data for the constituents of selected regions.                                        |
|  | <a href="#">Implied Overnight Rates (WIRP &lt;GO&gt;) in BQL</a> | A spreadsheet tutorial that shows how to use BQL to retrieve and analyze point-in-time market-implied policy rates found on the <i>World Interest Rate Probability</i> (WIRP) function. |

| File Type                                                                         | Name                        | Description                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Trade Flows</a> | A workflow tool demonstrating how to analyze global trade flows using IMF data via BQL, with detailed examples covering total trade, imports, exports, and trade balances including surpluses and deficits. |

### Calendar Tutorial and Templates

| File Type                                                                         | Name                                                                                                    | Description                                                                                                                                                        |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Eco Calendar BQL Tutorial</a>                                                               | A spreadsheet tutorial demonstrating how to download a calendar of economic releases to Excel using BQL, with examples demonstrating all the available parameters. |
|  | <a href="#">Eco Calendar BQL Templates</a><br><br><a href="#">Eco Calendar BQL Templates (Japanese)</a> | A spreadsheet consisting of ready-made economic calendar templates.                                                                                                |

### Economics Videos

| File Type                                                                           | Name                                                       | Description                                                                                                                                  |
|-------------------------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">BQL for Economics</a> (26:57)                  | A video tutorial that shows you how to analyze economic data with BQL.                                                                       |
|  | <a href="#">Quantifying the Economic Trade War</a> (19:13) | A video introduction to BQL for Economics that illustrates how you can create country risk scores and quantify exposure to trade war shocks. |
|  | <a href="#">Get Economic Release Calendars</a> (1:00)      | A video tutorial that shows you how to retrieve custom calendars of economic releases using BQL.                                             |

## BQL for FX

BQL is an efficient and flexible method for retrieving foreign exchange data from the Bloomberg database, such as FX forward curve pricing data.

### FX Docs and Media

The following table contains materials created by Bloomberg to assist you with analyzing FX data using BQL.

| File Type                                                                           | Name                                                             | Description                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">BQL FX Implied Volatility</a>                        | A spreadsheet tutorial including basic and advanced examples of retrieving FX implied volatility data, as well as a sample FX implied volatility dashboard driven by BQL.                      |
|    | <a href="#">FX Curves in BQL</a>                                 | A spreadsheet tutorial that shows how to retrieve FX curve data including outright, points, points per day, and implied yields.                                                                |
|    | <a href="#">Implied Overnight Rates (WIRP &lt;GO&gt;) in BQL</a> | A spreadsheet tutorial that shows how to use BQL to retrieve and analyze point-in-time market-implied policy rates found on the <i>World Interest Rate Probability</i> (WIRP) function.        |
|  | <a href="#">Currencies Universe Function</a>                     | A spreadsheet tutorial that shows how to use the <i>currencies()</i> universe function to retrieve and screen data for currencies based on specified criteria or Bloomberg's currency baskets. |

## BQL for Portfolios

BQL is an efficient and flexible method for retrieving a wide range of data about a portfolio, including portfolio members and positions/weights (depending on the portfolio type).

### Portfolios Docs and Media

The following tables contain spreadsheets, tutorials, and videos created by Bloomberg to assist you with performing data

analysis for portfolios using BQL.

### Portfolios Reference

| File Type                                                                         | Name                               | Description                                                                                                  |
|-----------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Function Reference</a> | A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL. |

### Portfolios Tutorials and Workflows

| File Type                                                                         | Name                               | Description                                                               |
|-----------------------------------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------|
|  | <a href="#">BQL for Portfolios</a> | A spreadsheet tutorial that shows how to access portfolio data using BQL. |

### BQL for AIM

BQL provides an efficient and flexible method of retrieving a wide range of data about your accounts in the Asset and Investment Manager (AIM).

BQL allows you to retrieve current day holdings for AIM Accounts, enriching the power and flexibility offered by BQL by complementing it with the intraday nature of the AIM Account holdings. A wide variety of Microsoft® Excel templates allow you to interact with AIM data to complement decision-making processes and display your AIM holdings in a customized way.

### AIM Docs and Media

The following tables contain Microsoft® Excel workbooks and templates created by Bloomberg to help you analyze AIM Account data using BQL.

The templates, in cell-referenced format, allow you to harness BQL functions, including *group()*, *sort()*, *groupSort()*, and *matches()*, to aggregate and filter across both AIM Accounts and Account Groups.

### AIM Reference

| File Type                                                                           | Name                               | Description                                                                                                  |
|-------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Function Reference</a> | A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL. |

## AIM Tutorials and Workflows

| File Type                                                                           | Name                                         | Description                                                                                                                                                                        |
|-------------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">AIM-BQL Quick Formula Guide</a>  | A spreadsheet tutorial introducing AIM syntax, BQL.QUERY(), and BQuant across equity, fixed income, and general use cases. Includes a list of functions and additional parameters. |
|    | <a href="#">FWP: Firm Wide Position</a>      | A spreadsheet template designed to replicate the <i>Firm-Wide Position</i> (FWP) function, exposing detailed position information for a specified Account Group.                   |
|    | <a href="#">Field Query (Acc)</a>            | A spreadsheet template that queries ten data items with column headers for a specified Account.                                                                                    |
|    | <a href="#">Field Query (Acc Group)</a>      | A spreadsheet template that queries ten data items with column headers for a specified Account Group.                                                                              |
|    | <a href="#">Sorting (Acc)</a>                | A spreadsheet template that sorts all holdings of an Account by a specified field from the <i>Field Search</i> (FLDS) function (e.g., GICS_SECTOR_NAME).                           |
|  | <a href="#">Sorting (Acc Group)</a>          | A spreadsheet template that sorts all holdings of an Account Group by a specified FLDS field (e.g., ACCOUNT_CODE).                                                                 |
|  | <a href="#">Group Aggregate (Acc)</a>        | A spreadsheet template that pulls a grouping level aggregate (e.g., by CLASSIFICATION_NAME) for five data items, with column headers, for a specified Account.                     |
|  | <a href="#">Group Aggregate (Acc Group)</a>  | A spreadsheet template that pulls a grouping level aggregate (e.g., by CLASSIFICATION_NAME) for five data items, with column headers, for a specified Account Group.               |
|  | <a href="#">Double Group Aggregate (Acc)</a> | A spreadsheet template that pulls a grouping level aggregate (e.g., by RTG_SP and YEAR(MATURITY)) for five data items, with column headers, for a specified Account.               |

| File Type                                                                           | Name                                                 | Description                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#"><u>Single Filter (Acc)</u></a>           | A spreadsheet template that filters all holdings of a specified Account by defining a data item filter and desired output. The example filters by all holdings with a Strategy TAG of 'ST1'.                                                                                                                  |
|    | <a href="#"><u>Single Filter with Sort (Acc)</u></a> | A spreadsheet template that filters and sorts all holdings of an Account using a specified data item filter, output, and sorting criteria. The example filters by all holdings with an ASSET_TYPE of 'Corporate Bond', sorted by Moody rating in descending order.                                            |
|    | <a href="#"><u>Double Filter (Acc)</u></a>           | A spreadsheet template that filters all holdings of a specified Account using two predefined data item filters. The example filters by all holdings with an ASSET_TYPE of 'Corporate Bond' and CRNCY of 'EUR'.                                                                                                |
|    | <a href="#"><u>Double Filter with Sort (Acc)</u></a> | A spreadsheet template that filters and sorts all holdings of a specified Account using two predefined data item filters, both filter outputs and sorting criteria. The example filters by all holdings with an ASSET_TYPE of 'Corporate Bond' and CRNCY of 'EUR', sorted by Moody rating in ascending order. |
|  | <a href="#"><u>Cash Filter (Acc Group)</u></a>       | A spreadsheet template that queries cash positions per Account for a specified currency within an Account Group.                                                                                                                                                                                              |
|  | <a href="#"><u>Cash: NAV-Cash MV (Acc Group)</u></a> | A spreadsheet template that calculates security market value per Account in a specified Account Group. The example calculation deducts Cash MV from the Fund MV.                                                                                                                                              |
|  | <a href="#"><u>Scoring Model (Acc)</u></a>           | A spreadsheet template that constructs a scoring model based on three predefined conditions (variables), with each condition warranting a score of 1, then filtering by all holdings of an Account that equal to a score of 3 (i.e., all three conditions are met).                                           |

## BQL for TOMS

TOMS BQL offers powerful querying capabilities to retrieve rich computed datasets from TOMS, Bloomberg's sell-side order management system.

TOMS BQL is a premium subscription and an add-on module to the TOMS suite of products. For more: [TRSK <GO> > TOMS BQL](#).

## BQL for ESG

BQL is an efficient and flexible method for retrieving a wide range of environmental, social, and governance (ESG) data from Bloomberg.

## ESG Docs and Media

The following tables contain spreadsheets, tutorials, and videos created by Bloomberg to assist you with performing data analysis for ESG using BQL.

### ESG Reference

| File Type                                                                         | Name                               | Description                                                                                                  |
|-----------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Function Reference</a> | A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL. |

### ESG Tutorials and Workflows

| File Type                                                                           | Name                                                        | Description                                                                                                                                      |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">BQL GHG Emissions</a>                           | A spreadsheet tutorial that shows you how to use GHG emissions data items in your BQL queries.                                                   |
|  | <a href="#">ESG Scores in BQL</a>                           | A spreadsheet template that shows detailed workflow examples using BQL sample queries with Bloomberg ESG score data items.                       |
|  | <a href="#">BQL for Green, Social, and Sustainable Debt</a> | A spreadsheet tutorial that shows you how to use BQL to analyze sustainable debt.                                                                |
|  | <a href="#">Sustainable Finance Revenue Exposure in BQL</a> | A spreadsheet tutorial that shows how to use BQL to analyze climate-related risks and transition opportunities within corporate revenue streams. |

ESG Universe Functions

The following topics provide information on universe functions for ESG data sets.

esgTicker()

In BQL() and BQL.Query() formulas in Excel, *esgTicker()* lets you translate a company's debt security into the equity ticker that will return ESG data for the company. In cases where a company's equity ticker does not return ESG data, *esgTicker()* searches the debt security's capital structure to find the equity ticker of the nearest parent company that returns ESG data.

The *esgTicker()* universe function identifies companies that appear on the Bloomberg ESG Data Index. The index tracks companies that disclose key performance indicators (KPI) that align with the ESG framework.

**Signature:** *esgTicker(symbols)*

where:

- symbols* is the ID of the debt security (e.g., '023135CB0 Corp').

Example Formulas:

Retrieve the equity ticker that returns ESG data for the company that issued the bond BK079386 Corp:

=BQL.Query("get(id()) for(esgTicker('BK079386 Corp'))",)

=BQL("esgTicker('BK079386 Corp')", "id()")

BQL for RMS

BQL is an efficient and flexible method for retrieving a wide range of Bloomberg Research Management Solutions (RMS) data, including note publication and readership history.

RMS Docs and Media

The following tables contain spreadsheets, tutorials, and videos created by Bloomberg to assist you with performing data analysis for RMS using BQL.

RMS Reference

| File Type                                                                           | Name                               | Description                                                                                                  |
|-------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Function Reference</a> | A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL. |

## RMS Tutorials and Workflows

| File Type                                                                         | Name                                                                  | Description                                                                                                       |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">BQL for RMS: In Depth</a>                                 | A spreadsheet that provides a formula guide and sample BQL queries for retrieving data related to research notes. |
|  | <a href="#">BQL for RMS Example Template: ESG Engagement Tracking</a> | A spreadsheet that demonstrates how to capture note metadata for tracking ESG engagement.                         |

## RMS Data Items

The following RMS data items allow you to specify the content you want to retrieve for a given RMS note. The note must be shared with an RMS community in order for BQL to retrieve data.

| Mnemonics                                               | Description                                                                                                                     | Example                                                              |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <code>note_brokers()</code>                             | Brokers or research providers tagged to the note.                                                                               | <code>note_brokers() == 'JP Morgan'</code>                           |
| <code>note_communities()</code>                         | Name of community that a note has been shared with.                                                                             | <code>note_communities() == 'US Equity Team'</code>                  |
| <code>note_corp_tickers()</code>                        | Corporate ticker(s) tagged to a note.                                                                                           | <code>note_corp_tickers() == 'IBM Corp'</code>                       |
| <code>note_countries_of_risk()</code>                   | Country/Countries of risk tagged to a note.                                                                                     | <code>note_countries_of_risk() == 'US'</code>                        |
| <code>note_creator()</code>                             | Author of a note.                                                                                                               | <code>note_creator() == 'Nick Joseph'</code>                         |
| <code>note_custom_taglist("CUSTOM TAGLIST NAME")</code> | Custom taglist values tagged to a note. These tags are created via TAGS <GO>.                                                   | <code>note_custom_taglist('Note Type') == 'Investment Update'</code> |
| <code>note_custom_tags()</code>                         | Custom tag values tagged to a note that do not exist in a tag list, typically applied in the form of hashtag (e.g., #Earnings). | <code>note_custom_tags() == 'Earnings'</code>                        |

| Mnemonics                           | Description                                    | Example                                                                                                                                                                                                                                                                                |
|-------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>note_event_date()</code>      | The event date of a note.                      | <code>note_event_date() == '2/15/2022'</code>                                                                                                                                                                                                                                          |
| <code>note_event_types()</code>     | The type of event.                             | <code>Note_event_types() == 'Analyst Call(1-1)'</code>                                                                                                                                                                                                                                 |
| <code>note_headline()</code>        | The title or headline of a note.               | <code>note_headline() == 'IBM Meeting Note'</code>                                                                                                                                                                                                                                     |
| <code>note_industries()</code>      | Industries tagged to a note.                   | <code>note_industries() == 'IT Services'</code>                                                                                                                                                                                                                                        |
| <code>note_industry_groups()</code> | Industry Groups tagged to a note.              | <code>note_industry_groups() == 'Technology Services'</code>                                                                                                                                                                                                                           |
| <code>note_last_updated()</code>    | Last update date of a note.                    | <code>note_last_updated() &gt;= -12m</code>                                                                                                                                                                                                                                            |
| <code>note_people()</code>          | People tagged to a note.                       | <code>note_people() == 'Elon Musk'</code>                                                                                                                                                                                                                                              |
| <code>note_primary_tickers()</code> | The primary ticker of a note.                  | <code>note_primary_tickers() == 'AAPL US Equity'</code>                                                                                                                                                                                                                                |
| <code>note_reads()</code>           | The total number of reads a note has received. | <code>note_reads().INTERACTION_ID == 'c204005d-abfa-4d83-9a92-fb9a5300dbb7'</code><br><br><code>note_reads().INTERACTION_BUYSIDE_CONTACT == 'Moritz Hamann'</code><br><br><code>note_reads().INTERACTION_DATE == '04/04/2022'</code><br><br><code>note_reads() == '6.56999E+18'</code> |
| <code>note_securities()</code>      | Securities tagged to a note.                   | <code>note_securities() == 'IBM US Equity'</code>                                                                                                                                                                                                                                      |
| <code>note_teams()</code>           | The RESM team of the author of the note.       | <code>note_teams() == 'Large Cap Team'</code>                                                                                                                                                                                                                                          |

**Example Formulas:**

- Retrieve all creators of notes about IBM US Equity:  
`get(note_creator()) for(filter(NotesUniv(type = RESEARCH), note_securities() == "IBM US Equity"))`

```
=BQL.Query("get(note_creator()) for(filter(NotesUniv(type = RESEARCH), note_securities() == 'IBM US Equity'))")
```

- Retrieve all notes with MyCustomTaglist tags displayed (or a blank if none present):  
get(note\_custom\_taglist("RESA Note Type")) for(NotesUniv(type = RESEARCH))

```
=BQL.Query("get(note_custom_taglist('RESA Note Type')) for(NotesUniv(type = RESEARCH))")
```

- Retrieve all notes with the MyTag tag from the MyCustomTaglist taglist:  
get(id) for(filter(NotesUniv(type = RESEARCH), note\_custom\_taglist("RESA Note Type") == "Investment Update"))

```
=BQL.Query("get(id) for(filter(NotesUniv(type = RESEARCH), note_custom_taglist('RESA Note Type') == 'Investment Update'))")
```

- Show all stocks that have research published in the last year from the community RESA DEMO:  
get(id) for(filter(NotesUniv(type = RESEARCH), note\_last\_updated() >= -1y AND note\_communities() == "RESA DEMO"))

```
=BQL.Query("get(id) for(filter(NotesUniv(type = RESEARCH), note_last_updated() >= -1y AND note_communities() == 'RESA DEMO'))")
```

- Show all stocks that have research published in the last year, filtered by note creator Nick Joseph:  
get(first(group(note\_securities(), note\_securities())) for(filter(NotesUniv(type = RESEARCH), note\_last\_updated() >= -1y AND note\_creator() == "Nick Joseph"))

```
=BQL.Query("get(first(group(note_securities(), note_securities())) for(filter(NotesUniv(type = RESEARCH), note_last_updated() >= -1y AND note_creator() == 'Nick Joseph'))")
```

- Count the number of notes for all securities within my portfolio (including zeros):  
let(#note\_universe=NotesUniv(type = RESEARCH);

```
#condition1=note_last_updated() >= -14m;
```

```
#condition2 = note_communities() == 'RESA DEMO';
```

```
#condition3 = note_securities() != NA;
```

```
#note_universe_final=filter(#note_universe, #condition1 and #condition2 and #condition3);
```

```
#portfolio_of_names=members('U7310996-13',type=PORT);
```

```
#names=value(note_securities(),#note_universe_final);
```

```
#name_count=groupsort(count(matches(#names,#names==ID)),order=asc);
```

```
get(#name_count)
```

```
for(translatesymbols(#portfolio_of_names,targetidtype='composite'))
```

- Give me the count of notes for all securities in the Dow Jones:  
get(note\_securities()) for(Notes(Members('INDU Index')))

- Show the author, headline, and number of reads for every note created in the last year:

```
=BQL.Query("get(note_creator(), note_headline(), count(note_reads)) for(filter(notesuniv(type=RESEARCH),
note_last_updated() >= -1y))")
```

## BQL for Alternative Data

BQL is an efficient and flexible method for retrieving alternative data from the Bloomberg database, including Bloomberg Second Measure transaction data and Placer.ai foot traffic data, also available in the *Alternative Data Company Analysis* (ALTD) function.

### Alternative Data Docs and Media

The following table contains materials created by Bloomberg to assist you with analyzing alternative data using BQL.

| File Type                                                                         | Name                                            | Description                                                                                                                        |
|-----------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">BQL for Alternative Data (ALTD)</a> | A spreadsheet tutorial that shows how to retrieve and analyze alternative data in Excel using BQL.                                 |
|  | <a href="#">Alternative Data BQL Templates</a>  | A spreadsheet consisting of ready-made alternative data templates.                                                                 |
|  | <a href="#">ALTD KPI Estimates in BQL</a>       | A spreadsheet tutorial that shows how to use ALTD fields for forecasting public company financials and key performance indicators. |

## BQL for Commodities

BQL is an efficient and flexible method for retrieving a wide range of commodities data from the Bloomberg database, including commodity curves and Commitment of Traders (COT) positioning data.

### Commodities Docs and Media

The following tables contain materials created by Bloomberg to assist you with analyzing commodities data using BQL.

#### Commodities Reference

| File Type                                                                           | Name                               | Description                                                                                                  |
|-------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
|  | <a href="#">Function Reference</a> | A reference guide to functions you can use to perform calculations, statistical analysis, and more with BQL. |

## Commodities Tutorials and Workflows

| File Type                                                                           | Name                                                            | Description                                                                                                                                                                  |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">BQL for Commodities</a>                             | A spreadsheet tutorial that provides an overview of using BQL to analyze commodities, including query construction and futures chains.                                       |
|    | <a href="#">Commodity Curvemembers in BQL Tutorial</a>          | A spreadsheet tutorial that shows how to retrieve and analyze pricing data for commodity curves using BQL.                                                                   |
|    | <a href="#">COT in BQL Tutorial</a>                             | A spreadsheet tutorial that shows how to download Commitment of Traders (COT) positioning data to Excel using BQL, with examples demonstrating all the available parameters. |
|    | <a href="#">COT Workflows in BQL</a>                            | A workflow tool demonstrating how to retrieve COT data using BQL, with detailed workflow examples of seasonality charts, correlation analysis, and relative value z-scores.  |
|   | <a href="#">COT BQL Instruction Manual</a>                      | A guide that provides comprehensive instructions for using the COT BQL dataset through BQL.                                                                                  |
|  | <a href="#">BQL for Commodity Product Universe</a>              | A spreadsheet tutorial that shows how to retrieve pricing and reference data for the commodity product universe in BQL.                                                      |
|  | <a href="#">BQL for Commodity Product Universe Training Doc</a> | A reference guide to the <i>commodities()</i> universe, including available data items, Commodity Product ID retrieval, and practical examples.                              |

## BQL for Derivatives

BQL is an efficient and flexible method for retrieving a wide range of derivatives data from the Bloomberg database, such as historical options data.

### Derivatives Docs and Media

The following table contains materials created by Bloomberg to assist you with analyzing derivatives data using BQL.

| File Type                                                                         | Name                                          | Description                                                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">BQL Historical Equity Options</a> | A spreadsheet tutorial that shows how to analyze historical data for exchange-listed equity and equity index options using BQL. |

## Excel Formula Reference

### BQL() Formulas

The following topics explain how to use the BQL() syntax to return current and historical data across asset classes from the Bloomberg Cloud.

### BQL() Syntax

The syntax for BQL() formulas follows the standard Bloomberg API syntax: ticker(s) (aka the *Universe*), data item(s) (aka the *Expression*), optional parameters, and optional local variables (aka *Local Variables*).

Each input can either be a literal value or a cell reference. For more information on cell referencing: [DAPI Help Page > Cell Referencing](#).

**Syntax:** =BQL("Universe", "Expression", "Optional Parameters", "Optional Local Variables")

- **Universe** can contain one or more comma-separated tickers or the *members()* function and an index ticker, which lets you create a universe containing all the members of the specified index.
- **Expression** can contain one or more comma-separated data items and, if desired, optional parameters that control the format or calculation of the data item.
- **Optional Parameters** that apply to all specified data items are each surrounded by their own set of quotation marks and separated by commas.
- **Optional Local Variables** can contain one or more variable name and value pairs. Each name and value pair is surrounded by its own set of quotation marks and separated by commas. Each name begins with #.

For example:

- To retrieve the current last prices for IBM and Microsoft:

=BQL("IBM US Equity, MSFT US Equity", "px\_last")

- To retrieve a historical time series of last prices for IBM over the last month, converted to EUR:

=BQL("IBM US Equity", "px\_last", "dates=range(-1M, OD)", "currency=EUR")

or

=BQL("IBM US Equity", "px\_last(dates=range(-1M, OD), currency=EUR)")

- To retrieve a historical time series of last prices and low prices for IBM over the last month, converted to EUR:  
=BQL("IBM US Equity", "px\_last, px\_low", "dates=range(-1M, OD)", "currency=EUR")
- To retrieve a historical time series of last prices and low prices for IBM over the last month using #lastPrice as a variable for last price and #lowPrice as a variable for low price:  
=BQL("IBM US Equity", "#lastPrice, #lowPrice", "dates=range(-1M, OD)", "#lastPrice=px\_last", "#lowPrice=px\_low")

**Note:** BQL() formulas are case-insensitive.

## Current/Historical Data

Using a BQL() formula, you can easily download a current data point or a historical time series of data for a security.

- **Current:** To download the current last price for a security, you just need the ticker and the data item:  
=BQL("IBM US Equity", "px\_last")
- **Historical Data Point:** To download a single historical price for a security, you need the ticker, data item, and the [As Of Date](#) parameter. You can enter a relative date (e.g., -2D) or an absolute date (in the YYYY-MM-DD format):  
=BQL("IBM US Equity", "px\_last(dates=-1Y)")  
  
=BQL("IBM US Equity", "px\_last(dates=2014-01-07)")
- **Historical Data Series:** To download a historical time series of price for a security, use the *Dates* parameter with the *range()* function to enter the absolute or relative date range:  
=BQL("IBM US Equity", "px\_last(dates=range(2014-01-07, 2014-01-15))")  
  
=BQL("IBM US Equity", "px\_last(dates=range(-1M, OD))")

**Note:** The relative date "OD" indicates today.

Additional parameters let you control how historical data is downloaded. For example, [Periodicity](#) (per) and [Fill When Data is Missing](#) (fill) let you choose the sampling frequency of the data and fill in values for non-trading days:

=BQL("IBM US Equity", "px\_last(dates=range(2014-01-01, 2016-01-15), per=M, fill=PREV)")

## Index Members

*members()* lets you create a universe in your BQL() formula by decomposing one or more indices into a list of their constituents.

If you specify a index ticker in your universe without using *members()*, you retrieve data for the index itself. For example, to retrieve the last price of the Euro Stoxx 50 Index:

```
=BQL("SX5E Index", "px_last")
```

However, if you want to retrieve the last price of each index member, just add *members()* to your universe:

```
=BQL("members(['SX5E Index'])", "px_last")
```

## BQL.Query() Formulas

The following topics explain how to use the BQL.Query() syntax to return current and historical data across asset classes from the Bloomberg Cloud.

### BQL.Query() Syntax

BQL.Query() formulas allow you to use raw BQL strings to retrieve data from Bloomberg.

The syntax for BQL.Query() formulas is unlike other Bloomberg API formulas. BQL.Query uses a *get()*, a *for()*, an optional *with()*, and an optional *let()* clause instead of the standard Bloomberg API "ticker", "field", "optional parameter" syntax.

**Syntax:** =BQL.Query("<let(> get() for() <with(>")

The *get()* clause specifies the type of data you want to download (aka the *Expression*) and the *for()* clause specifies the securities from which you want to download the data (aka the *Universe*). The *with()* clause lets you specify optional parameters that apply to all data items in the query (aka *Global Parameters*). The *let()* clause assigns variable names to expression values that you can reuse throughout your formula (aka *Local Variables*).

- **let()** (Optional) (*Local Variables*) can contain one or more variable name and value pairs. Each name and value pair begins with # and ends with a semi-colon (e.g., #lastPrice=px\_last(dates=range(-2M, OD));). For more: [let\(\) Clause](#).
- **get()** (the *Expression*) can contain one or more comma-separated data items. Optional parameters that control the format or calculation of a specific data item are enclosed within parentheses adjacent to the data item and separated by commas (e.g., px\_last(dates=-1M)). For more: [get\(\) Clause](#).
- **for()** (the *Universe*) can contain one or more comma-separated tickers within square brackets, each surrounded by a set of single quotation marks (e.g., ['IBM US Equity', 'AAPL US Equity']). It can also contain the *members()* function and an index ticker, which creates a universe that includes all members of the specified index (e.g., members(['UKX Index'])). For more: [for\(\) Clause](#).
- **with()** (Optional) (*Global Parameters*) can contain parameters that control the format or calculation of all data items in your BQL.Query formula and are comma-separated (e.g., dates=-1M, per=W). For more: [with\(\) Clause](#).

For example:

- To retrieve the current last prices for IBM and Microsoft:  
=BQL.Query("get(px\_last) for(['IBM US Equity', 'MSFT US Equity'])")
- To retrieve a historical time series of last prices for IBM over the last month, converted to EUR:  
=BQL.Query("get(px\_last(dates=range(-1M, OD), currency=EUR)) for(['IBM US Equity'])")
- To retrieve a historical time series of last prices and low prices for IBM over the last month, converted to EUR:  
=BQL.Query("get(px\_last, px\_low) for(['IBM US Equity']) with(dates=range(-1M, OD), currency=EUR)")

- To retrieve a historical time series of last prices and low prices for IBM over the last month using #lastPrice as a variable for last price and #lowPrice as a variable for low price:  
`=BQL.Query("let(#lastPrice=px_last; #lowPrice=px_low;) get(#lastPrice, #lowPrice) for(['IBM US Equity']) with(dates=range(-1M, OD))")`

**Note:** BQL.Query() formulas are case-insensitive.

## get() Clause

The `get()` clause in a BQL.Query() formula lets you specify the type of data you want to download.

The minimum you can specify within the `get()` clause is a BQL data item. For example, `px_last` (*Last Price*), `px_volume` (*Volume*), and `bics_level_1_sector_name` (*BICS Level 1 Sector Name*).

Within the clause, you can also specify different combinations of optional parameters for each data item to obtain customized data.

For example:

- Data Items:** To download the last prices for IBM and Microsoft:  
`=BQL.Query("get(px_last) for(['IBM US Equity', 'MSFT US Equity'])")`
- Parameters:** To download a time series of daily prices for IBM between January 7, 2014 and January 15, 2014:  
`=BQL.Query("get(px_last(dates=range(2014-01-07, 2014-01-15))) for(['IBM US Equity'])")`

**Note:** `get()` clauses can contain individual cell or cell range references, literal references (as in the examples above), or a combination of both. For more on cell referencing: [DAPI Help Page > Cell Referencing](#).

## let() Clause

The `let()` clause in a BQL.Query() formula lets you assign variable names to expression values, which you can use repeatedly throughout your formula and update all instances at once.

You can specify one or more variable and value pairs. Each variable and value pair must end with a semi-colon and each variable must begin with #.

For example:

- Single Variable:**  
`=BQL.Query("let(#marketCap=eqy_sh_out*px_last;) get(#marketCap/groupAvg(#marketCap, bics_level_1_sector_name)) for(members(['UKX Index']))")`
- Multiple Variables:**  
`=BQL.Query("let(#name=name; #price=px_last().value; #zChange=abs(groupZScore(day_to_day_total_return(fill=PREV)).value);) get(dropNA(matches(#zChange, #zChange >= 1.5), remove_id=TRUE)) for(members(['UKX Index']))")`

**Note:** `let()` clauses can contain individual cell or cell range references, literal references (as in the examples above), or a combination of both. For more on cell referencing local variables: [Cell Referencing](#).

## for() Clause

The *for()* clause in a BQL.Query() formula lets you specify the universe of tickers you want to analyze.

Each ticker you specify must be enclosed in single quotation marks, e.g., 'IBM US Equity'. The entire list of one or more tickers must be enclosed within a set of square brackets, e.g., ['IBM US Equity', 'MSFT US Equity', 'AAPL US Equity'].

For example:

- **Single Security:**  
=BQL.Query("get(px\_last) for(['IBM US Equity'])")
- **Multiple Securities:**  
=BQL.Query("get(px\_last) for(['IBM US Equity', 'MSFT US Equity', 'AAPL US Equity'])")
- **Members of an Index:**  
=BQL.Query("get(px\_last) for(members(['UKX Index']))")

For more on *members()*: [Index Members](#).

**Note:** *for()* clauses can contain individual cell or cell range references, literal references (as in the examples above), or a combination of both. For more on cell referencing: [DAPI Help Page > Cell Referencing](#).

## with() Clause

The *with()* clause in a BQL.Query() formula lets you specify global optional parameters that control the format or calculation of all data items in your formula at once.

You can specify one or more parameters. Each parameter and value set must be comma-separated.

For example:

- **Global Parameters:**  
=BQL.Query("get(px\_last, px\_low) for(['IBM US Equity']) with(currency=EUR, dates=range(-1M, 0D))")
- **Global Parameters and Data Item-Specific Parameters:**  
=BQL.Query("get(px\_last(ca\_adj=RAW), px\_low) for(['IBM US Equity']) with(currency=EUR, dates=range(-1M, 0D))")

**Note:** *with()* clauses can contain individual cell or cell range references, literal references (as in the examples above), or a combination of both. For more on cell referencing: [DAPI Help Page > Cell Referencing](#).

## Current/Historical Data

Using a BQL.Query() formula, you can easily download a current data point or a historical time series of data for a security.

- **Current:** To download the current last price for a security, you just need the ticker and the data item:  
=BQL.Query("get(px\_last) for(['IBM US Equity'])")
- **Historical Data Point:** To download a single historical price for a security, you need the ticker, data item, and the [As Of Date](#) parameter. You can enter a relative date (e.g., -2D) or an absolute date (in the YYYY-MM-DD format):

```
=BQL.Query("get(px_last(dates=-1Y)) for(['IBM US Equity']))")
```

```
=BQL.Query("get(px_last(dates=2014-01-07)) for(['IBM US Equity']))")
```

- **Historical Data Series:** To download a historical time series of price for a security, use the *Dates* parameter with the *range()* function to enter the absolute or relative date range:

```
=BQL.Query("get(px_last(dates=range(2014-01-07, 2014-01-15))) for(['IBM US Equity']))")
```

```
=BQL.Query("get(px_last(dates=range(-1M, 0D))) for(['IBM US Equity']))")
```

**Note:** The relative date "0D" indicates today.

Additional parameters let you control how historical data is downloaded. For example, [Periodicity](#) (per) and [Fill When Data is Missing](#) (fill) let you choose the sampling frequency of the data and fill in values for non-trading days:

```
=BQL.Query("get(px_last(dates=range(2014-01-01, 2016-01-15), per=M, fill=PREV)) for(['IBM US Equity']))")
```

## Index Members

*members()* lets you create a universe in your BQL.Query() formula by decomposing one or more indices into a list of their constituents.

If you specify a index ticker in your universe without using *members()*, you retrieve data for the index itself. For example, to retrieve the last price of the Euro Stoxx 50 Index:

```
=BQL.Query("get(px_last) for(['SX5E Index']))")
```

However, if you want to retrieve the last price of each index member, just add *members()* to your universe:

```
=BQL.Query("get(px_last) for(members(['SX5E Index']))")
```

## Cell Referencing

When creating a BQL() or BQL.Query() formula that includes multiple instances of the same expression, you can save time by cell referencing a local variable. Referencing a local variable allows you to use an expression several times throughout your formula and update all the instances at once by changing the referenced cell value. You can include one or more local variables in your formula by entering a cell reference or a cell reference range.

To reference one or more local variables, type the variable name(s) and value(s) in adjacent cells and then reference the cells or cell range at the end of your formula.

**[Hint]** To add a cell reference to a formula, you can simply click the corresponding cell while typing the formula.

For example:

- Referencing individual cells in BQL.Query() formulas:  

```
=BQL.Query("get(#eps) for(['IBM US Equity']), A1,B1)
```

|   | A                                                      | B                                      | C |
|---|--------------------------------------------------------|----------------------------------------|---|
| 1 | #eps                                                   | is_eps(fpt=A,<br>dates=range(-5D, 0D)) |   |
| 2 | =BQL.Query("get(#eps) for(['IBM US Equity'])", A1, B1) |                                        |   |

Cell A1 contains the name of the variable (#eps) and cell B1 contains the expression (is\_eps, fpt=A, dates=range(-5D, 0D)).

- Referencing a range of cells in BQL.Query() formulas:  
=BQL.Query("get(#appleReturn-avg(#spxReturn)/std(#spxReturn)) for(['SPX Index'])", A1:B2)

|   | A                                                                                         | B                                                  | C | D |
|---|-------------------------------------------------------------------------------------------|----------------------------------------------------|---|---|
| 1 | #appleReturn                                                                              | value(day_to_day_total_return, ['AAPL US Equity']) |   |   |
| 2 | #spxReturn                                                                                | day_to_day_total_return(dates=range(-1Y, 0D))      |   |   |
| 3 | =BQL.Query("get(#appleReturn-avg(#spxReturn)/std(#spxReturn)) for(['SPX Index'])", A1:B2) |                                                    |   |   |

Cell A1 contains the name of the first variable (#appleReturn), cell B1 contains the first expression (value(day\_to\_day\_total\_return, ['AAPL US Equity'])), cell A2 contains the name of the second variable (#spxReturn), and cell B2 contains the second expression (day\_to\_day\_total\_return(dates=range(-1Y, 0D))).

- Referencing individual cells in BQL() formulas:  
=BQL("IBM US Equity", "#eps", A1, B1)

|   | A                                     | B                                      |
|---|---------------------------------------|----------------------------------------|
| 1 | #eps                                  | is_eps(fpt=A,<br>dates=range(-5D, 0D)) |
| 2 | =BQL("IBM US Equity", "#eps", A1, B1) |                                        |

Cell A1 contains the name of the variable (#eps) and cell B1 contains the expression (is\_eps(fpt=A, dates=range(-5D, 0D))).

- Referencing a range of cells in BQL() formulas:  
=BQL("SPX Index", "#appleReturn-avg(#spxReturn)/std(#spxReturn)", A1:B2)

|   | A                                                                        | B                                                  | C |
|---|--------------------------------------------------------------------------|----------------------------------------------------|---|
| 1 | #appleReturn                                                             | value(day_to_day_total_return, ['AAPL US Equity']) |   |
| 2 | #spxReturn                                                               | day_to_day_total_return(dates=range(-1Y, 0D))      |   |
| 3 | =BQL("SPX Index", "#appleReturn-avg(#spxReturn)/std(#spxReturn)", A1:B2) |                                                    |   |

Cell A1 contains the name of the first variable (#appleReturn), cell B1 contains the first expression (value(day\_to\_day\_total\_return, ['AAPL US Equity'])), cell A2 contains the name of the second variable (#spxReturn), and cell B2 contains the second expression (day\_to\_day\_total\_return(dates=range(-1Y, 0D))).

For more: [DAPI Help Page > Cell Referencing](#).

## Display Parameters

Display parameters let you change how the data returned by your BQL() or BQL.Query() formula is arranged in your spreadsheet:

### Show Dates

For BQL() and BQL.Query() formulas in Excel, the *Show Dates* (showdates) parameter lets you choose whether or not to return the dates associated with your data.

**Parameter Mnemonic:** showdates

**Valid Values:**

| Value      | Description                                                |
|------------|------------------------------------------------------------|
| False or N | Dates are not shown. <b>(Default for one data point)</b>   |
| True or Y  | Dates are shown. <b>(Default for multiple data points)</b> |

**Example Formulas:**

- Retrieve the last price of IBM and hide its as of date:  
 =BQL("IBM US Equity", "px\_last")  
  
 =BQL.Query("get(px\_last) for(['IBM US Equity'])")
- Retrieve the last price of IBM and show its as of date:  
 =BQL("IBM US Equity", "px\_last", "showdates=True")  
  
 =BQL.Query("get(px\_last) for(['IBM US Equity'])", "showdates=True")

### Show Headers

For BQL() and BQL.Query() formulas in Excel, the *Show Headers* (showheaders) parameter lets you choose whether or not to return column headers associated with your data.

**Parameter Mnemonic:** showheaders

**Valid Values:**

| Value      | Description                   |
|------------|-------------------------------|
| False or N | Column headers are not shown. |

| Value     | Description                                |
|-----------|--------------------------------------------|
| True or Y | Column headers are shown. <b>(Default)</b> |

**Example Formulas:**

- Retrieve the daily last price of IBM over the last month and show the column headers for the downloaded data:  
`=BQL("IBM US Equity", "px_last(dates=range(-1M, 0D))")`  
`=BQL.Query("get(px_last(dates=range(-1M, 0D))) for(['IBM US Equity'])")`
- Retrieve the daily last price of IBM over the last month and hide the column headers for the downloaded data:  
`=BQL("IBM US Equity", "px_last(dates=range(-1M, 0D))", "showheaders=False")`  
`=BQL.Query("get(px_last(dates=range(-1M, 0D))) for(['IBM US Equity']), "showheaders=False")`

**Show IDs**

For BQL() and BQL.Query() formulas in Excel with more than one ID (e.g., ticker) in the universe, the *Show IDs* (showids) parameter allows you to show or hide the IDs of your universe members in the output of your formula.

**Parameter Mnemonic:** showids

**Valid Values:**

| Value      | Description                     |
|------------|---------------------------------|
| False or N | IDs are not shown.              |
| True or Y  | IDs are shown. <b>(Default)</b> |

**Example Formulas:**

- Retrieve the daily last price of FTSE 100 Index members over the last month and show the security IDs in the output:  
`=BQL("members('UKX Index')", "px_last(dates=range(-1M, 0D))")`  
`=BQL.Query("get(px_last(dates=range(-1M, 0D))) for(members(['UKX Index']))")`
- Retrieve the daily last price of IBM over the last month and hide the security IDs in the output:  
`=BQL("IBM US Equity", "px_last(dates=range(-1M, 0D))", "showids=False")`  
`=BQL.Query("get(px_last(dates=range(-1M, 0D))) for(['IBM US Equity']), "showids=False")`

**Show Query**

For BQL() and BQL.Query() formulas in Excel, the *Show Query* (showquery) parameter allows you to show your formula as a raw

BQL query string instead of downloading data to your spreadsheet.

**Parameter Mnemonic:** showquery

**Valid Values:**

| Value      | Description                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------------------|
| False or N | Data is retrieved based on your BQL formula. <b>(Default)</b>                                                                          |
| True or Y  | Your BQL formula is displayed as a raw BQL query string. For example:<br><br>get(px_last(dates=range(-1M, OD))) for(['IBM US Equity']) |

**Example Formulas:**

- Retrieve the daily last price of IBM over the last month:  
=BQL("IBM US Equity", "px\_last(dates=range(-1M, OD))")  
  
=BQL.Query("get(px\_last(dates=range(-1M, OD))) for(['IBM US Equity'])")
- Show the raw BQL query string that retrieves the daily last price of IBM over the last month:  
=BQL("IBM US Equity", "px\_last(dates=range(-1M, OD))", "showquery=True")  
  
=BQL.Query("get(px\_last(dates=range(-1M, OD))) for(['IBM US Equity'])", "showquery=True")

## Transpose Axes

For BQL() and BQL.Query() formulas in Excel, the *Transpose Axes* (transpose) parameter lets you choose whether to return data tables vertically or horizontally.

**Parameter Mnemonic:** transpose

**Valid Values:**

| Value      | Description                                       |
|------------|---------------------------------------------------|
| False or N | Return data in a vertical table. <b>(Default)</b> |
| True or Y  | Return data in a horizontal table.                |

**Example Formulas:**

- Retrieve the daily last price of IBM over the last month and arrange the data vertically:  
=BQL("IBM US Equity", "px\_last(dates=range(-1M, OD))")  
  
=BQL.Query("get(px\_last(dates=range(-1M, OD))) for(['IBM US Equity'])")

- Retrieve the daily last price of IBM over the last month and arrange the data horizontally:  
`=BQL("IBM US Equity", "px_last(dates=range(-1M, OD))", "transpose=True")`  
  
`=BQL.Query("get(px_last(dates=range(-1M, OD))) for(['IBM US Equity'])", "transpose=True")`

## Sort Dates

For BQL() and BQL.Query() formulas in Excel, the *Sort Dates* (xlsort) parameter lets you choose whether to display dates in ascending or descending order when you are retrieving a time series of values.

**Parameter Mnemonic:** xlsort

**Valid Values:**

| Value | Description                                                                  |
|-------|------------------------------------------------------------------------------|
| ASC   | The list of values starts with the least recent date first. <b>(Default)</b> |
| DESC  | The list of values starts with the most recent date first.                   |

**Example Formulas:**

- Retrieve the last price of IBM over the last month, starting with the least recent date:  
`=BQL("IBM US Equity", "px_last(dates=range(-1M, OD))")`  
  
`=BQL.Query("get(px_last(dates=range(-1M, OD))) for(['IBM US Equity'])")`
- Retrieve the last price of IBM, starting with the most recent date:  
`=BQL("IBM US Equity", "px_last(dates=range(-1M, OD))", "xlsort=DESC")`  
  
`=BQL.Query("get(px_last(dates=range(-1M, OD))) for(['IBM US Equity']), "xlsort=DESC")`

## Group By Fields

For BQL() and BQL.Query() formulas in Excel, the *Group By Fields* (groupbyfields) parameter lets you choose whether results are grouped by identifier first and then by fields, or by fields first and then by identifier.

**Parameter Mnemonic:** groupbyfields

**Valid Values:**

| Value      | Description                                                |
|------------|------------------------------------------------------------|
| False or N | Results are arranged by identifier first. <b>(Default)</b> |

| Value     | Description                          |
|-----------|--------------------------------------|
| True or Y | Results are arranged by field first. |

**Example Formulas:**

- Retrieve the seven-day bid and ask prices of IBM and Vodafone, arranging results by ticker then field:  
`=BQL("IBM US Equity, VOD LN Equity", "px_bid, px_ask", "dates=range(-7D, -0D)")`  
`=BQL.Query("get(px_bid,px_last) for(['IBM US Equity', 'VOD LN Equity']) with (dates=range(-7D,- 0D))")`
- Retrieve the seven-day bid and ask prices of IBM and Vodafone, arranging results by field then ticker:  
`=BQL("IBM US Equity, VOD LN Equity", "px_bid, px_ask", "dates=range(-7D, -0D)", "groupbyfields=True")`  
`=BQL.Query("get(px_bid,px_last) for(['IBM US Equity', 'VOD LN Equity']) with (dates=range(-7D, -0D))", "groupbyfields=true")`

**Other Formulas**

Other formulas in Microsoft® Excel allow you to easily convert inputs (e.g., dates or securities) into formats that can be ingested in a BQL() or BQL.Query() formula.

**BQL.List()**

The BQL.List() formula allows you to convert one or more securities into the correct BQL list format, so you can reference the list in a BQL.Query() formula. Note that a BQL.List() output cannot be referenced in a BQL() formula, which automatically creates a list from the tickers it contains.

For example, the following formula creates a list that contains IBM, Microsoft, and Apple:

```
=BQL.List("IBM US Equity, MSFT US Equity, AAPL US Equity")
```

**Note:** The value within a BQL.List() formula can be a cell reference or literal values. For more on cell referencing: [DAPI Help Page > Cell Referencing](#).

**BQL.Date()**

The BQL.Date() formula allows you to convert a Microsoft® Excel date to the correct BQL format, so the output can be referenced in a BQL() or BQL.Query() formula. The only required value is an Excel date or the BToday() Bloomberg API formula.

For example, the following formula transforms the current date into the BQL format (YYYY-MM-DD):

```
=BQL.Date(BToday())
```

**Note:** The value within a BQL.Date() formula can be a cell reference or specified value (as in the example above). For more on cell referencing: [DAPI Help Page > Cell Referencing](#).

## Error Codes

The following table displays the most common error codes you may encounter while using BQL. Please contact the Help Desk for further assistance.

| Error                                                | Possible Cause                                                                                                                                                                                    | Resolution                                                                                                                                                                          |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| #N/A BQL Compute Capacity<br><br>#N/A Query Review   | You have exceeded your monthly BQL compute allocation for BQL requests to download Bloomberg data via the Desktop API. Your monthly BQL compute allocation resets at the beginning of each month. | Contact the Bloomberg Help Desk by pressing the <Help> key twice. For information on receiving live help while using Excel: <a href="#">DAPI Help Page &gt; Getting Live Help</a> . |
| #N/A BQL Query Timed Out<br><br>#N/A Query Timed Out | Your BQL query has exceeded the BQL response time limit.                                                                                                                                          | Optimize your BQL query to take less time to complete.                                                                                                                              |
| #N/A Daily Capacity                                  | You have exceeded the daily capacity for requests to download Bloomberg data via the Desktop API.<br><br>The daily capacity resets at the beginning of each day.                                  | For more: <a href="#">DAPI Help Page &gt; Daily Capacity Best Practices</a> .                                                                                                       |
| #N/A Review                                          | Your workflow requires a review by Bloomberg.                                                                                                                                                     | Contact the Bloomberg Help Desk by pressing the <Help> key twice. For information on receiving live help while using Excel: <a href="#">DAPI Help Page &gt; Getting Live Help</a> . |

## BQL News and Updates

### BQL Spotlights

The following link displays BQL Spotlights on BLP Tips for BQL Language on NSE <GO>. BQL Spotlights are monthly

spreadsheets that highlight using BQL to analyze timely topics in Excel.

To see recently published BQL Spotlight articles, click [BQL Spotlights](#) .

**Note:** BLP Tips for BQL Language contains a running list of updates to BQL, including articles and tutorials that explain how to use BQL to analyze current market trends. To subscribe to BLP Tips for BQL Language, click [NI BLPBQL <GO>](#) and select **Actions > Set Alert Delivery**.

## BQL Spotlights Archive

The following tables display BQL Spotlights articles published prior to June 2021.

### 2021

| Type                                                                                | Title                         | Topics                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------|
|    | <a href="#">April 2021</a>    | Peers; SFDR; Short interest tracker                                                          |
|    | <a href="#">March 2021</a>    | Deep estimates; RSI screener; Board tenure                                                   |
|    | <a href="#">February 2021</a> | Tease trends out of ESG ETFs; ESG equity ETFs; Universe analysis; Top funds; ESG performance |
|  | <a href="#">January 2021</a>  | 2020 Summary and 2021 Preview; All Spotlights 2020                                           |

### 2020

| Type                                                                                | Title                                                   | Topics                                                                |
|-------------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------|
|  | <a href="#">August 2020</a>                             | Rating outlook and Altman Z-score; Fund holdings comparison; Surprise |
|  | <a href="#">July 2020</a>                               | Fund holdings; Airline performance during Covid-19; Pandemic bonds    |
|  | <a href="#">April 2020</a>                              | Bond ETF NAV discount; Sustainable debt; Fallen angels                |
|  | <a href="#">March 2020 Special Edition: Coronavirus</a> | More ways to analyze Covid-19's development and impact                |
|  | <a href="#">March 2020</a>                              | Earnings date screen; Coronavirus impact; any(), all(), and in()      |

| Type                                                                              | Title                         | Topics                                                                                                            |
|-----------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">February 2020</a> | Custom peers; Real Effective Exchange Rate (REER); Bond ETF flows                                                 |
|  | <a href="#">January 2020</a>  | Election monitor; Bond liquidity analysis via TRACE; Credit Default Swaps (CDS); Spreadsheet Builder enhancements |

## 2019

| Type                                                                                | Title                          | Topics                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">December 2019</a>  | Fund family metrics; Cumulative fund flows; Bond total return; ESG analysis (from Bloomberg Markets magazine)                                                                            |
|    | <a href="#">November 2019</a>  | Spread evolution; Highest/lowest rating; Index valuation and Brexit; BQL Builder update                                                                                                  |
|    | <a href="#">October 2019</a>   | ESG data; Green bonds issuance; Issuer ESG screening                                                                                                                                     |
|   | <a href="#">September 2019</a> | Stock upside potential; Combining PMI and equity data; Winsorization                                                                                                                     |
|  | <a href="#">August 2019</a>    | Issuer navigation; Mapping equity data to FI data; Handling unavailable data                                                                                                             |
|  | <a href="#">July 2019</a>      | All year-to-date BQL Spotlights, seminars, and Functions for the Market (FFM) stories                                                                                                    |
|  | <a href="#">June 2019</a>      | New fund metrics; Probability of default; Sector/industry classifications using classification_name(); BQL with worksheets (W <GO>); BQL with existing searches (SRCH <GO> and EQS <GO>) |
|  | <a href="#">May 2019</a>       | Case Study: Explore Chinese Bonds with top down analysis, index changes, and liquidity screens                                                                                           |
|  | <a href="#">April 2019</a>     | Earnings revision momentum; Green bond issuance; Spread watch list                                                                                                                       |
|  | <a href="#">March 2019</a>     | Fixed income index analysis; Bond/oil correlation; ETF dashboard                                                                                                                         |

| Type                                                                              | Title                         | Topics                                                                                   |
|-----------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------|
|  | <a href="#">February 2019</a> | Sector correlation matrix; Custom correlation matrix; Debt chains via debt(); BQL.List() |

### BQL Queries of the Week

The following link displays Queries of The Week (QoTW) articles on BLP Tips for BQL Language on NSE <GO>. QoTW articles show you tips and tricks for constructing custom analysis in Excel with BQL.

To see recently published QoTW articles, click [BQL Queries of the Week](#) .

**Note:** BLP Tips for BQL Language contains a running list of updates to BQL, including articles and tutorials that explain how to use BQL to analyze current market trends. To subscribe to BLP Tips for BQL Language, click [NI BLPBQL <GO>](#) and select **Actions > Set Alert Delivery**.

### Functions For the Market

The following link displays Functions for the Market (FFM) articles on NSE <GO>. FFM articles take a deep-dive approach to analyzing current events with BQL.

To see recently published BQL-related FFM articles, click [BQL Functions For the Market](#) .

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